

IMPERIAL

Department of Aeronautics



June 2026

UG Final Year Internal Project List

*Projects with a * next to the project code have been identified as requiring student effort to be evenly distributed between both the autumn and spring/summer terms. Students should keep this in mind when selecting projects and choosing optional modules. Projects with (S) after the Project Title indicates space related projects.*

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Aliabadi, Ferri Prof.

Project no: *FA01*
Project title: Environmental effects on damage detection using Structural Health Monitoring techniques (S) - 1 project available

Supervisor: Aliabadi, Ferri Prof.
Co-supervisor(s):
Category: Experimental;Computational
Software:
Confidential: No

Structural Health Monitoring (SHM) techniques are well established for impact and damage detection at laboratory conditions where the environmental factors are controlled. However, applying the SHM methodologies to real aircraft structure under real load conditions and environmental effects might not result in correct diagnosis of the structure. Therefore it is very important to study the effect of variable such as temperature and humidity on the wave propagation in composite structures. Most of the damage detection techniques based on guided waves are developed on the basis of comparing the response of the structure in its pristine state to the damaged state. This means that it is very important to study the influence of the environment on the sensor data when they are recorded under different conditions to avoid mis-detections or false alarms due to environmental factors. In addition pre-loading of the structures can also change the propagational properties of the baseline signals. Therefore the aim of the project is to identify the influence of the environmental effects together with the impact of the pre-loading on the wave propagation and possibly develop baseline free methods for damage detection in composite aircraft panels.

Aliabadi, Ferri Prof.

Project no: *FA02*
Project title: Passive sensing under operational condition, varying impactor size and shape - 1 project available

Supervisor: Aliabadi, Ferri Prof.
Co-supervisor(s):
Category: Experimental;Computational
Software:
Confidential: No

Passive sensing as a structural health monitoring technique results in impact detection and characterisation. Impacts above certain energy will cause barely visible impact damage (BVID) in composite. By having sensors permanently mounted on the structure, when there is an impact even, the generated waves are recorded the sensors. By post processing the recorded signals impact can be detected, localized and also characterized. There are various algorithms for detecting impact such as creating meta-models or transfer function method. However, their applicability under real operational conditions of the aircraft is questionable. The aim of this project is investigate the performance of passive sensing algorithms under various operational conditions, varying impactor size and shape and to assess their reliability.

Aliabadi, Ferri Prof.

Project no: FA03 *

Project title: XFEM/Boundary Element Analysis of Crack Growth (S) - 1 project available

Supervisor: Aliabadi, Ferri Prof.

Co-

supervisor(s):

Category: Computational;Theoretical

Software:

Confidential: No

Catastrophic fracture failure of engineering structures is caused by cracks that extend beyond a safe size. In this project crack growth processes are simulated with the boundary element method. Boundary element method is an alternative method to the more established finite element method. The attraction of the boundary element method is largely attributed to the reduction in the dimensionality of the problem; for two-dimensional problems, only the line-boundary of the domain needs to be discretized into elements, and for three-dimensional problems only the surface of the problem needs to be discretized. This means that, compared to the FEM, a boundary element analysis results in a substantial reduction in modelling effort. Boundary element has been particularly successful for crack growth analysis. The aim of the project is to extend the current in-house code to include thermal, centrifugal and other types of body forces.

A good level Abaqus Knowledge is essential.

Amato, Davide Dr

Project no: DA01 *

Project title: Long-term attitude evolution of space debris (S) - 1 project available

Supervisor: Amato, Davide Dr

Co-supervisor(s):

Category: Theoretical; Numerical; Analysis; Survey

Software: Any of Julia ecosystem, Python, MATLAB with Aerospace Toolbox

Confidential: No

The rapid growth of space debris orbiting the Earth poses a significant risk to the safety and sustainability of space activities due to the risk of in-space collisions, which would create more debris in a collisional cascade, commonly called Kessler's syndrome. To remediate the risk proposed by space debris, Active Debris Removal (ADR) missions have been proposed, in which a spacecraft (such as a "space tug") interacts with a large piece of debris to deorbit it or to move it into a safe, graveyard orbit. One of the primary design drivers for ADR missions is the attitude motion of the debris, as debris rotating with large angular velocities significantly complicate the design of ADR missions.

The goal of the project is to understand the long-term (over a scale of decades) attitude motion of space debris, especially in terms of their angular velocity. The supervisor and the student will identify several objects of interest for the risk that they pose to active spacecraft (such as SL-16 rocket boosters or the derelict ENVISAT satellite [1], along with a generic box-with-wings satellite), and establish whether they are subject to secular torques that would result in a long-term spin up through analytical and numerical means [2]. The results will be confirmed and validated through numerical simulations of the couple attitude and orbital motion through a suitable numerical method.

An outstanding background in orbital mechanics and a willingness to go beyond the module material, especially towards learning Julia, Python, and Git, are essential to carry out this project successfully. Knowledge of Julia, Python, and Git is a plus.

References:

[1] D. McKnight et al., 'Identifying the 50 statistically-most-concerning derelict objects in LEO', Acta Astronautica, vol. 181, pp. 282–291, Apr. 2021, doi: 10.1016/j.actaastro.2021.01.021.

[2] H. Schaub and J. L. Junkins, Analytical mechanics of space systems, Third edition. in AIAA education series. Reston, Virginia: American Institute of Aeronautics and Astronautics, Inc, 2014.

Amato, Davide Dr

Project no: DA02 *

Project title: Astrometry and photometry of space objects with the Compact Imperial College London OPTical Sensor (CICLOPS) (S) - 1 project available

Supervisor: Amato, Davide Dr

Co-

supervisor(s):

Category: Experimental: Ready-made experiment; Survey; Analysis; Computational

Software: Python ecosystem; MATLAB; SharpCap; 10Micron Model Creator; MountWizzard4

Confidential: No

A key challenge in Space Situational Awareness (SSA) is the lack of accurate and timely data on trajectories and brightness of space objects. It is therefore extremely useful to have access to optical observations of space objects through telescopes [1]. The Computational Astrodynamics group in the Department has been operating a small portable telescope for the discovery and tracking of resident space objects (i.e. artificial satellites and space debris), CICLOPS (Compact Imperial College London Optical Sensor) [2].

The CICLOPS telescope acquires images of satellite objects orbiting in the night sky. These must be reduced (i.e. processed) in order to extract trajectory data (astrometry) and brightness data (photometry). In this project, you will develop control and data processing pipelines for the operation of the telescope. This will involve:

- surveying literature and software available for telescope control and image processing in amateur astronomy;
- refining and validating pointing model (mount) calibration and optical calibration procedures;
- developing code and software to perform accurate pointing and tracking of objects from known trajectories;
- developing automatic image acquisition pipelines;
- developing an automatic image processing pipeline for data reduction;
- computing predicted sky positions and visibility of objects in Geostationary (GEO) and Low Earth Orbit (LEO);
- carry out observations, astrometry, and photometry of objects in GEO and LEO.

****IMPORTANT**:** The project will involve observation campaigns of satellites in Low and Geostationary Orbits. Telescope observations can only be carried out at nighttime, therefore you will be required to sporadically work out of hours. Observations can only be planned with short notice (~2 days notice) due to variable weather conditions. This project requires availability during out of hours and at short notice.

References:

[1] R.D. Coder, A.D. Jaunzemis, M.V. Mathew, J.L. Worthy, and M.J. Holzinger, "Georgia Tech Space Object Research Telescope", Journal of Spacecraft and Rockets 54(6), 2017, doi: 10.2514/1.A33852.

[2] E. Kaygorodov, M. Hjelmstad Rattigan, M.I. Hallgarten La Casta, and D. Amato, "Development and operation of CICLOPS, the Compact Imperial College London Optical Sensor", 9th Space Debris Conference, European Space Agency, Bonn, Germany, 2025.

[3] T. Schildknecht, "Optical surveys for space debris", The Astronomy and Astrophysics Review 14(1), 2007, doi: 10.1007/s00159-006-0003-9.

Amato, Davide Dr

Project no: *DA03 **

Project title: Dynamics and control of Very Low Earth Orbit spacecraft (S) - 1 project available

Supervisor: Amato, Davide Dr

Co-supervisor(s): Knoll, Aaron Dr

Category: Literature; Numerical; Design; Theoretical

Software: Julia ecosystem; MATLAB

Confidential: No

Very Low Earth Orbit (altitude between 120 and 250 km) is becoming a disruptive option for Earth observation and low-latency telecommunication spacecraft. Lower altitudes significantly improve resolution of EO payloads, reduces path delay and guarantees rapid post-mission decay. Until recently, drag made continuous operations impractical. That barrier is being removed by advanced propulsion concepts such as air-breathing electric propulsion (ABEP), which harvests atmospheric molecules to generate low-thrust drag compensation.

You will formulate a preliminary dynamics model for the simulation of VLEO orbits, along with a control algorithm based on previous work using Lyapunov control. The objective is to keep the spacecraft within a prescribed altitude band, avoiding accidental re-entry and guaranteeing the performance of the electric propulsion system.

In addition, you will investigate the potential of performing aeroassisted manoeuvres in VLEO: the idea consists of exploiting aerodynamic forces to perform manoeuvres that would not be feasible using purely the onboard propulsion system. This approach will be based on extending previous work in the department on using aerodynamic lift to change the orbital plane of the spacecraft and achieve the capability of rapidly passing over any point of the Earth at short notice, for instance to support emergency relief operations in disaster areas. The developed control algorithm will be a key enabler for upcoming VLEO missions. An outstanding background in orbital mechanics and a willingness to go beyond the module material is required. A solid background in control and interest in advanced control is also a requirement. Knowledge of the Julia programming language is a plus.

Amato, Davide Dr

Project no: *DA04 **

Project title: Accurate trajectory predictions of satellite constellations in Low Earth Orb (S) - 1 project available

Supervisor: Amato, Davide Dr
Co-supervisor(s):
Category: Computational; Literature; Analysis
Software:
Confidential: No

The near-Earth space environment is undergoing tremendous expansion driven by the launch and orbiting of large satellite constellations in Low Earth Orbit (LEO). Current plans for deployment of these constellations (such as Starlink, Kuiper, Qianfan, Guowang) imply that a scenario with more than 50 000 active satellites by 2036 is likely. To put this in context, this number is 10 times larger than all the satellites launched into orbit from the dawn of the space age until 2018.

It is difficult to predict trajectories of LEO satellites because of uncertainty in their physical parameters (size, shape, attitude, ...) and because we ignore their manoeuvre plans, which are necessary to maintain their orbits and are not communicated publicly by mission operators. Without accurate trajectory predictions we cannot warn other satellites of potential collisions; in addition, if satellites passes in the sky are not well predicted they blind astronomical telescopes on the ground, resulting in the loss of precious scientific data.

In this project, you will develop new methods for the accurate prediction of trajectories of large satellite constellations in Low Earth Orbit over time spans of up to 10 days. You will achieve this by addressing and mitigating both measurement uncertainty (noise in satellite observations) as well as process uncertainty, that is lack of acceleration knowledge due to unpredicted manoeuvres and unmodelled physics. You will study and implement high-fidelity models for natural accelerations (due to the Earth's asphericity, aerodynamics, and radiation pressure) as well as data-driven models to predict manoeuvres. You will estimate unknown spacecraft parameters through batch least-squares regression or Kalman filtering. You will compare your solutions to real data on satellite trajectories published in the US Space Catalogue, and to industry-standard trajectory prediction tools such as SGP4.

This project requires a strong background in Orbital Mechanics. A good background in Spacecraft Systems, Control Systems, and Optimisation is a plus. Willingness to learn the Julia programming language is also required.

Amato, Davide Dr

Project no: *DA05 **
Project title: Differentiable orbit propagation (S) - 1 project available

Supervisor: Amato, Davide Dr
Co-supervisor(s): Fung, Lloyd Dr
Category: Theoretical; Numerical; Literature
Software: Julia ecosystem

Confidential: No

In trajectory estimation and optimisation, we often need partial derivatives of a spacecraft state (that is, current position and velocity) with respect to some set of parameters such as initial conditions (that is, position and velocity in the past) or coefficients expressing physical properties (area, drag coefficient, ...). These partial derivatives tell us how sensitive the trajectory is to the selected parameter and are used to compute the parameters that provide the “best fit” for a given criterion. For example, by quantifying how sensitive the trajectory is to the drag coefficient, we can find the value of the drag coefficient that best explains observations of a spacecraft. Partial derivatives are also required to quantify how trajectory uncertainty evolves in time: by knowing how sensitive a trajectory is to changes in the state, we can compute how uncertain our future trajectory will be due to errors in the initial conditions. Computing partial derivatives can be done analytically only for simple problems; even in these, higher order partial derivatives result in expressions that are cumbersome and tedious to manipulate. Most often, partials are obtained through some form of finite differencing, which is however computationally expensive and only provides an approximation to the true values of the partials. Modern approaches to computing partials use Automatic Differentiation (AD), which consists of the calculation of partial derivatives through the systematic application of the chain rule. AD techniques and packages are very mature due to their usage in the training of machine learning models.

In this project, you will develop a new orbit prediction algorithm to propagate partial derivatives to spacecraft parameters using AD techniques. You will implement the AD techniques within an existing orbit propagation package, *Thalassa.jl*, which has implemented a high-fidelity physical model. You will demonstrate the power of AD techniques in challenging scenarios of practical utility, such as making robust trajectory predictions under uncertainty and improve estimates of collision probabilities between spacecraft. An excellent background in Computing and Numerical Methods 2 is essential. A solid background in Orbital Mechanics is a plus. Knowledge of AD techniques will also be covered in Fundamentals of Scientific Machine Learning. The project requires willingness to learn and code in the Julia language.

Amato, Davide Dr

Project no: *DA06 **

Project title: Fast orbit prediction through spectral methods (S) - 1 project available

Supervisor: Amato, Davide Dr

Co-supervisor(s):

Category: Numerical;Theoretical

Software:

Confidential: No

Orbit propagation is the process of predicting a spacecraft trajectory in the future from its initial position and velocity at a given time. This is a process of fundamental importance in astrodynamics and to enable autonomous operations in spacecraft guidance, navigation, and control.

Many different orbit propagation methods have been developed in the last 150 years. Coarsely speaking, any orbit propagation method involves a trade-off between accuracy and computational cost: the more accurate the solution, the higher the computational cost required to compute it. The main objective in orbit propagation is to maximise the propagation efficiency, i.e. to minimise the cost required to achieve a given accuracy. In the Computational Astrodynamics group of the Department of Aeronautics, we are developing new, highly efficient orbit propagation methods maximising propagation efficiency. The methods are specifically aimed at trajectory predictions of space debris and onboard predictions, applications in which it is important to compute trajectories of moderate accuracy with very low computational cost.

In this project, you will develop a new Chebyshev extrapolation method. The core idea is to approximate the evolution of the spacecraft trajectory within a fixed timespan through a Chebyshev expansion, which is analogous to a Fourier series. The coefficients of the expansion are then used to extrapolate the trajectory for all future times by enforcing continuity in the solution and in its derivatives. Conceptually, this is similar to fitting a low-order polynomial through data and perform extrapolation: the future evolution is predicted through a reduced-order model from past data.

You will compare the computational efficiency of this approach to state-of-the-art analytical methods for orbit propagation, including the analytical solution to Keplerian motion and SGP4, the industry standard for fast orbit propagation.

This project requires a solid background in Computing and Numerical Methods. A solid background, and an interest, in Orbital Mechanics are a plus.

Armanini, Sophie Dr

Project no: SA01

Project title: Take-off and landing control for a flapping-wing robot - 1 project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s):

Category: Experimental;Analysis;Design

Software: MATLAB, Simulink, optionally MSC Adams

Confidential: No

Bioinspired flapping-wing micro aerial vehicles (MAVs) are highly promising for many applications due to their extraordinary flight capabilities and their lightweight, part-soft structure, which makes them very safe. Real-world usage of these robots, however, requires a higher endurance, and autonomous take-off and landing capabilities.

The aim of this project is to investigate perched landing and take-off strategies (i.e. manoeuvres and control approaches) for bioinspired MAVs, building on previous research in the lab. The proposed methods will be tested using an existing simulation environment, as well as on an existing robotic prototype in the aerial robotics flight arena.

This project requires familiarity with flight dynamics and control, as well as an interest in hands-on drone development, hardware integration and flight testing.

Armanini, Sophie Dr

Project no: SA02

Project title: Control of a tilt-wing flapping-wing micro aerial vehicle
- 1 project available

Supervisor: Armanini, Sophie Dr

Co-

supervisor(s):

Category: Experimental;Design;Computational

Software:

Confidential: No

This project aims to design, implement and test a controller for a hummingbird-inspired flapping-wing micro aerial vehicle that can perform effectively in different flight modes (i.e., hover, high-speed directional flight, and glide). For this, the controller design will need to be validated and optimised using simulation tools and the flying hardware in our Flight Lab. Flapping-wing MAVs are unconventional aerial drones that utilise unsteady aerodynamic phenomena inspired by birds and insects. This approach provides efficient flight at small sizes, leading to new applications in the field of aerial robotics.

Our custom robot design features two separately motorised wings that can be tilted to a large degree independently. This novel configuration allows for the exploration of a wide range of possible control architectures. Furthermore, tilting wings enable the MAV to efficiently perform at different flight modes with exceptional manoeuvrability.

This project requires a good understanding of basic flight control.

Armanini, Sophie Dr

Project no: SA03

Project title: Design study for a multimodal tiltwing aerial robot - 1 project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s):

Category: Design;Experimental;Analysis

Software:

Confidential: No

This project aims to investigate and test new design and actuation strategies for a hummingbird-inspired flapping-wing micro aerial vehicle that can operate in different flight modes (i.e., hover, high-speed directional flight, and glide) and transition seamlessly between these. For this, areas of optimisation potential will be identified and solutions will be designed, manufactured and tested with an existing experimental setup. Optionally, solutions will be flight-tested in our Flight Lab.

Flapping-wing MAVs are unconventional aerial robots that fly using unsteady aerodynamic phenomena inspired by birds and insects. This approach provides efficient flight at small scales, leading to new applications in the field of aerial robotics.

Our custom robot design features two separately motorised wings that can be independently tilted by large angles. This novel configuration enables various flight modes and entails a broad solution space for design optimisation, at both global and component level (e.g. wings, actuation configuration, etc.)

Armanini, Sophie Dr

Project no: SA04

Project title: A Novel Modular Serial-Linkage Robot for High-Confinement Navigation - 1 project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s):

Category: Design;Computational;Analysis;Experimental

Software:

Confidential: No

Navigating amphibious, swamp and aquatic environments is still a critical challenge where traditional robotic systems often fail. This project will contribute to the design, modelling and testing of a novel bioinspired modular serial-linkage robot, specifically engineered for robust exploration in complex transition areas (from land to wetland and water). By using the high-redundancy kinematics of serpentine locomotion, the system achieves a multi-modal “slithering” gait that allows it to traverse cluttered terrains, dense vegetation, and various fluid-to-solid transitions where conventional robots get stuck.

The platform is designed with a reconfigurable, segment-based architecture, allowing the robot’s total length and degrees of freedom to be modified according to specific environmental constraints. This modularity integrates distributed sensing into an intelligent mobile system. Instead of relying on a fixed frame, the system uses its adaptable, multi-

segmented body to maintain a minimal footprint while navigating and sensing the most inaccessible "nooks and crannies" of both artificial infrastructures and sensitive ecological zones. The work contributes to the field of bioinspired kinematics and multi-modal robotics, offering a scalable and resilient solution for the inspection of hazardous, unstructured or unstable confined spaces.

The student project may focus on design exploration, modelling and simulation, and/or early-stage testing.

Armanini, Sophie Dr

Project no: SA05

Project title: HydroJumper: A Hybrid Aerial-Aquatic Platform for Resilient Wetland Exploration - 1 project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s):

Category: Design;Experimental;Computational;Analysis

Software:

Confidential: No

Navigating highly biodiverse swamp and wetland environments remains a frontier challenge where traditional wheeled or legged robots fail due to unstable mud-water interfaces. This project proposes HydroJumper, a novel hybrid aerial-aquatic robotic platform designed for intermittent jumping locomotion in complex transition zones. By integrating a micro-UAV thrust module with a passive variable-geometry landing skirt, the system overcomes the high energetic cost of continuous flight while avoiding the mechanical entrapment common in viscous swamp terrains.

The platform uses a unique passive morphing mechanism inspired by jellyfish kinematics to modulate drag. During the impulsive ascent phase, the skirt collapses to a minimum-drag configuration, allowing the UAV to efficiently pull the system from sticky substrates. Upon descent, the skirt autonomously deploys to a high-drag state, providing passive stabilization and impact reduction without the need for active control or complex actuators. This approach combines the manoeuvrability of aerial systems with the durability required for long-term ecological monitoring in harsh, unstructured wetlands. By combining hybrid locomotion and passive morphing structures, the proposed robot enables resilient sensing in delicate ecological regions.

This student project may focus on design, modelling, simulation and/or testing aspects, depending on the background of the student.

Armanini, Sophie Dr

Project no: SA06

Project title: Control strategies for dragonfly-inspired tailless microgliders - 1 project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s):

Category: Analysis;Computational;Design

Software:

Confidential: No

Dragonflies have remarkable flight capabilities: they are highly agile and efficient, have an extremely wide flight envelope, and can perform a wide variety of complex flight manoeuvres. They are also one of the few insects capable of unpowered gliding flight as well as active flapping flight. Based on initial dragonfly-inspired microglider designs, this project will investigate tailless gliding flight and unconventional actuation, stabilisation and control strategies, e.g. exploring the use of abdomen motion and mass shifts. The project may also investigate strategies to increase energy-efficiency and performance by transitioning between different flight modes. The developed approaches will be modelled and tested in simulation. Optionally, they may also be implemented on an existing robotic prototype and flight tested.

Armanini, Sophie Dr

Project no: SA07

Project title: Investigating formation flight for multirotor drones - 1 project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s): Rigas, Georgios Dr

Category: Experimental;Analysis

Software:

Confidential: No

In nature, birds and fish routinely move in coordinated formations. This enables them to conserve energy and mitigate gust effects by exploiting flow wakes. Translating these behaviours into aviation can reduce energy use and load fluctuations for aerial vehicles. While coordinated formation flight has been explored for fixed-wing aircraft, mainly in the

military context, it remains underexplored for small-scale multirotor drones. As drone applications are growing rapidly, energy-efficient operation is increasingly important. Effective formation flight may allow for lower power consumption, hence longer endurance/range for small drones, as well as improved resistance to gusts. The goal of this project is to investigate and characterise formation flight for small drones, especially multirotor configurations. Wind-tunnel testing will be used to measure the energy and load benefits of small formations (2-5 vehicles) compared to the solo flight case, under a variety of flight conditions, and while varying formation parameters (e.g. spacing, phase alignment, role-rotation frequency, etc.). Existing drones from the Aerial Robotics Lab will be used as test platforms. Both tethered and free-flight experimental setups will be considered.

Armanini, Sophie Dr

Project no: SA08

Project title: A framework for heterogeneous aerial robotic swarms - project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s):

Category: Computational;Analysis;Survey

Software: Python or MATLAB; potentially robotics simulation software e.g. Gazebo

Confidential: No

This project will investigate the potential of mixed-vehicle aerial robotic swarms, e.g. combining fixed-wing, multirotor, VTOL and/or flapping-wing vehicles, to improve overall versatility, performance and/or energy-efficiency. Starting from homogeneous aerial robotic swarm results in the literature, and simple dynamics models, this project will contribute towards a simulation framework to explore how vehicles with different dynamics can synergise, coordinate, and share information.

Armanini, Sophie Dr

Project no: SA09

Project title: Inflight icing effects on eVTOL air vehicles - 1 project available

Supervisor: Armanini, Sophie Dr

Co-supervisor(s):

Category: Computational;Analysis;Survey

Software:

Confidential: No

Inflight icing can lead to significant deterioration in aerodynamic performance and control authority, with potentially catastrophic consequences. Despite decades of research, icing remains a challenge in aviation, and poses a particularly hazard for small-scale vehicles operating at low altitudes, where ice is more likely to accrete.

As electric vertical take-off and landing (eVTOL) aircraft and drones emerge as promising platforms for advanced air mobility (AAM), understanding the effects of icing on these vehicles is essential. eVTOL aircraft are affected by icing in different/additional ways, compared to conventional rotorcraft or fixed-wing aircraft -- due to their wide flight envelope (including hover, cruise and transition regimes), their distinctive designs (e.g. multirotor or tilting-based), and their electric propulsion systems. This project aims to identify and characterise the effects of ice accretion on the performance and dynamics of eVTOL aircraft, in order to determine critical icing scenarios and inform ice protection strategies for next-generation AAM air vehicles.

Bilotti, Emiliano Dr

Project no: *EB01 **

Project title: Recyclable Fibre Reinforced Polymer (FRP) Composites for Wind Turbine Blades and Aircrafts (S) - 1 project available

Supervisor: Bilotti, Emiliano Dr

Co-supervisor(s): Pinho, Silvestre Prof.

Category: Experimental; Experimental: Structures

Software:

Confidential: No

Achieving NetZero by 2050 will require a significant increase in wind energy capacity over the next decades. For instance, if we are to power the current aviation industry with green hydrogen produced by wind turbines, this alone would require that we increase by about 700% the wind energy currently produced. Each wind turbine blade contains more than one ton of polymeric matrix composite, where the matrix is a thermoset which cannot be easily recycled. Hence the only current end-of-life option for composites wind turbines blades is landfilling. This project will evaluate the suitability of thermoplastic (TP) polymer matrix composites for recyclable wind turbines. A similar approach would be suitable also for some aircraft structures. Thermoplastic polymers are more amenable of being recycled as they can be 'melted'/liquified and solidified several times, without significantly changing their properties. TP are also generally tougher than thermoset matrices. Vice versa, the Young's modulus and resistance to creep and of TP are typically lower than thermoset. Moreover, TP/continuous fibres composites can be more difficult to manufacture due to higher viscosity and higher processing temperature.

In this project, you will:

- i) Review the literature on the potentials and limitations of thermoplastics FRP for wind-turbine blades and aircraft;
- ii) Select the most suitable materials (e.g. thermoplastic polymer matrix and reinforcing fibres) and manufacturing method.
- iii) Prepare and test TP composites and benchmark it against typical GFRP for wind turbines.
- iv) Evaluate the feasibility of TP composites for welding/repairing.
- v) If time allows, evaluate feasibility and efficacy of different recycling processes.

Bilotti, Emiliano Dr

Project no: *EB02 **

Project title: Learning from Nature: from silk fibre to high performance materials - 1 project available

Supervisor: Bilotti, Emiliano Dr

Co-supervisor(s):

Category: Experimental; Experimental: Structures

Software:

Confidential: No

Natural silk has been used in daily life for a long time. Thousands of years ago, the ancient Chinese raised silkworms and extracted silk fibres to make luxury clothes, exported across continents. Ancient Greeks reported the use of natural silk for wounds healing.

However, synthetic polymer materials have gradually replaced natural silks over the last 60 years because of their scalability and wider availability, lower costs and good chemical stability.

Recent progresses in the fields of synthetic biology and genetic engineering have reignited interest in silk, and, in particular, spider silk, which has become among the most promising natural materials. Drag-line spider silk, for instance, shows unique combinations of high thermal conductivity, high strength (0.9-1.4 GPa), high toughness (160-250 MJ/m³) and high energy absorption.

As part of this project, you will investigate manufacturing processes to convert silk fibres into high-performance composite materials. You will find inspiration in the hierarchical (nano)structures present in natural materials like spider silk – but also bone, nacre, enamel or tree – to develop the next generation of biomimetic high-performance sustainable materials.

Bilotti, Emiliano Dr

Project no: **EB03 ***

Project title: Edible Electronics (S) - 1 project available

Supervisor: Bilotti, Emiliano Dr

Co-

supervisor(s):

Category: Experimental

Software:

Confidential: No

Electronic devices composed of 'green' materials that can be gradually degraded with time in the body, are called edible electronics. They can offer various range of applications from smart tags for food aging monitoring towards internal human body health tracking, which could precisely deliver the medications into the target tissues.

Although, the technological desire for edible electronics is great, the field is at its infancy. As they are in contact with the human body, there are great challenges to be considered for real applications. One of these challenges is in the very restricted spectrum of materials that can be safely used. Most of the materials in normal electronics are unsuitable for edible applications, due to cytotoxicity, non-biocompatibility, and instability in contact with human body fluids in Gastrointestinal (GI) tract, for instance.

This project will offer you to be a pioneer of this new exciting field. You will select suitable materials, manufacture simple edible electronic and, ideally, develop a proof-of-concept, which could range from smart edible packaging to non-invasive diagnostic and environmental remediation solutions.

Bilotti, Emiliano Dr

Project no: **EB04 ***

Project title: Smart Bioresorbable Soft Composites Materials for Safer Tumour Ablation - 1 project available

Supervisor: Bilotti, Emiliano Dr

Co-

supervisor(s):

Category: Experimental

Software:

Confidential: Yes

Interested in working at the frontiers of Engineering and Medicine and making a difference in cancer treatment? Join this adventurous project, in collaboration with the Royal Marsden Hospital (RMH) / Institute of Cancer Research (ICR), aimed at making minimally invasive tumour ablation procedures safer and more effective.

Tumour ablation is a minimally-invasive 'pinhole' surgical operation, used in multiple organs, where needles are placed percutaneously, directly into tumours, to destroy them with different energy sources (e.g. microwaves). Ablation is less invasive than a normal surgery, with shorter recovery time, less pain, lower cost and fewer complications for the patient. However, the risk of permanent injury and severe consequences exists if the ablation zone is not finely controlled and healthy tissues surrounding the tumour are inadvertently affected during ablation. Solving this shortcoming would make tumour ablation more effective and more widely adopted.

As part of this project, you will design, make and characterise new injectable bioresorbable soft (hydrogel) composites materials that, when injected, and after passed through a needle lumen (<3 mm), will expand to exert a sufficient force to displace healthy tissues away from the tumour.

Bilotti, Emiliano Dr

Project no: *EB05 **

Project title: Smart Polymer Nanocomposites for Soft Robotics Skins (S) - 1 project available

Supervisor: Bilotti, Emiliano Dr

Co-

supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

Tactile sensitive soft electronic skins have great potential to increase robot dexterity in unstructured

and semi-structured environments, especially when challenging objects (e.g. delicate, deformable,

damp) have to be handled. Despite the recent progress in the field, current tactile sensitive skins into

robots still presents a number of challenges.

This project will develop novel stretchable conductive polymer composites to be integrated in robotic

arms as tactile sensing skins. Different materials and sensing technologies will be explored, including

electrical resistive sensing and magnetic sensing.

Key performance parameters will be: (i) sensitivity, (ii) repeatability and (iii) sampling rate, (iv) coverage

area, (v) spatial resolution and (vi) temporal resolution.

In this project, you will:

- i) Review the literature on the potentials and limitations of polymer and polymer nanocomposites for soft robotic skins;
- ii) Select the most suitable materials (e.g. thermoplastic elastomers and conductive particles) and manufacturing method;
- iii) Prepare and test smart polymer nanocomposites films for mechanical, electrical and sensing properties and benchmark it against state-of-art;
- iv) If time allows, integrate the smart polymer nanocomposites manufactured and tested in a robotic hand.

Bilotti, Emiliano Dr

Project no: *EB06 **

Project title: Upcycling Leather Waste: A Sustainable Solution for high-performance materials in transport, interior design and fashion - 1 project available

Supervisor: Bilotti, Emiliano Dr

Co-supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

The leather industry produces over 800,000 tonnes of pre-consumer waste globally each year. Most of

this material, despite being high-quality, is discarded due to the lack of effective recycling techniques.

This project aims to change that by transforming leather waste into high-performance, leather-like

materials, free from any synthetic binders or additives.

In collaboration Generation Phoenix Ltd, you will pioneer a new generation of sustainable materials

made entirely from waste leather fibres, in high-demand sectors such as transport, interior design and fashion.

Unlike existing recycled leather products which rely heavily on synthetic polymer additives, binders, and coatings, this project will use a novel method to reconstitute defibrillated collagen microfibrils from leather waste into a robust, flexible, and biodegradable leather-like material.

Bruce, Paul Dr

Project no: *PB01*

Project title: Experimental study of SBLI (S) - 1 project available

Supervisor: Bruce, Paul Dr

Co-

supervisor(s):

Category: Experimental: Manufacturing intensive

Software:

Confidential: No

Shock wave - boundary layer interaction (SBLI) is an important and complex flow feature encountered in practically all high-speed vehicles and flow systems. This project builds on the material covered in your Aerothermodynamics course (AERO70005) with the aim of pushing the boundary of our understanding of SBLI. The scope of the project will align with one of several ongoing research efforts in the department that span the following broad topic areas: (1) impact of surface flexibility on SBLI; (2) prediction of heat transfer in a hypersonic SBLI; (3) shock-train development and unsteadiness; (4) SBLI flow control. The project will include the design and execution of wind tunnel experiments in the department's supersonic wind tunnel and collaboration with other researchers in the high-speed aerodynamics group at Imperial.

Bruce, Paul Dr

Project no: *PB02*

Project title: Computational study of SBLI (S) - 1 project available

Supervisor: Bruce, Paul Dr

Co-

supervisor(s):

Category: Computational

Software: Star-CCM+ and/or Nektar++

Confidential: No

Shock wave - boundary layer interaction (SBLI) is an important and complex flow feature encountered in practically all high-speed vehicles and flow systems. This project builds on the material covered in your Aerothermodynamics course (AERO70005) with the aim of pushing the boundary of our understanding of SBLI. The scope of the project will align with one of several ongoing research efforts in the department that span the following broad topic areas: (1) impact of surface flexibility on SBLI; (2) prediction of heat transfer in a hypersonic SBLI; (3) shock-train development and unsteadiness; (4) SBLI flow control. The project will involve performing simulations using CFD software (PyFR and/or Star-CCM+) and collaboration with other researchers in the high-speed aerodynamics group at Imperial.

Bruce, Paul Dr

Project no: *PB03*

Project title: Aerodynamics in high-performance sports - 1 project available

Supervisor: Bruce, Paul Dr

Co-

supervisor(s):

Category: Computational;Analysis;Literature

Software: Star-CCM+ (possibly), Matlab

Confidential: No

This project focuses on the important role of aerodynamics in competitive high-performance sports such as (but not limited to) cycling and rowing. The project specification is broad and flexible (negotiable) depending on the interests of the selected student, who will ideally have a strong interest in a sport such as cycling or rowing. This project aims to advance the state-of-the-art in sports aerodynamics and – uniquely – explore the relationship between aerodynamics and athlete physiology and race strategy. The project will comprise some or all of the following tasks for a given sport: (1) perform a literature review to quantify the importance of aerodynamics and the state-of-the-art in technologies previously employed to maximise aerodynamic performance gains, (2) construct a simplified engineering model (or adapt an existing one) to accurately capture the critical aerodynamics-related physics, (3) identify appropriate tools for the measurement and estimation of relevant aerodynamic quantities (such as drag) using (e.g.) real-world data, wind tunnel experiments, reduced-order models, CFD, etc., (4) use insight gained from tasks (1)-(3) to develop optimal strategies for monitoring and improving performance.

Bruce, Paul Dr

Project no: *PB04 **

Project title: Experimental study of the dynamic stability of re-entry vehicles (S)
- 1 project available

Supervisor: Bruce, Paul Dr

Co-

supervisor(s):

Category: Experimental: Manufacturing intensive

Software:

Confidential: No

The dynamic stability of re-entry vehicles at moderate to low supersonic and transonic Mach numbers is often overlooked in their design but can be critical to mission performance. This project will utilise a novel custom-built rig to explore the dynamic behaviour of (conventional and unconventional geometry) re-entry vehicles at transonic and supersonic conditions with 1, 2 and 3 degrees of freedom in Imperial's supersonic wind tunnel. The primary aim of the project is to provide high quality dynamic experimental data that will (1) improve our understanding of the physics that govern re-entry vehicle stability, (2) quantify the stability of unconventional re-entry vehicles including deployable architectures, and (3) validate existing computational results from URANS and LES. This project is closely aligned to ongoing research at Imperial and a high degree of interaction and collaboration is expected.

Bruce, Paul Dr

Project no: *PB05*

Project title: Computational study of re-entry vehicle dynamic stability (S) - 1 project available

Supervisor: Bruce, Paul Dr

Co-

supervisor(s):

Category: Computational

Software: Star-CCM+ and Nektar++

Confidential: No

The dynamic stability of re-entry vehicles at moderate to low supersonic and transonic Mach numbers is often overlooked in their design but can be critical to mission performance. This project will utilise computational methods (Star-CCM+) to explore the dynamic behaviour of (conventional and unconventional geometry) re-entry vehicles at transonic and supersonic conditions. The aims of the project are (1) to explore the complex physics that govern the dynamic stability of conventional and novel re-entry vehicle architectures; and (2) quantify the importance of small-scale surface roughness and geometric imperfections on vehicle stability. This project is closely aligned to ongoing research at Imperial and a high degree of interaction and collaboration is expected.

Bruce, Paul Dr

Project no: *PB06*

Project title: Experimental study of hypersonic boundary layer transition control (S) - 1 project available

Supervisor: Bruce, Paul Dr

Co-supervisor(s):

Category: Experimental: Ready-made experiment

Software:

Confidential: No

Laminar-turbulent transition in hypersonic flow causes an increase in aerodynamic heating, and transition control is crucial to suppress the heat flux. This project builds on ongoing experimental work at Imperial College London to explore the application of surfaces with non-uniform temperature distributions to control instabilities and transition in hypersonic boundary layers. This project will utilise Imperial's supersonic wind tunnel. The aims of this project are flexible, but it is envisaged contributions will be made through one or more of the following tasks: (1) design, development and instrumentation of a test article suitable for Imperial's supersonic wind tunnel; (2) conduction of wind tunnel experiments using appropriate methodology; (3) practical considerations of how to overcome concept drawbacks such as manufacturing imperfections, introduction of thermal stresses, surface coating requirements, etc.

Bruce, Paul Dr

Project no: *PB07*

Project title: Computational study of hypersonic boundary layer transition control (S) - 1 project available

Supervisor: Bruce, Paul Dr

Co-supervisor(s):

Category: Computational

Software: In-house DNS code

Confidential: No

Laminar-turbulent transition in hypersonic flow causes an increase in aerodynamic heating, and transition control is crucial to suppress the heat flux. This project builds on ongoing work

at Imperial College London to explore the application of surfaces with non-uniform temperature distributions to control instabilities and transition in hypersonic boundary layers and ultimately provide hypersonic vehicle designers with a novel and highly tuneable capability for suppressing or promoting hypersonic boundary layer transition. The aims of this project are flexible but it is envisaged contributions will be made through one or more of the following tasks: (1) conduction of direct numerical simulation (DNS) of a hypersonic boundary layer with different surface configurations, (2) stability analysis of existing and new DNS results to assess boundary layer stability and control effectiveness, (3) identification of optimal control configurations; and (4) tailoring of optimal configurations for more complex geometries and flow cases.

Bruce, Paul Dr

Project no: *PB08*

Project title: Improved reduced order models for the rapid prediction of vehicle aerothermodynamic performance at hypersonic flight conditions (S) - 1 project available

Supervisor: Bruce, Paul Dr

Co-supervisor(s):

Category: Theoretical;Computational

Software: Python

Confidential: No

PyHEAT3D is a Python-based engineering level code developed at Imperial for the estimation of preliminary heat transfer rates and heat loads experienced by 3D heat shields during atmospheric entry. This project aims to further extend the capabilities of this tool and apply it to relevant applications. The following tasks and activities are envisaged: (1) literature review of the re-entry environment and current modelling approaches; (2) validation of PyHEAT3D for test cases spanning a range of Mach numbers; (3) extension of the capabilities and utility of PyHEAT3D to consider more complex geometry bodies and/or at a wider range of Mach numbers; (4) interaction with other ongoing numerical (CFD) and experimental research into aerothermodynamics of re-entry vehicles in the high-speed aerodynamics group at Imperial.

Buxton, Oliver Prof.

Project no: *OB01 **

Project title: The aerodynamics of swing bowling in cricket - 1 project available

Supervisor: Buxton, Oliver Prof.

Co-

supervisor(s):

Category: Experimental

Software:

Confidential: No

Swing bowling in cricket is still considered to be something of a mystery. Swing is achieved when seam bowlers, typically bowling in excess of c. 70 mph, are able to get the ball to follow a curved trajectory. However, there are two types of swing: conventional swing and reverse swing. Conventional swing typically happens with a new cricket ball and is presumed to occur due to asymmetry in the flow past the ball as the seam (the stitching that holds the two halves of the ball together) is tilted towards one side. Reverse swing, on the other hand, occurs when the ball becomes old. Players shine one side of the ball and allow the other side to become rough. It is presumed that the rough side of the ball trips the boundary layer meaning that the flow remains attached for longer on the rough side of the ball, leading to a later separation and hence greater pressure recovery on the rough side of the ball leading to reduced drag and a lift force that leads the ball to follow a curved trajectory towards the rough side of the ball. There are, however, many variables associated to whether the ball will swing or not including i) humidity, ii) roughness of the rough side/shininess of the shiny side, iii) prominence of the seam, iv) pace, v) rotational speed of the ball, vi) level of background turbulence etc. In this project the student will construct a rig to mount a cricket ball (both new and old) into the wind tunnel. The ball will be rotated at a carefully controlled speed to simulate the action of seam bowling. We will then make hot-wire/pressure measurements to record the velocity/pressure field around the ball in response to the different factors listed previously that affect swing bowling to try and de-mystify this important aspect of cricket. If successful it is hoped that we will be able to present our results to potentially interested professional cricket teams.

Buxton, Oliver Prof.

Project no: *OB02 **

Project title: Applying turbulence analysis to data from multiple probes orbiting the Sun - Earth L1 Lagrange point (S) - 1 project available

Supervisor: Buxton, Oliver Prof.

Co- Horbury, Tim Prof. (Physics)

supervisor(s):

Category: Theoretical;Analysis

Software:

Confidential: No

Recently IMAP became the sixth probe to enter orbit about the Earth – Sun’s L1 Lagrange point, upstream of the Earth, to study the solar wind and interplanetary magnetic field. The solar wind is a highly turbulent plasma that flows from the Sun and fills the heliosphere. When the wind and magnetic field arrive at the Earth, they can drive “space weather” with significant societal impacts both on Earth (power grid, navigation and radio disruption) and in space (satellite and astronaut harm).

The breadth of length scales present in the heliospheric magnetic field is vast, ranging from an outer-scale that is commensurate with the diameter of the Sun to $O(10\text{m})$ at which individual particle dynamics dominate. Between the outer scale and the proton gyroradius, $O(100\text{km})$ close to Earth) the plasma can be treated as a continuum that behaves as a compressible turbulent flow described by the Navier-Stokes (NS) equations. Accordingly, the heliospheric plasma adheres to many of the phenomenologies of classical NS turbulence, including an inertial range in the spectrum of turbulent kinetic energy that obeys a $-5/3$ power law, and a mean cascade of kinetic energy from large to small scales. This turbulence is at an extremely high Reynolds number and is anisotropic due to the Sun’s fluctuating magnetic field making it a reliable analogue to oceanic/atmospheric turbulence on Earth. Measuring high Reynolds number, anisotropic turbulence on Earth is virtually impossible, hindering our understanding of a multitude of geodynamics.

Multi-point statistics of turbulence in space plasmas have been previously performed in the near-Earth environment, however studies have so far focused on scales where the plasma behaviour deviates from classical NS turbulence. The arrival of IMAP into orbit completes the fleet of six probes and unlocks satellite separations that are larger (10^4 - 10^6 km) than previously available, opening up the study of interplanetary space turbulence in a range where the medium behaves much more closely to classic fluids. This provides new convergence-science possibilities to analyse this rich data set through the lens of these multi-point statistics developed for classic fluids.

We will analyse particularly interesting cases of data from the satellite fleet, including where the satellites are parallel, and perpendicular to the magnetic field. Our analysis techniques will include multi-point structure functions and volume reconstruction methods (e.g. Grad-Shafranov). The analysis techniques that we will develop in this project will help to prepare for the HelioSwarm mission (launch 2029) with 9 measurement points.

Buxton, Oliver Prof.

Project no: OB03 *

Project title: Lagrangian analysis of entrainment into a turbulent cylinder wake
- 1 project available

Supervisor: Buxton, Oliver Prof.

Co-supervisor(s):

Category: Theoretical;Analysis

Software:

Confidential: No

Turbulent flows grow through the process of entrainment which encompasses the transport of background fluid into the primary turbulent flow, crossing the bounding interface that separates the primary flow from the background. Entrainment has typically been studied i) in the special context of the background being non-turbulent/irrotational and ii) from an Eulerian perspective.

Our research group has spent the past 6 years or so expanding our understanding of entrainment from a background that is also turbulent. This encompasses the vast majority of industrial and environmental flows, such as the spreading of a volcanic plume exposed to atmospheric turbulence or the growth of a turbulent boundary layer over a gas-turbine blade that is exposed to the turbulent outflow from the combustor.

We have also recently undertaken some cutting-edge experiments acquiring Lagrangian particle tracking (LPT) data in which we can observe the actual process of entrainment and its temporal dependence. In particular we have recently acquired data in which the wake of a cylinder, exposed to a turbulent background, is observed through LPT experiments. The objective of this project will be to analyse this data set in order to further our understanding of the physics of turbulent entrainment. Given the novelty of both i) consideration of turbulent/turbulent entrainment and ii) the Lagrangian nature of the data set there is a good chance that a high-quality project could lead to a journal publication.

Buxton, Oliver Prof.

Project no: **OB04 ***

Project title: Engulfment in turbulent jets - 1 project available

Supervisor: Buxton, Oliver Prof.

Co-supervisor(s):

Category: Theoretical;Analysis

Software:

Confidential: No

Turbulent flows, including turbulent jets, grow through the process of entrainment which encompasses the transport of background fluid into the primary turbulent flow, crossing the bounding interface that separates the primary flow from the background. Entrainment can broadly be broken down into small-scale processes (“nibbling”) and large scale processes (“engulfment”).

Turbulent jets feature “doughnut-like” coherent motions in the near field due to classical shear-layer instabilities. These coherent motions are thought to be responsible for driving engulfment in the near field of the jet, thereby enhancing the entrainment (and spreading) rate of the jet. However, it has proven difficult, up until now, to determine the relative

importance of these coherent motions to the rate at which background fluid is engulfed, particularly in the case where the background flow is also turbulent.

Our research group recently underwent a research visit to McGill University in Canada in order to undertake some experiments that will hopefully answer this question. Two different jets were placed into a turbulent background, created by the McGill random jet array (RJA), and their spreading was measured by using planar laser induced fluorescence (PLIF). Both jets had precisely the same mass flow rate, and hence identical Reynolds numbers, however one jet was equipped with a fractal orifice that suppresses the coherent motions in the near field of the jet. We are therefore able to directly compare two jets at identical Reynolds numbers with one jet featuring large-scale coherence in the near field and one not. The student undertaking this project is expected to analyse this data set and thereby determine the role that these coherent motions play in driving engulfment in turbulent jets. If rapid progress is made it may be possible to extend this analysis to other turbulent flows such as wakes to try and identify universal aspects of this phenomenon.

Cantwell, Chris Dr

Project no: CC01

Project title: High-performance numerical algorithms for next-generation computer architectures - 2 projects available

Supervisor: Cantwell, Chris Dr

Co-

supervisor(s):

Category: Computational; Numerical

Software: Nektar++ (www.nektar.info)

Confidential: No

High-performance computing is now used extensively for simulating a broad range of physical processes from fluid dynamics and material properties through to chemical reactions and the behaviour of complex biological systems. The challenging nature of performing high-fidelity simulations, due to their huge computational requirements, necessitate the use of codes, such as the high-order finite element code Nektar++, which are highly optimised for performance. Achieving this requires understanding the capabilities of the computer architecture on which it will run and tailoring both the numerical algorithms and the software implementation to align with and exploit the hardware configuration. Projects are available to develop aspects of Nektar++ to target the next generation of GPU-enabled supercomputers, ranging from exploring the low-level exploitation of the latest generation of CPU and GPU architectures, through to higher-level multi-node parallelism of algorithms. Knowledge of C++ is essential. It is recommended that the student has a strong understanding of the course on High-performance Computing.

Cantwell, Chris Dr

Project no: CC02
Project title: High-performance software for modelling plasma kinetics in tokamaks - 1 project available

Supervisor: Cantwell, Chris Dr
Co-supervisor(s):
Category: Computational; Numerical
Software: Nektar++

Confidential: No

Nuclear fusion technology has the potential to solve many of the worlds current energy needs in a safe and environmentally friendly way. However, designing a suitable reactor (a tokamak) which can harness energy is a major scientific and engineering challenge. Computational modelling is the key tool which will enable us to realise a working and commercially viable tokamak. One of the challenges is how to model the dynamics of plasma inside the reactor. Particle methods are commonly used to solve plasma kinetic models as they are cheap to run. However, in some important regimes, such as near the wall of the reactor where there are low-density contaminants that need to be captured, statistical noise limits their use and a full continuum representation of the Vlasov-Maxwell equations needs to be used. Projects are available, aligned with on-going work with the UK Atomic Energy Authority, in developing computationally tractable code within Nektar++ to solve these equations, as well as build on recent efforts to leverage GPUs within the code. These projects have a strong software engineering focus and knowledge of C++ is essential. It is recommended that the student has a strong understanding of the course on High-performance Computing.

Cantwell, Chris Dr

Project no: CC03
Project title: AI-driven optimisation of Nektar++ - 1 project available

Supervisor: Cantwell, Chris Dr
Co-supervisor(s):
Category: Computational; Numerical
Software: Nektar++, PyTorch

Confidential: No

Nektar++ is a high-order finite element solver for performing high-fidelity simulations of processes governed by differential equations. The high-order representation, range of element shapes and structured nature of the different finite element operators exposes a rich space of numerical implementations. While, for a given operator, they each achieve the same result, they have different performance characteristics on different computer architectures, due to differences in FLOPS and memory access patterns. Consequently, selecting the most effective combination of implementation strategies for a given problem becomes an insurmountable (and unwanted) challenge for the user.

This project aims to automate the algorithm selection by training a machine learning model to map a given problem definition and computer architecture to the optimal combination of algorithm implementations to use. Different machine learning approaches will be considered and tested. The resulting model will be directly integrated into the Nektar++ framework so that inference can be performed directly when launching simulations.

Cantwell, Chris Dr

Project no: CC04

Project title: Software Engineering in the open-source CFD package Nektar++
- 2 projects available

Supervisor: Cantwell, Chris Dr
Co-supervisor(s): Sherwin, Spencer Prof.

Category: Computational

Software: Nektar++, Python

Confidential: No

Software engineering is an integral part of any computational package in order to support correctness, robustness and reproducibility of results. For example, to improve the robustness and flexibility of the Nektar++ (www.nektar.info) flow solver and to help provide an easier way to understand the fundamentals of spectral/hp element methods, we have been developing a Python interface. We also maintain a Continual Integration environments within our GitLab instance (<https://gitlab.nektar.info>) which helps us ensure that new developments are robust and do not break existing functionality. However we do not currently have a robust performance monitoring framework. This project is therefore focussed on the software engineering aspects of the code development and can be tailored depending on mutual discussion on your interests and the needs of the project.

Chen, Tian Dr

Project no: *TC01 **

Project title: Microfabrication of bistable deployable structures - 1 project available

Supervisor: Chen, Tian Dr

Co-supervisor(s):

Category: Experimental: Manufacturing intensive

Software:

Confidential: No

Deployable structures are mechanical systems that transform from a compact stowed configuration to a larger operational one, and they are critical in applications ranging from spacecraft antennas and solar arrays to biomedical implants. Recent work has shown that wafer-scale microfabrication offers a powerful route to producing planar precursors that mechanically deploy into complex 3D geometries, combining the precision of semiconductor processing with the programmability of structural mechanics. However, exploiting microfabrication for deployable structures at Imperial requires first establishing a baseline process flow using the College cleanroom facilities.

This project will familiarise the student with Imperial's microfabrication capabilities and use them to produce a set of bistable deployable micro-structures. The student will design planar precursors with engineered stress or geometric features that snap into a stable 3D configuration upon release, fabricate them using photolithography and etching, and characterise their deployment behaviour. The project will produce both a documented process recipe for use by future students and a working demonstrator of micro-scale bistable deployment.

A willingness to work in a cleanroom environment is essential, along with basic proficiency in CAD and Python. No prior microfabrication experience is required.

For inspiration, watch some previous results from our group:

<https://www.youtube.com/watch?v=vrOjy-v5JgQ>.

Chen, Tian Dr

Project no: *TC02 **

Project title: Robotic extrusion deposition for automated damage repair - 1 project available

Supervisor: Chen, Tian Dr

Co-supervisor(s):

Category: Experimental;Computational;Design

Software:

Confidential: No

Automated repair of damaged structural surfaces — aerospace components, industrial tooling, or infrastructure — is a long-standing goal of advanced manufacturing. The core challenge is the tight coupling between sensing, motion planning, and material deposition: the robot must identify the damaged region, generate an appropriate toolpath, and deposit material in a way that conforms to the existing geometry.

This project will develop the hardware and control pipeline for a robotic extrusion-deposition system aimed at local surface repair. The student will interface a robotic arm with an extrusion head, develop motion-control software that follows a specified toolpath, and integrate the system end-to-end with input from a 3D scan of a damaged region. A simplified curvature-based metric (provided) will be used to identify the repair region, and the student will demonstrate closed-loop deposition on a set of test specimens with engineered defects. The project suits a student with strong interests in robotics, mechatronics, and hands-on system integration, and requires good Python skills. Prior exposure to ROS is helpful but not required.

Chen, Tian Dr

Project no: TC03 *

Project title: Robotic ceramic shaping as kinetic art - 1 project available

Supervisor: Chen, Tian Dr

Co-

supervisor(s):

Category: Experimental;Design

Software:

Confidential: No

Robotic manipulation of soft, yield-stress materials such as clay sits at an unusual intersection of mechanics, control, and craft. It is also a domain in which contemporary artists — most famously in works such as Sun Yuan and Peng Yu's *Can't Help Myself* — have used robots not to perform a utilitarian task, but to stage an open-ended, absurd, or meditative action. This project embraces that framing: a robot that engages repeatedly and imperfectly with a ceramic medium, producing an artefact whose value lies in the process rather than the end product.

This project will design, build, and operate a robotic system that performs a repeated shaping or throwing action on clay. The student will interface a robotic arm with a simple end effector, program a control routine that executes an open-ended shaping behaviour, and document the resulting artefacts and process. The engineering contributions are the hardware integration, the motion-control routine, and an analysis of how the clay's rheology constrains the achievable motions; the broader output is a physical art piece that can be exhibited. The project suits a student with strong mechatronics and coding skills who is interested in the intersection of engineering and art. A willingness to iterate experimentally and tolerate messy outcomes is essential.

A robot that repeatedly engages with a deformable medium — clay, for example — requires a perception system that tracks the state of the material and adapts its actions accordingly. When the goal is an open-ended, aesthetically driven interaction rather than a specified end shape, the perception and control problem is particularly interesting: the robot is not pursuing a target but reacting to what is there, producing behaviour that is responsive without being goal-directed in a conventional sense.

This project will develop the perception and behavioural-control layer for a robotic ceramic art installation. The student will integrate a camera system with a robotic arm performing shaping operations on clay, train a computer-vision pipeline to track the geometry and deformation of the clay in real time, and design a behavioural control loop in which the robot's actions are modulated by what it perceives. The work builds on a parallel hardware-focused project (3a) and offers the opportunity to contribute to an exhibited artefact as well as a technical thesis.

The project requires strong programming skills in Python, familiarity with computer vision and/or deep learning, and an interest in robotics. Willingness to engage with the artistic framing of the project is essential.

Chen, Tian Dr

Project no: *TC04 **

Project title: Design and characterisation of a locally interacting reconfigurable module
1 project available

Supervisor: Chen, Tian Dr

Co-

supervisor(s):

Category: Experimental;Design;Analysis

Software:

Confidential: No

Modular reconfigurable materials — imagine a collection of small units that can rearrange themselves into different global configurations — have long been a goal of robotics and programmable-matter research, with the microbots of Big Hero 6 serving as a popular cultural reference. Achieving this in practice requires modules that can couple to their neighbours, change their coupling state on command, and slide or rotate relative to each other under some form of local interaction. This project focuses on the fundamental building block: a single module and its interaction with one or two neighbours.

This project will design, fabricate, and characterise a reconfigurable module with a controllable local interaction. The student will propose and prototype a coupling mechanism (magnetic, mechanical, electro-adhesive, or other), characterise its engagement and release behaviour, and demonstrate at least one reconfiguration action — typically sliding, rotation,

or snap-attach — between a pair of modules. The project will produce a working prototype, a characterised coupling mechanism, and a proposed scaling strategy for larger assemblies. The project suits a student with strong design and fabrication instincts and an interest in mechatronics. Prior exposure to CAD and basic electronics is expected.

Chen, Tian Dr

Project no: *TC05 **

Project title: Frictional characterisation of 3D-printed textile filaments - 1 project available

Supervisor: Chen, Tian Dr

Co-

supervisor(s):

Category: Experimental;Analysis

Software:

Confidential: No

Textiles — both traditional (knits, weaves) and non-traditional (chainmails, crochets) — derive much of their mechanical behaviour from frictional interactions between the filaments that make them up. Additive manufacturing opens the possibility of designing filaments with programmed surface textures, which could in turn programme the frictional and therefore mechanical response of an entire textile. Understanding this connection requires accurate characterisation of filament-scale friction, including how it depends on filament surface topography at the microscopic scale.

This project will characterise frictional behaviour of 3D-printed filaments with a range of deliberately varied surface textures. The student will design and print filament specimens with controlled surface features, develop a benchtop friction-testing setup suitable for filament-on-filament contact, and quantify the dependence of friction on texture, load, and sliding direction. The resulting dataset will form the foundation for subsequent textile-design work.

The project suits a student with interests in experimental mechanics, additive manufacturing, and tribology. Good programming skills in Python are expected for data processing.

Chen, Tian Dr

Project no: *TC06 **

Project title: Robotic deployable meta-surface from a triangular grid with extensible edges - 1 project available

Supervisor: Chen, Tian Dr

Co-

supervisor(s):

Category: Experimental;Design

Software:

Confidential: No

Transforming a flat sheet into a target three-dimensional shape by prescribing local length changes is a classical problem in differential geometry — and one with direct implications for deployable structures. A triangular grid whose edges can individually extend or contract provides an experimental realisation of this idea: by commanding each edge to a prescribed length, an initially planar grid deploys into a target 3D surface. When the prescribed lengths are derived from a conformal map, the deployed shape approximates a target surface in a geometrically controlled way.

This project will build and demonstrate a robotic deployable meta-surface based on a triangular grid with individually actuated edges. The student will design a modular triangular unit with an extensible edge actuator (linear actuator, shape-memory alloy, or equivalent), assemble a small grid of these units, and implement open-loop control that commands each edge to a specified length. A conformal-map-derived length field (provided or computed using existing tools) will be used to deploy the grid into a target surface. The project will produce a working demonstrator and a characterised mapping from commanded lengths to deployed geometry.

The project suits a student with strong design, fabrication, and coding skills. Prior exposure to robotics or Grasshopper is advantageous but not required.

Chen, Tian Dr

Project no: *TC07 **

Project title: Design and fabrication of frictional interlocking bistable plates - 1 project available

Supervisor: Chen, Tian Dr

Co-

supervisor(s):

Category: Experimental;Design;Numerical

Software:

Confidential: No

Interlocking plate assemblies — in which flat elements are arranged so that frictional and geometric contact produces a coherent structural response — offer a route to reconfigurable, demountable, and topologically interlocked structures. When the interlocking is engineered together with a bistable mechanical response, the assembly acquires discrete stable configurations, switchable by applied load, that could be exploited for deployment, locking, or energy absorption.

This project will design, fabricate, and characterise a set of frictional interlocking plates with an engineered bistable response. The student will iterate on plate geometry and contact features, fabricate specimens using 3D printing or laser cutting, and characterise the snap-through behaviour and the conditions under which interlocking is preserved across snap events. The project will produce a documented design with a characterised force–displacement response and a demonstration of controlled switching between stable configurations.

The project suits a student with strong design and hands-on instincts. Good CAD skills are expected; prior experience with mechanical testing is beneficial.

Chernyshenko, Sergei Prof.

Project no: SC01

Project title: Sum-of-squares of polynomials optimisation in fluid dynamics and related problems. (S) - 6 projects available

Supervisor: Chernyshenko, Sergei Prof.

Co-supervisor(s):

Category: Computational;Theoretical

Software: Matlab

Confidential: No

This is a group of related projects, offered to several students in parallel. All these projects are based on the same mathematical idea of an auxiliary function and in many cases rely on the same software (the Matlab toolbox YALMIP, or Python library SumOfSquares.py, or similar) implementing the sum of squares of polynomials/semialgebraic optimisation technique. The idea and the software can be applied to a large variety of problems. The students begin with studying this mathematical idea and the software and then choose a specific problem from the list of offered problems, or suggest their own problem, to which they will apply the idea and possibly the software. This then become the student's project. Strong mathematical background will be needed. Some topics might also require programming skills.

Before going any further, note that the heavily mathematical nature of these projects dictates deep concentration on the work, which has to be done full time without distractions and has, therefore, to be mostly concentrated in one term only. Highly important: do not select this project if you will not be able to devote yourself fully to this project in the spring term (this depends on your choice of optional modules).

If you can work full time on the project in the spring term, the next step is to watch a few short videos about this group of projects. The videos are highlighted in yellow at <https://www.imperial.ac.uk/aeronautics/fluidynamics/ChernyshenkoResearch/misc.php> (If you like to watch videos, here is far a more serious video on this topic: at https://www.youtube.com/watch?v=NrF7n3MyCy4&list=PLf_ipOSbWC86n18q4JMn_1J04S90FpdeE&index=8)

Next, read a short introduction to auxiliary functions at

<https://www.imperial.ac.uk/aeronautics/fluidynamics/Projects/IntroToAuxiliaryFunctions.pdf>

This introduction, its content, and its style will give you an idea of what this group of projects is like. In specific projects this can be complemented with programming, but there are also purely theoretical, pen-and-paper versions.

In addition to the topics covered in the videos, namely bounds for time averages, finding orbits in space, and accelerating averaging, more will be offered. For example, consider a pen-and-paper project on bounds for current in tokamaks and/or Hall thrusters, which would continue the work of previous students, see the report by ACM MSc 2022 project report by J.Gong at <https://imperiallondon.sharepoint.com/:b:/r/sites/foe/aero/student-reports/MSc%20Advanced%20Computational%20Methods/2022%20Project%20Reports/GONG%20Jiayi.pdf?csf=1&web=1>). There will be more. Another good sample is the FYP project report by M.Baber, 2014, at

<https://imperiallondon.sharepoint.com/:b:/r/sites/foe/aero/student-reports/Undergraduate%20Reports/Final%20Year%20Projects/2014%20Final%20Year%20Projects/BABER,%20Monib.pdf?csf=1&web=1&e=bict6b>

This project is aligned with ongoing research in the department.

Doan, Anh Khoa Dr

Project no: *AKD01*

Project title: AI-based identification of large-scale structures in turbulence - 1 project available

Supervisor: Doan, Anh Khoa Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical

Software: Python, matlab, paraview

Confidential: No

The understanding of turbulence remains challenging given its chaotic and multiscale (in space and time) nature. Despite this complexity, many attempts have been made at describing turbulence as a hierarchy of coherent structures (so-called eddies) of different scales whose morphology and spatial clustering depend on the Reynolds number and who interact in a statistical manner explained via the concept of Kolmogorov's energy cascade. In this description, energy cascades from the large scale eddies down to the smaller ones, all the way down to the Kolmogorov scale where energy is dissipated through viscous effects. However, a consistent identification of these coherent structures is still lacking.

The objective of this project is to leverage recent developments in Artificial Intelligence to enable the automated detection of such coherent structures in flows which would enable

their tracking and the understanding of the physical process leading to their formation. This will be explored on a series of turbulent flows datasets.

This project is ideal for students with a strong interest in flows, applied mathematics, or machine learning. Prior exposure to numerical methods will be helpful but not strictly required.

Doan, Anh Khoa Dr

Project no: *AKD02*

Project title: Machine learning based detection of flashback in hydrogen combustor - 1 project available

Supervisor: Doan, Anh Khoa Dr

Co-

supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical

Software: Python, tensorflow/pytorch, matlab

Confidential: No

To move towards a sustainable aviation, increased attention is dedicated to the use of hydrogen as fuel for gas turbine. However, due to its high reactivity, clean hydrogen-based combustors are more prone to flashback, which is a rare transient event where the flame suddenly moves back into the injector.

In this project, you will aim to develop machine learning (ML) algorithms to identify precursors to flashback in hydrogen combustor. Precursors are measurable quantities that indicate that the flashback will occur, providing a horizon to react and prevent it. This will be developed using an experimental dataset of sensors on a combustor which exhibited flashback for different operating conditions. A combination of dimensionality reduction techniques with anomaly detection approaches will be explored.

This project is ideal for students with a strong interest in reacting flows, applied mathematics, or machine learning. Prior exposure to numerical methods would be advantageous.

Doan, Anh Khoa Dr

Project no: *AKD03*

Project title: Hybrid machine learning/differentiable solver approach to flow control - 2 projects available

Supervisor: Doan, Anh Khoa Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical

Software: Python, pytorch

Confidential: No

The control of turbulent flow is very challenging due to the chaotic nature of flows and the nonlinear influence that flow actuators can impart on the flow. Typical approaches have relied on the combination of system identification and control strategies but have only performed well in a linear context. Alternatively, recent works have explored the use of deep reinforcement learning for such a task, but such an approach can be extremely costly. In this project, we will explore the use of a differentiable flow solver combined with a deep learning framework to control the behaviour of flows. Such a combination would use the differentiable capability of the flow solver to provide the gradients of the controller parameters with respect to the objective function over multiple timesteps, enabling an acceleration of the training of the controller and ensuring a stable control over longer period. This will be tested on typical flow control application.

This project is ideal for students with a strong interest in turbulence, applied mathematics, or machine learning. Prior exposure to machine learning and numerical methods is desirable for this project.

Doan, Anh Khoa Dr

Project no: *AKD04*

Project title: Reduced-order modeling for unsteady reactive flows - 1 project available

Supervisor: Doan, Anh Khoa Dr

Co-supervisor(s):

Category: Computational;Analysis;Numerical

Software:

Confidential: No

Turbulent reacting flows are complex multiphysics systems which play a vital role in the transportation, and energy production sector. The simulations of these reacting flows is however computationally expensive due to the very large number of degrees of freedom and the complex chemistry that occurs therein.

To address this, this project will explore aim to develop reduced-order model (ROMs) of such unsteady reacting flows. Specifically, machine learning techniques will be leveraged for both the encoding/decoding of the full reacting flow states and time-evolution/prediction of the encoded flow state.

This will be applied on datasets of reacting flows under unsteady excitation. Investigations will focus on the dataset requirements to obtain a ROM able to generalize to other operating/excitation conditions and on the accuracy of the prediction of the ROM. This project is ideal for students with a strong interest in reacting flows, applied mathematics, or machine learning. Prior exposure to numerical methods would be advantageous.

Doan, Anh Khoa Dr

Project no: *AKD05*
Project title: Machine learning-based modelling of hydrogen reacting flows - 1 project available

Supervisor: Doan, Anh Khoa Dr
Co-supervisor(s): Magri, Luca Prof.
Category: Computational;Analysis;Numerical
Software: fortran, c, python
Confidential: No

The transition towards a sustainable transportation and energy generation sector is requiring the investigation of alternate sustainable fuels, such as hydrogen. These investigations are often carried out through numerical simulations of hydrogen-based turbulent reacting flows, but these flows are complex multiphysics systems involving nonlinear interactions between turbulence and chemistry. These characteristics make them particularly challenging and expensive to simulate - a complexity which is compounded when considering hydrogen owing to its particular extreme reacting properties.

In this project, you will aim to develop machine learning (ML)-based models for hydrogen-based turbulent reacting flows. From high fidelity datasets simulations, you will identify features relevant for the modelling of hydrogen reaction and train models able to predict relevant quantities of interest (such as heat release rate, emission factors, ...).

This project is ideal for students with a strong interest in reacting flows, applied mathematics, or machine learning. Prior exposure to numerical methods would be advantageous.

Doorly, Denis Prof.

Project no: *DD01 **

Project title: Machine learning/quantum machine learning for disease screening (S) - 2 projects available

Supervisor: Doorly, Denis Prof.

Co-

supervisor(s):

Category: Computational; Numerical

Software: Matlab, Qiskit library, Anaconda, Python

Confidential: No

This project will investigate the use of machine learning/quantum machine learning (ML/QML) to screen for disease from image data. It will may focus on ophthalmic image data - in which case it may start from a previous project, but could alternatively use CT airway data.

For those interested in QML, it is advisable they have gained experience through completing the short online course based on the Qiskit open simulator - see <https://qiskit.org/learn/course/machine-learning-course/> It will best suit a person with good mathematical/computational skills who is also a capable self-learner.

Doorly, Denis Prof.

Project no: *DD02 **

Project title: Modelling exhaled aerosol spreading and containment - 2 projects available

Supervisor: Doorly, Denis Prof.

Co-

supervisor(s):

Category: Experimental; Computational

Software: Matlab, Star CCM+

Confidential: No

Up to 2 projects (1 MSc, 1 UG) can be offered on aerosol transport in external flows, where the aerosols are generated by exhalation. One of these may be primarily computational and one could be experimentally focussed. The objective of these projects is to characterise the spread of aerosols of different sizes, produced for example by laser surgery, to investigate means to contain or limit aerosol spread using screens and suction. The problem of trapping of a jet by a shield and applied suction is actually a fundamental problem in continuum fluid mechanics. Whereas the free jet, the jet in cross-flow and to a lesser extent the impacting jet have all been subject to many investigations, the effect of combined suction and impaction has not been greatly studied.

It is expected that the computational project will rely mostly on Star-CCM, though the use of one of the department's in-house flow solvers may be considered for comparison if a student is particularly interested. Although primarily computational, study of previous works on

aerosol dispersion in turbulent plumes is an important component of the project. The experimental project will rely on simple but clearly defined configurations to facilitate comparison with computations. Given the broad nature of the topic, the projects is well-suited for a pair students to work on in complementary areas, but either part could be handled independently.

Eastham, Sebastian Dr

Project no: SE01
Project title: Adaptive simulation of aircraft to support projections of a net-zero fleet - 1 project available

Supervisor: Eastham, Sebastian Dr
Co- Levis, Errikos Dr

supervisor(s):

Category: Computational;Theoretical;Numerical

Software: C++ or Julia

Confidential: No

Estimates of the environmental impacts of aviation (past, current, and future) rely on accurate estimates of the location, quantity, and timing of aircraft emissions. This in turn requires accurate models of aircraft performance. While economic models can provide estimates of demand for every airport pair, a flexible aircraft performance model is needed to determine the emissions along that route.

This project seeks a student who will apply the principles of aircraft design and performance to develop a flexible aircraft performance simulator. The simulator will need to be capable (to first order) of generating plausible aircraft performance models given simple constraints (e.g. range, fuel efficiency, payload) and then applying that model to estimate the rate of fuel burn at all stages along a simulated flight for multiple different airport pairs. A successful project will be able to reproduce publicly-available performance data for a set of representative aircraft, capturing broad trends in performance based on mission requirements.

Eastham, Sebastian Dr

Project no: SE02
Project title: Long-term evolution of high altitude exhaust plumes (S) - 1 project available

Supervisor: Eastham, Sebastian Dr

Co-supervisor(s):

Category: Computational; Numerical

Software: C++ or Julia

Confidential: No

The number of launch vehicles is increasing rapidly year by year, but very little is known regarding the emissions associated with these launches. A specific challenge is that there is very little observational data available, and there are few open-source options available which model the long-term dynamics of the exhaust plume. This makes quantifying the emissions (and therefore environmental impacts) from launches challenging.

This project aims to advance our understanding of rocket exhaust plumes, considering specifically how they will stretch, deform, and dilute at different altitudes. This project is intentionally open-ended, and you will have the freedom to experiment with different approaches at different levels of fidelity. However the end goal will be to generate a parameterisation of plume dilution rate as a function of emission altitude, time of year, and (if emitting aerosols) composition. Results will need to be compared against literature results and, where possible, observations which may become available during the course of the project.

Eastham, Sebastian Dr

Project no: SE03

Project title: Tracking the environmental dispersion of aircraft exhaust - 1 project available

Supervisor: Eastham, Sebastian Dr

Co-supervisor(s):

Category: Computational; Numerical

Software: C++, Julia, or Fortran

Confidential: No

Aviation emissions contribute to growing environmental impacts, including ozone loss, air quality degradation, and climate change. However, some of these impacts can be greater or smaller depending on exactly where the emissions go in the atmosphere. It may be possible to optimize flight paths so that the overall effect of aviation on the environment is significantly reduced, by (for example) choosing flight paths where the emissions will soon be washed out of the atmosphere by precipitation. However, this requires the development of global atmospheric models which can rapidly simulate the path, chemistry, and fate of emissions. This project will require the development of exactly such a model. Starting from an existing framework which simulates the chemistry and trajectory of emissions from an aircraft, you will implement a representation of aerosol formation and washout along the trajectory to allow an improved understanding of the potential health consequences of aviation

emissions. Combining this with a model of planetary boundary layer mixing, you will then be able to compare the results of your Lagrangian model against those from a conventional Eulerian model of aircraft exhaust. The end result will be a comparison of how the emissions from all flights over Europe affect our climate and our air quality, using two independent modelling approaches.

Eastham, Sebastian Dr

Project no: SE04 *

Project title: Appreciating clouds - 1 project available

Supervisor: Eastham, Sebastian Dr

Co-

supervisor(s):

Category: Computational;Analysis;Theoretical

Software: C++ or C# most likely, although any language is permissible as long as the result is sufficiently performant.

Confidential: No

Condensation trails (contrails) forming in aircraft exhaust are estimated to be one of the three largest contributors to aviation's overall climate impact, potentially exceeding even that from aircraft CO₂. However they are very poorly understood, largely due to the difficulty of developing verifiable models of contrail evolution over time. This challenge is made harder by the fact that contrails are hard to visualise; although their development is strongly dependent on their complex 3D structure, our intuitions about them are largely developed based on only a 2D image: viewed from below.

This project will require the student to develop a visual model of a contrail which allows it to be viewed in 3D. The simulation will need to take 3D information about contrail ice water content and properties, and use state-of-the-science radiative transfer calculations to simulate the path of light through the contrail to the viewer. Creativity will be strongly encouraged; for example, a productive approach may be to apply ray-tracing algorithms with a publicly-available game engine. If successful, the project will allow researchers (and the public) to better experience how a contrail forms, develops, and sublimates - and to understand what the different contrail features they see from the ground actually mean.

Eastham, Sebastian Dr

Project no: SE05

Project title: Speeding up contrail simulation - 1 project available

Supervisor: Eastham, Sebastian Dr
Co-supervisor(s):
Category: Computational
Software: APCEMM contrail model
Confidential: No

The net climate impacts of the aerospace sector are dominated by CO₂, nitrogen oxides, and condensation trails (contrails). Of these three factors, contrails are the least well understood - in part due to the difficulty of accurately simulating their behaviour and impacts. This project will seek to improve the performance and accuracy of the contrail model APCEMM, specifically by investigating its (non-)convergence with respect to ice crystal radius resolution, time step length, and spatial resolution. In this project you will learn to install, run, and modify the C++-based APCEMM contrail simulator; you will investigate convergence of the solution with respect to key simulation parameters; and you will quantify the degree to which the solution and its convergence vary with changes in meteorological and aircraft parameters. Success of the project will be defined by achieving at least a 35% improvement in runtime, verifying that any changes in integral quantities relative to a reference implementation are either less than 5% or justified based on improvements in the underlying science.

Eastham, Sebastian Dr

Project no: SE06
Project title: CFD of rocket exhaust (S) - 1 project available

Supervisor: Eastham, Sebastian Dr
Co-supervisor(s):
Category: Computational; Numerical
Software: CFD package (eg PyFR)
Confidential: No

The number of launch vehicles is increasing rapidly year by year, but very little is known regarding the emissions associated with these launches. A specific challenge is that there is very little observational data available, and there are few open-source options available which model the exhaust plume. This makes quantifying the emissions (and therefore environmental impacts) from launches challenging. One of the key limitations is the lack of open-source data on the temperature distribution - and therefore chemical behaviour - of the plume.

This project aims to fill that gap by performing a series of open-source simulations of the flow in a typical launch vehicle exhaust plume. You will need to familiarise yourself with an appropriate CFD package such as PyFR, develop meshes and boundary conditions relevant to recent rocket launches, and execute high-resolution 2D and/or 3D simulations. Results

will need to be compared against literature results and, where possible, observations which may become available during the course of the project.

Eastham, Sebastian Dr

Project no: SE07

Project title: Novel analytical solutions for contrail evolution - 1 project available

Supervisor: Eastham, Sebastian Dr

Co-

supervisor(s):

Category: Computational;Theoretical

Software: Python

Confidential: No

Condensation trails (contrails) are thought to be a significant contributor to aviation's climate impacts, but predicting their evolution over time faces a challenge of scales. Over 24 hours a single contrail can spread from a width of a few meters up to tens or hundreds of kilometers, with the behaviour dictated by complex microphysical relations which can be expensive to accurately represent in computational models. With millions of kilometres of contrail formed every day, this poses a significant challenge. It is thus far not clear how to simplify models of contrail evolution without fatally compromising their accuracy.

In this project, you will seek to define a new, analytical model of contrail formation. You will need to familiarise yourself with both the theory and practice of ice cloud physics and broader contrail modelling, at scales from single ice crystals up to the global atmosphere. A successful analytical model could dramatically accelerate efforts to understand and mitigate one of aviation's largest climate impacts.

Eastham, Sebastian Dr

Project no: SE08

Project title: Stratospheric aerosol injection and chemistry - 1 project available

Supervisor: Eastham, Sebastian Dr

Co-

supervisor(s):

Category: Computational;Numerical;Analysis

Software:

Confidential: No

With energy consumption outpacing efficiency improvements, climate change continues to grow and is projected to continue doing so for the foreseeable future. Alongside mitigation, climate interventions such as stratospheric aerosol injection (SAI) are therefore becoming more heavily scrutinised as real possibilities which may be able to help mitigate some of the worst impacts of climate change while carbon emissions are drawn down. Most assessments of SAI, however, neglect the role that background chemistry - and the emissions from aircraft exhaust - might have in mitigating or enhancing the effectiveness and side effects of SAI.

This project will be dedicated to evaluating the effect that background chemistry and deployment platform emissions might have on the environmental effects of SAI. You will either develop a representation of the aircraft emissions plume or apply a global-scale model of atmospheric chemistry to evaluate these impacts, with a focus on a) the role of aircraft nitrogen oxides and b) the effect of background emissions on the development of the stratospheric aerosol. This project has the potential to inform ongoing scientific and political debate regarding the risks of such an unprecedented intervention.

Fasel, Urban Dr

Project no: *UF01 **

Project title: Exploiting natural dynamics for efficient bio-inspired motion - 2 projects available

Supervisor: Fasel, Urban Dr

Co-supervisor(s):

Category: Experimental;Design;Analysis

Software:

Confidential: No

Throughout nature, many different systems, from human walking to cheetah sprinting, achieve remarkable energy efficiency by exploiting their natural dynamics. Instead of relying on high-gain control to suppress nonlinearities, these systems leverage passive dynamics and compliance to produce efficient, repeatable motion.

In this project, you will explore how such principles can be applied to engineered systems. You will work with an existing hardware platform designed to benchmark walking robots. The project may involve testing and comparing control strategies such as Model Predictive Control (MPC), Hybrid Zero Dynamics (HZD), and Zero Moment Point (ZMP), as well as incorporating compliant elements (e.g. springs or other flexible components) to enhance energy efficiency.

There is also scope to develop a computational pipeline for design optimisation, depending on your interests.

This project is well suited to students interested in control theory, dynamical systems, and robotics. Familiarity with CAD is recommended.

Designing a system to exhibit favorable natural dynamics:

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=8710301&tag=1>

Fasel, Urban Dr

Project no: *UF02 **

Project title: Robust Co-Design for Aerial Robots - 1 project available

Supervisor: Fasel, Urban Dr

Co-supervisor(s):

Category: Theoretical;Analysis;Numerical;Computational

Software:

Confidential: No

Designing autonomous aerial robots typically separates physical design (e.g. wing shape) from control. Co-design brings these together, enabling better-performing and more efficient systems.

In this project, you will work on improving a recently developed robust co-design framework for agile fixed-wing aerial robots that accounts for uncertainty and wind disturbances during optimisation. The current approach relies on trajectory planning combined with Monte Carlo simulations and LQR feedback control, which is computationally expensive and limits scalability.

Your goal will be to investigate more efficient alternatives, such as differential flatness-based trajectory planning and dynamic tube Model Predictive Control (MPC), and evaluate their performance within the co-design framework.

This project is ideal for students interested in optimisation, control theory, and autonomous systems. Some familiarity with advanced control methods is recommended.

Robust co-design: <https://arxiv.org/pdf/2603.11130>

Differential flatness: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10224280>

Dynamic tube MPC: <https://arxiv.org/abs/1907.06553>

Fasel, Urban Dr

Project no: *UF03 **

Project title: Design of Adaptive Wings for Agile Aerial Robots - 2 projects available

Supervisor: Fasel, Urban Dr

Co-supervisor(s):

Category: Experimental;Design;Analysis

Software:

Confidential: No

Avian-inspired morphing wings offer a promising way to improve the efficiency and manoeuvrability of autonomous aerial vehicles. Birds, for example, actively change their wing shape to hover in strong winds, respond rapidly to disturbances, and perform highly agile manoeuvres.

In this project, you will explore how similar capabilities can be achieved in engineered systems. Building on an existing indoor fixed-wing glider platform, you will design and prototype morphing wing mechanisms and develop control strategies to enable active, bird-like shape adaptation during flight. The resulting designs will be implemented and tested on the indoor glider.

This project combines wing design, control, and experimentation, and is well suited to students interested in aerial robotics and optimisation. Familiarity with CAD and an interest in flight testing are recommended.

Avian-inspired morphing wings:

<https://www.sciencedirect.com/science/article/pii/S0376042122000173>

Fasel, Urban Dr

Project no: *UF04 **

Project title: Co-Design for Wind-Resilient Lighter-Than-Air Systems - 1 project available

Supervisor: Fasel, Urban Dr

Co-

supervisor(s):

Category: Design;Experimental;Literature;Analysis

Software:

Confidential: No

Lighter-than-air (LTA) robots, such as blimps, are inherently energy-efficient and safe, making them attractive for indoor and urban applications. However, their low airspeed makes them highly sensitive to wind disturbances, creating a significant gap between simulated performance and real-world behaviour.

In this project, you will investigate how co-design optimisation can be used to improve the real-world robustness of LTA systems. Rather than developing models from scratch, you will build on existing LTA platforms, models, and controllers, and focus on bridging the gap between simulation and hardware.

The project will involve evaluating and refining optimisation strategies under realistic wind disturbances, and potentially validating results through experiments on a physical LTA platform. A key objective is to understand how design and control choices transfer from simulation to real-world performance.

This project is ideal for students interested in robotics experimentation, control, and optimisation. It will involve a combination of simulation and hands-on testing.

Fasel, Urban Dr

Project no: *UF05 **

Project title: Multi-Agent Co-Design Optimisation for Lighter-Than-Air Systems - 1 proje available

Supervisor: Fasel, Urban Dr

Co-

supervisor(s):

Category: Theoretical;Computational;Numerical;Design

Software:

Confidential: No

Lighter-than-air (LTA) vehicles, such as blimps and balloon-based robots, offer a safe and energy-efficient alternative for urban aerial applications. However, their low airspeed makes them highly sensitive to wind disturbances, posing significant challenges for reliable operation in urban environments.

In this project, you will develop multi-agent co-design optimisation approaches to LTA systems. The goal is to jointly optimise vehicle design and control strategies for fleets of aerial robots operating in urban conditions. Key challenges include robustness to environmental disturbances, coordination between multiple agents, and scalability of the optimisation framework.

Depending on your interests, the project may involve developing simplified models of LTA dynamics, implementing multi-agent control strategies, or incorporating realistic wind and urban environment effects into the optimisation process.

This project is well suited to students interested in optimisation, control, and autonomous systems. Experience in aerial or multi-agent robotics is recommended.

Co-design for safe multi-agent navigation: <https://arxiv.org/pdf/2604.06972>

Fumarola, Isabella Dr

Project no: *IF01 **

Project title: Flow around a kirigami sheet - 2 projects available

Supervisor: Fumarola, Isabella Dr

Co-

supervisor(s):

Category:

Experimental: Ready-made
experiment;Experimental;Design;Analysis;Literature

Software:

Confidential:

No

A Kirigami, which literally means “paper with cuts,” consists of a thin sheet of material featuring cuts and creases that enable them to adopt complex three-dimensional shapes when subjected to external forces. In recent years, kirigami structures have attracted significant attention across various engineering fields, with many potential applications still to be explored.

One particularly promising area is flow control. When placed in a fluid flow, the kirigami structures can deform in response to the flow conditions, generating a range of wake patterns. This adaptability makes them especially attractive for passive flow control applications.

In this project, building on the analysis of volumetric PTV measurements of the wake behind different existing kirigami configurations (<https://doi.org/10.1016/j.eml.2024.102198>), a new series of wind tunnel experiments will be designed and conducted. The study will focus on two main aspects: (i) characterising the fluid–structure interaction (FSI) governing kirigami deployment in air and comparing it with previous experiments conducted in water, and (ii) investigating the complex wake dynamics generated downstream of the kirigami structures.

Fumarola, Isabella Dr

Project no:

IF02 *

Project title:

Aerodynamic of a Boxfish - 1 project available

Supervisor:

Fumarola, Isabella Dr

Co-

supervisor(s):

Category:

Experimental;Design;Analysis;Literature

Software:

Confidential:

No

The boxfish is a common tropical species that inhabits shallow waters. It is characterised by a distinctive bluff body with a large cross-sectional area, in contrast to the streamlined shapes typical of most fish. Despite this, it exhibits remarkable hydrodynamic performance. These features make the boxfish an interesting subject for the study of drag in bluff bodies. Part of its efficiency appears to be linked to the vortical structures that develop along the edges of its box-like geometry (<https://hal.science/hal-03654472v1>).

In this project, a model of a boxfish, based on a laser scan of a real specimen, will be 3D-printed and instrumented with pressure taps. Pressure and velocity measurements will be

conducted to fully characterise the flow field and its temporal evolution, with the aim of uncovering new insights into this complex aerodynamic problem.

Fumarola, Isabella Dr

Project no: *IF03 **

Project title: Travelling Surface Waves for Turbulent Skin-Friction Reduction - 1 project available

Supervisor: Fumarola, Isabella Dr

Co-supervisor(s):

Category: Experimental;Experimental: Structures;Design

Software:

Confidential: No

Generating travelling surface waves beneath a turbulent boundary layer is one of the most promising techniques for reducing turbulent skin friction. However, several key questions remain regarding the effectiveness of this mechanism at high Reynolds numbers, representative of flight conditions. Recent work carried out at Melbourne suggests the existence of a parameter space suitable for operation in this regime (DOI: <https://doi.org/10.1017/jfm.2023.498>).

Another major challenge lies in the practical generation of spanwise velocity waves at the wall. To address this, experiments have been conducted over several years using a silicone skin deformed by a kagome lattice across different flow regimes (<https://link.springer.com/article/10.1007/s10494-018-9926-2>; <https://link.springer.com/article/10.1007/s10494-024-00531-9>). While this approach offers significant flexibility in laboratory settings, it remains too fragile for realistic applications. We are currently exploring the use of engineered metamaterials (artificial materials with tailored properties, such as spatially varying density and bulk modulus) to generate two-dimensional surface acoustic waves. These are particularly suited to higher Reynolds number conditions, where wave amplitudes are small, $\mathcal{O}(\text{mm})$, and frequencies are high, $\mathcal{O}(\text{kHz})$.

The design phase, involving topology optimisation of the metamaterial microstructure for prescribed forcing amplitudes and frequencies, has been completed. The next stage of the project will focus on manufacturing a prototype and conducting bench-top experiments to characterise its structural response prior to wind tunnel testing.

Fumarola, Isabella Dr

Project no: *IF04 **

Project title: Control System for a Magnetic Suspension Balance System (MSBS) (S) - 1 project available

Supervisor: Fumarola, Isabella Dr

Co-

supervisor(s):

Category: Experimental;Design;Literature

Software: Labview

Confidential: No

The Magnetic Suspension Balance System (MSBS) is a force measurement system for wind tunnel applications based on the magnetic levitation of the test model. It provides a unique opportunity to perform sting-free measurements of time-dependent aerodynamic forces acting on the model.

By embedding permanent magnets within the model and controlling the magnetic field generated by external coils, it is also possible to actively adjust the model position. This enables experiments on spinning models or configurations undergoing complex prescribed motions within the same facility.

At Imperial College London, a new MSBS for low-speed applications is currently under development. The project is in the design phase, with a one-degree-of-freedom (1-DoF) system expected to become available soon.

This project will focus on the development of control algorithms for model positioning, which are a critical component of the feedback loop used to regulate the coil currents and, in turn, infer the aerodynamic forces. Different machine vision approaches will be explored for real-time position tracking.

The 1-DoF system will be used to test and validate the control strategies, while a preliminary feasibility study will be conducted for a full 6-DoF implementation.

Further readings: <https://doi.org/10.1109/TIM.2025.3629858>

Fung, Lloyd Dr

Project no: *LF01*

Project title: Physics-informed data-driven flow construction from particle tracks - 1 project available

Supervisor: Fung, Lloyd Dr

Co-

supervisor(s): Magri, Luca Prof.

Category: Computational;Analysis;Numerical

Software: JULIA (Waterlily.jl, Turing.jl) or JAX (in Python)

Confidential: No

Particle Tracking Velocimetry (PTV) is a non-invasive optical technique that tracks the movement of microscopic tracer particles introduced into a fluid to quantify the flow field. Unlike Particle Image Velocimetry (PIV), an Eulerian technique that deduces local flow fields at fixed locations using cross-correlation, Particle Tracking Velocimetry (PTV) tracks particles in a Lagrangian framework to estimate flow velocity. The particle density employed in PTV for flow visualisation is frequently restricted by the tracking algorithm's capacity to monitor and distinguish individual particles in three-dimensional space from the two-dimensional images projected onto the image sensor. If the seeding density is too high, the algorithm will be unable to differentiate the particles. However, if it is too low, the signal may be insufficiently dense to derive an accurate flow measurement. Recent advancements, such as the Vortex-in-cell (VIC) approach, aim to address sparse observations by interpolating velocity across particles in space or time, utilising established physics, i.e., the Navier-Stokes equation, to interpolate the velocity field.

We propose that, rather than optimising for minimal discrepancies between the velocity field derived from PTV and the simulation, as done in the VIC method, we can simulate the particle trajectory with the flow and assimilate the particle tracking data by minimising the differences in particle trajectories between the measurements and the simulation. By utilising auto-differentiation in differentiable solvers like Waterlily.jl and the probabilistic programming framework Turing.jl, we can streamline our workflow, transforming a previously complicated endeavour into a rather straightforward script. Due to the considerable challenges associated with direct data assimilation in fully turbulent flow, we shall first deal with simpler problems, specifically the wake flow behind a bluff body at low Reynolds numbers.

Students undertaking this research do not require extensive knowledge of PIV or PTV; nonetheless, a solid comprehension of fluid dynamics and CFD will be beneficial. They will acquire knowledge on JULIA, automatic differentiation, and probabilistic programming during the project, which will be highly beneficial to students who are interested in a career in scientific machine learning. This project is programming-intensive and will necessitate students to acquire proficiency in a new programming language, JULIA.

Fung, Lloyd Dr

Project no: *LF02*

Project title: Data-driven modeling of micro-plastic and phytoplankton sinking in a tank
- 1 project available

Supervisor: Fung, Lloyd Dr

Co-

supervisor(s):

Category: Theoretical;Analysis;Computational;Experimental: Ready-made experiment

Software: MATLAB (Bayesian-SINDy) OR Python (JAX and PySR)

Confidential: No

Every year, humans generate hundreds of millions of tonnes of plastic waste, with around 20 million tonnes entering the ocean, where they may degrade into microplastics and harm the marine ecosystem. However, only a small portion of the 20 million tonnes have been spotted on the ocean's surface, raising questions about their whereabouts. On the other hand, phytoplankton in the ocean are responsible for half the carbon sequestration on earth. They sequester carbon by photosynthesis and then sink into the ocean deep. Hence, their average sinking rate directly dictates the rate of sequestration.

Predicting the sinking trajectory of microplastics or phytoplankton in water can be treated as a simple drag problem in fluid dynamics, but the arbitrary and vast variety of densities, sizes, and shapes has made the problem intractable analytically and computationally. This project aims to relieve this difficulty by learning an empirical but interpretable equation for objects of arbitrary shape from a large amount of simulated data using recent advancements in symbolic regression and Bayesian-driven sparse regression, machine learning techniques that can recover interpretable equations from data.

Students embarking on this project will be expected to collect experiment data of the trajectory and flow around a sinking objects of arbitrary shape in a tank. They will then use the latest symbolic regression packages, such as Bayesian-SINDy or PySR, to learn a common model equation from the data. The project requires the student to be able to both conduct a simple experiment, collect data, and analyse the data using existing machine learning packages.

Fung, Lloyd Dr

Project no: *LF03*

Project title: Inverting a pole on a cart to benchmark reinforcement and active symbolic learning - 1 project available

Supervisor: Fung, Lloyd Dr

Co-

supervisor(s):

Category: Experimental: Ready-made experiment; Computational; Literature; Analysis

Software: Software: MATLAB, Python (Gym and PyTorch)

Confidential: No

The pendulum on a cart is a classical benchmarking task for reinforcement learning (RL). The problem's goal is for a learning algorithm to learn about the cart-pole system and control it through some actuation means, e.g., moving the cart left or right in this problem, to attain a goal, like inverting the pendulum in the cart-pole system. Classical RL techniques like soft actor-critic were demonstrated to be effective in performing this task, but the model learnt is not interpretable, putting doubt on its usability in real-world scenarios like flight control. In

the meantime, traditional control strategies involve first identifying the system with a model and then manually designing a control algorithm based on the identified model, which is tedious and time-consuming.

Recent advancement in automatic model identification of dynamics, such as SINDy, has offered an alternative approach that balances the interpretable model identification and the autonomy of modern machine learning approaches. However, the control strategy remains somewhat manual, as the system identification stage remains separate from the control stage. This project aims to combine the two stages into a streamlined and automated process to yield a completely automatic model discovery and control. Furthermore, we will explore the possibility of using an active learning strategy in (Bayesian-)SINDy to help the system explore state spaces that have not yet been quantified by the model, improving its data efficiency.

Students embarking on this project will gain experience in reinforcement learning and classical control strategies such as PID, MPC or LQR, as well as automatic model identification algorithms such as SINDy, Ensemble-SINDy and Bayesian-SINDy. The student is expected to be familiar with control theory and classical (open-loop and close-loop) feedback control, as well as concepts such as Lyapunov stability. If time allows, the student may also have the opportunity to implement their resulting control and learning algorithm in a real-world Cart Pole system built by previous student projects.

Fung, Lloyd Dr

Project no: **LF04**

Project title: Complexity Scaling Law of Physics-Informed Neural Networks in Representing PDE Solutions - 1 project available

Supervisor: Fung, Lloyd Dr

Co-supervisor(s):

Category: Computational;Theoretical;Numerical

Software: Python, JAX

Confidential: No

It is well known that some PDE solutions become more complex — i.e. requiring more degrees of freedom/memory to represent — as their parameters change. Most famously, the complexity of the solution to the Navier-Stokes equations scales with the Reynolds number (Re), due to the larger range of length scales that must be resolved as the Reynolds number increases. The implication of this result is that direct numerical simulation of the equations has a computational cost that scales with Re^4 , rendering DNS impractical in many aeronautical applications.

Meanwhile, Physics-Informed Neural Networks have been touted as a potentially breakthrough machine learning method that can improve the computational efficiency of solving PDEs. While much debate has centred on whether PINNs perform better than

traditional Finite Element Methods (FEM), little attention has been paid to how the size of the neural network scales with the parameters in PDEs that control the complexity of the solution. While PINNs may not always perform better than FEM on small problems, the fact that neural networks can overcome the curse of dimensionality through their depth gives hope that PINNs might be better at compressing data with high degrees of freedom. The project aims to investigate this particular problem by demonstrating how the PINN representation of a PDE solution, upon training to a certain accuracy, scales with the parameters that control its complexity. The project is divided into two parts. In the first part, the student will train a standard NN to fit existing PDE solutions of increasing complexity, and observe the scaling law of the NN fit. Special techniques may be required to train such fits to numerical precision. In the second part, the student will learn how to train PINNs without a data loss term (also known as the NeuralPDE or Deep Galerkin Method), so as to remove the reliance on pre-existing data. Students embarking on the project should have taken at least some machine learning modules, and have a basic understanding of neural networks and optimisation. Prior knowledge of JAX is preferable.

Gouder, Kevin Dr

Project no: *KG01 **
Project title: Transverse Correlation of Forces on a Circular Cylinder in an Atmospheric Boundary Layer (ABL) - 1 project available
Supervisor: Gouder, Kevin Dr
Co-supervisor(s): Graham, Michael Prof.
Category: Experimental
Software:
Confidential: No

This project intends to answer questions about how a body in wind, a circular cylinder in this particular case, experiences wind loading when a) the oncoming flow is turbulent, and b) the oncoming flow is turbulent and there is a superimposed wind shear along the cylinder's central axis. Transverse (i.e. along the cylinder's central axis) 2-point correlations of measured turbulent velocity of the oncoming flow, in the absence of the cylinder, are made. In addition, then, 2-point correlations of measured forces on the cylinder, through integration of belts of pressure taps, are also made. Measurements better our understanding of loads from incident turbulent flows; widely-used quasi-steady theory makes an assumption that the corresponding velocity and force correlations are the same. This experimental aerodynamics project makes use of a slot in the 10x5 wind tunnel, set up with one of the Atmospheric Boundary Layers (ABL) already in the facility database. The loading behaviour is of extensive academic and industrial interest, while the 2-point velocity correlations are very valuable in an ongoing effort for better ABL modelling.

Gouder, Kevin Dr

Project no: **KG02 ***

Project title: The effect of wind shields on bridge flutter and self-excitation - 1 project available

Supervisor: Gouder, Kevin Dr

Co-

supervisor(s):

Category: Experimental

Software:

Confidential: No

This project investigates the effect of wind shields installed on bridge decks on the bridge's flutter speed and self-excitation response. Wind shields are installed as a wind mitigation device on bridges, sometimes post-bridge completion, completely altering the dynamic behaviour of the original deck. This has severe impacts on bridge safety and vehicle comfort. In this experiment, a section model of a bridge deck is mounted across the wind tunnel on a system of springs, giving the deck the scaled structural dynamic properties. Accelerometers and load cells already on the existing rig measure the deck response. Wind shields description varies across height, depth, porosity, slat-type... we will focus on a subset parameter systematic study, in discussion with an industrial partner with close association to the 10x5 Wind Tunnel. Data from this study contributes to and informs the next version of the design code, which to date does not address bridges with wind shields.

Gouder, Kevin Dr

Project no: **KG03 ***

Project title: Boeing Common Research Model in High-Lift configuration (CRM-HL) - 2 projects - 2 projects available

Supervisor: Gouder, Kevin Dr

Co-

supervisor(s):

Category: Experimental

Software:

Confidential: No

The CRM-HL is a global initiative being led by Boeing and NASA. A generic commercial transport aircraft with conventional high-lift system has been designed and the geometry has

been made open source. The grand vision is that labs around the world would build and test the same common geometry provided, and make the data open access, for tunnel-tunnel and tunnel-CFD comparisons and validations. Data would improve our understanding of high-lift flow physics, and help establish and validate more advanced computational methods.

In collaboration with Boeing UK, we have started building a semi-span, 4% scale model for testing in the 10'x5' wind tunnel. The first round of projects have been tackling the conversion of the full-scale geometry into a wind tunnel model, installation of the model on a 6-axes balance and model motion system, and provision of symmetry plane boundary layer suction. The two projects now being offered will acquire and analyse direct load measurement and on-body / off-body flow data, and explore the effects that treatments of the symmetry plane (e.g. no treatment : where upstream tunnel floor boundary layer spillage onto the fuselage and wing leading edge is known to be detrimental; with peniche; without peniche; with boundary layer suction...) have on data reliability.

Gouder, Kevin Dr

Project no: *KG04 **

Project title: Addition of Complex Terrain to the Atmospheric Boundary Layer in the 10'x Large Test Section: Setup and Characterisation - 1 project available

Supervisor: Gouder, Kevin Dr

Co-supervisor(s): Graham, Michael Prof.

Category: Experimental

Software:

Confidential:

No

In a previous project, we set up and characterised a Marine Atmospheric Boundary Layer (ABL) in the 10'x5' wind tunnel larger test section. The Earth-normal distribution of the mean velocity, turbulence intensity, von Karman lengthscale, and spectrum were measured. We have also recently built a motorised turntable, and we are currently experimenting with a mini-wind farm of 6 instrumented wind turbines under coordinated control, with the marine ABL as inflow. This project builds upon these previous efforts. We will consider a wind farm in a coastal zone, or just off the coast in a hilly setting. We intend to place 3D-printed topographical models (e.g. hills, ridges, or cliffs) on the turntable, upstream of one or more of the instrumented wind turbines. The marine ABL will interact with the topography before it proceeds to interact with the wind turbines. We will study how the terrain-induced turbulence and speed-up effects (that will be characterised in this project) impact the structural loading and power output of the turbines (as measured using the existing instrumented wind turbines). This is critical in cases of micro-siting in hilly or mountainous regions.

Gouder, Kevin Dr

Project no: *KG05 **

Project title: Detection of streaks from sparse measurements in bypass transition - 1 project available

Supervisor: Gouder, Kevin Dr

Co-

supervisor(s):

Category: Analysis

Software:

Confidential: No

In the presence of a freestream flow with turbulence intensity exceeding approximately 1%, streaks form naturally and stochastically in the underlying, otherwise Blasius, boundary layer. Literature, such as JFM (2016) vol 801, pp 43-64, have aimed, through a Machine Learning (ML) framework, to identify the streaks that induce breakdown to turbulence in bypass transition, using data from Direct Numerical Simulation (DNS). Identifying streaks that are likely to cause bypass transition is important in a control strategy where turbulence needs to be delayed (e.g. maintaining a laminar flow over a gas turbine blade for longer). Streak identification could, in such an example, form part of a flow control system where the most dangerous (most likely to cause bypass transition) streaks are targeted and weakened before they become unstable. In the example from literature mentioned above, the input to the ML were geometric properties of streaks. In a physical experiment, 1) sensing of streaks is likely to be limited to wall-based sensors (i.e. we do not have direct access to the streak properties away from the wall), and 2) the physical size, and frequency response, of these sensors are likely to introduce spatial and temporal limitations, in contrast to DNS, where access to the time series everywhere in the flow is available. Therefore an experimental implementation of a streak detection technique needs to be able to make identifications from sparse data input. In this project, we will make use of an existing DNS database, and we will explore ways of detecting dangerous streaks solely from wall measurements, such as through, but not necessarily limited to ML training. We could then implement this strategy and observe the extent of success of the identification versus the DNS full data. The identification strategy could be implemented into a physical experiment in future projects.

Greenhalgh, Emile Prof.

Project no: *EG01 **

Project title: Multifunctional design & analysis for structural power composites in defence (S) - 1 project available

Supervisor: Greenhalgh, Emile Prof.

Co-supervisor(s): Nguyen, Sang Dr

Category: Theoretical;Design;Analysis

Software: Matlab and Abaqus

Confidential: No

Background: To date, structural power studies have focussed on aerospace and automotive applications, but we want to explore other areas where we could potentially apply structural power composites. We feel that one such field is defence. We now have a good methodology for multifunctional design of structural power in a range of aerospace and automotive applications and want to apply these methodologies to defence.

Research question: What are the potential benefits, hurdles, performance targets and a route map for adoption of structural power in defence?

Greenhalgh, Emile Prof.

Project no: *EG02 **

Project title: Development of the next generation Structural Power Composites (S) - 2 projects available

Supervisor: Greenhalgh, Emile Prof.

Co-supervisor(s): Nguyen, Sang Dr

Category: Experimental

Software:

Confidential: No

Background: Structural power composites has been recognised by the World Economic Forum as one of the key emerging technologies. The rewards of this technology are huge but there are some considerable technological challenges to be addressed. This is an open ended project focused on structural supercapacitors, and can be tailored to the particular interests of the student.

Further details of the technology can be found here (<https://www.imperial.ac.uk/structural-power-composites/>)

Greenhalgh, Emile Prof.

Project no: *EG03*

Project title: Multifunctional design for structural power in robotics (S) - 1 project available

Supervisor: Greenhalgh, Emile Prof.

Co-supervisor(s): Nguyen, Sang Dr

Category: Theoretical;Design;Computational;Numerical

Software: Matlab and Abaqus

Confidential: No

Background: To date, structural power studies have focussed on aerospace and automotive applications, but we want to explore other areas where we could potentially apply structural power such as robotics. We now have a good methodology for multifunctional design of structural power in a range of aerospace and automotive applications and want to apply these methodologies to robotics. Research question: What are the potential benefits, hurdles, performance targets and a route map for adoption of structural power in robotics?

Greenhalgh, Emile Prof.

Project no: *EG04 **

Project title: Use of AI for composite in-service failure investigation (S) - 1 project available

Supervisor: Greenhalgh, Emile Prof.

Co-supervisor(s): Panesar, Ajit Dr

Category: Experimental;Experimental: Ready-made experiment;Computational;Analysis;Numerical

Software:

Confidential: No

This project will explore the application for AI to be used in the interpretation of composite fracture surfaces, and as a potential tool to improve in-service accident investigations. It will entail training software to interpret information about crack mode and growth direction from optical micrographs.

Hewson, Robert Prof.

Project no: ***RH01***

Project title: Topology Optimisation For Fluids (S) - 2 projects available

Supervisor: Hewson, Robert Prof.

Co-supervisor(s):

Category: Computational;Theoretical;Design;Analysis;Numerical;Literature

Software: Python packages, most likely including JAX.

Confidential: No

Topology optimisation for fluid flow usually represents solid regions using a bulk resistance term, most commonly Brinkman penalisation, on a fixed mesh. That is the standard approach in fluid topology optimisation. More recent work has also explored immersed-boundary and embedded-boundary methods, where walls are represented inside a non-body-fitted mesh rather than by remeshing the geometry.

This project will investigate a different idea using the discontinuous Galerkin (DG) method. In DG, neighbouring elements communicate through numerical fluxes on their shared faces. The project will test whether a design variable can be used to smoothly change a face from behaving like a normal fluid interface to behaving like a no-slip wall. DG has already been used for fluid topology optimisation, but this direct flux-based wall interpolation appears to be much less explored than Brinkman-type methods.

The student will review the relevant literature, implement a simple 2D incompressible flow model, introduce a face-based design variable, and compare the resulting optimisation behaviour against a standard Brinkman formulation. The aim is to assess whether this DG-native approach can provide a simple and effective alternative for fluid topology optimisation.

Hewson, Robert Prof.

Project no: ***RH02***

Project title: Differentiable Thermo-Mechanical Modelling and Reduced-Order Distortion Prediction for Additive Manufacturing (S) - 1 project available

Supervisor: Hewson, Robert Prof.

Co-supervisor(s):

Category: Computational;Theoretical;Design;Analysis;Numerical

Software: Python packages, most likely including JAX.

Confidential: Yes

This project will develop a differentiable thermo-mechanical model for additive manufacturing using JAX-FEM, with a focus on efficient prediction of thermal distortion. The work will be based on a small-strain thermo-elastic formulation in which temperature

changes induce thermal strain, which in turn generates stress and deformation due to mechanical constraints during layer-by-layer deposition.

The student will implement an incremental model in which each layer is activated sequentially and subjected to a prescribed temperature evolution. The governing equations will be solved to obtain the displacement field, allowing prediction of distortion and residual stress as the structure builds and cools. The emphasis will be on understanding how thermal strain acts as a driving term in the mechanical equilibrium problem, rather than being directly converted into displacement.

Beyond the baseline model, the project will explore reduced-order approaches, particularly the inherent strain method, in which the full thermo-mechanical history is replaced by an equivalent strain field. The student will compare the full thermo-elastic simulation with this reduced model and assess its accuracy and computational efficiency.

A key aspect of the project is the use of automatic differentiation within JAX-FEM. This enables gradient-based calibration of model parameters or inference of effective inherent strain fields from target deformation data. The student will investigate how sensitivities of distortion with respect to process parameters or strain assumptions can be computed and exploited.

The outcome will be a computational framework capable of predicting distortion in a simplified additive manufacturing setting, together with a critical comparison between high-fidelity and reduced-order approaches, and an assessment of the benefits of differentiable modelling for calibration and optimisation.

Hewson, Robert Prof.

Project no: *RH03*

Project title: Topology Optimisation with Subtractive Manufacturing Constraints (S) - 1 project available

Supervisor: Hewson, Robert Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Design;Numerical;Analysis

Software: Python packages, most likely including JAX.

Confidential: No

Topology optimisation determines the optimal material layout within a design domain to maximise structural performance. However, most formulations produce complex geometries suited only to additive manufacturing. Subtractive processes such as milling, wire-EDM, waterjet cutting, remain the dominant industrial manufacturing route, offering superior surface quality, material integrity, and cost efficiency, yet impose fundamental geometric restrictions: material can only be removed via through-cuts from external faces.

This project will develop a topology optimisation framework where the geometry is constrained to be realisable by subtractive manufacturing. Cut profiles are defined by 2D

fields on each workpiece face, extruded along the face-normal direction; material survives only where all cuts retain it, guaranteeing manufacturability by construction. Smooth projection with continuation drives designs toward crisp boundaries.

The project will implement and verify the subtractive parameterisation within a finite element framework, solve benchmark structural problems (cantilever, MBB beam, L-bracket), and quantify the stiffness penalty relative to unconstrained designs. Extensions include investigating the effect of cutting direction (axis-aligned vs. diagonal), multiple load cases, and stress constraints.

The project combines structural mechanics, finite element analysis, and gradient-based optimisation using an existing Python codebase.

Hewson, Robert Prof.

Project no: *RH04*

Project title: Project Vortex Cell - Cell Design (S) - 1 project available

Supervisor: Hewson, Robert Prof.

Co-

supervisor(s):

Category: Theoretical;Computational;Design;Analysis;Numerical

Software: Python packages, most likely including JAX.

Confidential: No

This project focuses on the aerodynamic design optimisation of a wing profile incorporating a vortex cell, with the objective of achieving and maintaining a stable dividing streamline across the cavity opening. The dividing streamline effectively acts as a virtual moving wall, reducing downstream separation and improving aerodynamic performance, but only when the cavity geometry and flow conditions are carefully matched. The project will formulate a two-dimensional optimisation problem in which the airfoil shape and cavity geometry are parameterised and adjusted to satisfy flow constraints derived from high-Reynolds-number theory.

An adjoint-based framework will be used to compute sensitivities of aerodynamic objectives such as lift-to-drag ratio, while enforcing constraints on the stability and position of the trapped vortex. The project may also explore control co-design, where steady or distributed suction is introduced as an additional design variable to stabilise the vortex, reflecting the strong dependence of performance on active control observed in experiments. Sensitivity analysis will be used to understand robustness with respect to incidence and upstream boundary-layer conditions. The outcome will be a quantitative assessment of how tightly coupled geometry and control must be to realise performance gains, and whether a practical design space exists for stable, high-performance vortex-cell airfoils.

Hewson, Robert Prof.

Project no: *RH05*

Project title: Project Vortex Cell - Aircraft Integration (S) - 1 project available

Supervisor: Hewson, Robert Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Design;Analysis;Numerical

Software: Python packages, most likely including JAX.

Confidential: No

This project investigates whether vortex-cell-enabled thick wings could make hydrogen storage within the wing a viable alternative to conventional fuselage-based storage. Hydrogen aircraft concepts typically assume bulky tanks in the fuselage, driving larger airframes and reducing overall efficiency. This project challenges that assumption by exploring whether thicker wings, augmented by vortex cells to delay separation, could offset the aerodynamic penalties and enable distributed fuel storage.

The student will develop a conceptual aircraft design framework linking wing thickness, aerodynamic efficiency, structural volume and fuel storage requirements. Using simplified aerodynamic models informed by vortex-cell behaviour, including the limited operating envelope and sensitivity to control and flow conditions, the project will quantify trade-offs between increased drag from thicker wings and reduced fuselage penalties. Parametric studies will explore whether a “sweet spot” exists where overall aircraft performance improves, or whether the penalties associated with vortex-cell operation, including control power and instability, outweigh the benefits.

The outcome will be a system-level assessment of feasibility, providing clear numerical evidence on whether embedding hydrogen storage within vortex-cell-enabled wings is a viable design direction or fundamentally limited by aerodynamic and operational constraints.

Knoll, Aaron Dr

Project no: *AK01*

Project title: Numerical modelling of a novel plasma lens using negative ions (S) - 1 proj available

Supervisor: Knoll, Aaron Dr

Co-

supervisor(s):

Category: Computational;Theoretical

Software: PlasmaSim

(in-house code written in Julia)

Confidential: No

A plasma lens is a device that focuses a beam of charged particles using a plasma column. Such devices have applications in particle accelerators and for medical applications. The working principle is to create a region of trapped negatively charged particles, which exert a radial force on a beam of positively charged particles passing through them. The effect is to focus the beam of positively charged particles, analogous to an optical lens. Traditionally, the plasma lens is formed of electrons confined in a configuration of electric and magnetic fields. The aim of this project is to investigate a new approach based on negatively charged ions that are trapped using inertial electrostatic confinement.

This project will make use of an existing code, PlasmaSim, that is written in Julia language. No prior plasma physics experience is necessary, but I'm looking for students with an openness to learn new things. Through this project you will learn many practical skills that are specific to modelling plasmas numerically, so it's an ideal opportunity for those interested in becoming involved in the field of applied plasma physics.

Knoll, Aaron Dr

Project no: AK02

Project title: Design of a compact plasma toroid generator (S) - 1 project available

Supervisor: Knoll, Aaron Dr

Co-

supervisor(s):

Category: Theoretical;Design;Analysis

Software: Autodesk Fusion 360

Confidential: No

The pursuit of fusion energy through magnetic plasma confinement is an enduring challenge for humankind. While generating the required temperatures and pressures needed to produce fusion energy is theoretically possible, natural instabilities and turbulence occurring within the plasma allow the high energy particles to escape the magnetic field and rapidly drain energy from the system.

The goal of this project is to design an experimental test-bed to study the turbulence within a magnetically confined plasma, and investigate possible strategies to control (tame) this turbulence. This project will involve the development of a generator for a compact plasma toroid – a self-stable magnetically confined plasma structure, with a donut-like shape. This project builds on three previous MSc/MEng projects (2020/21, 2021/22, 2022/23, 2025/26).

Knoll, Aaron Dr

Project no: **AK03**

Project title: Design of a 3-axis Langmuir probe instrument for plasma thruster plume characterization (S) - 1 project available

Supervisor: Knoll, Aaron Dr

Co-supervisor(s):

Category: Theoretical;Design

Software: Autodesk Fusion 360LabView

Confidential: No

A Langmuir probe instrument is used to directly measure physical properties of a plasma including density, temperature and potential (voltage). To understand/verify the underlying physics of operation of plasma thrusters, it is useful to profile the plasma parameters of the thruster plume over a 3-dimensional volume. The purpose of this project is to construct a 3-axis translating Langmuir probe instrument capable of surveying the plasma properties with minimal disruption to the plasma thruster discharge. The translation mechanism will operate within a vacuum chamber environment, and interface with an external PC via LabView. This project builds on the design that was developed during an MSc project in 2019/20. It is anticipated that this instrument will play an essential role in future experimental activities conducted within Imperial's Plasma Propulsion Lab.

Knoll, Aaron Dr

Project no: **AK04**

Project title: Experimental testing of an E×B probe for plasma thruster plume characterization (S) - 1 project available

Supervisor: Knoll, Aaron Dr

Co-supervisor(s):

Category: Experimental;Experimental: Ready-made experiment;Analysis

Software: Matlab, LabView

Confidential: No

An E×B probe is an instrument used to measure the velocity and charge state of ions. While there are various methods for determining the average velocity of an ion beam, the advantage of the E×B probe is that it also distinguishes the velocity of each ion type (singly ionized, doubly ionized, triply ionized, etc.). The aim of this project is to experimentally qualify a new E×B probe that can be used to measure the plume of spacecraft plasma thruster (Electric Thruster) within Imperial's new large vacuum test facility: the Boltzmann Vacuum Chamber. This will involve the initial testing of the probe, as well as the development of a LabView

software interface to collect and analyse the data. It is anticipated that this instrument will play an essential role in future experimental activities conducted within Imperial's Plasma Propulsion Lab. This project builds on four prior MSc projects (2019-20, 2020-21, 2022-23, 2025-26) that involved the theory, sizing, CAD modelling and construction of the probe. No prior plasma physics experience is necessary, but I'm looking for students with an openness to learn new things. Through this project you will learn concepts and theory related to applied plasma physics, so it's an ideal opportunity for those interested in becoming involved in the field of spacecraft electric propulsion.

Knoll, Aaron Dr

Project no: **AK05**

Project title: Conceptual design of a hypersonic wind tunnel for very low earth orbit conditions (S) - 1 project available

Supervisor: Knoll, Aaron Dr

Co-

supervisor(s):

Category: Design;Analysis;Computational

Software: Julia, C++, Matlab

Confidential: No

There is a growing interest within the research communities and spacecraft industry to achieve air breathing electric propulsion at Very Low Earth Orbit (VLEO) conditions. A number of exciting propulsion concepts have already been proposed, which can sustain the orbital altitude of VLEO satellites using the rarified atmosphere in orbits between 250 to 350 km altitudes.

One of the key challenges in VLEO propulsion development is a lack of test facilities that replicate the conditions in a VLEO orbit. This project will investigate a "VLEO wind tunnel" that uses an plasma thruster to accelerate a population of ions to orbital velocities, followed by a charge neutralization and screening chamber to transform the charged ion beam into a neutral particle beam that will enter a test cell. The aim of the project is to create a conceptual design of a facility that can recreate the atmospheric composition, density, and velocity encountered in VLEO conditions.

Knoll, Aaron Dr

Project no: **AK06**

Project title: System-level trade study for a MicroGEO satellite using gridded ion thrusters (S) - 1 project available

Supervisor: Knoll, Aaron Dr

Co-

supervisor(s):

Category: Design; Analysis

Software:

Confidential: No

This project is done in collaboration with Thales Alenia Space UK, who helped prepare the project briefing below.

Introduction and Study Case

The GEOsat industry has a current value of €17.88 billion (projected to reach €22.54 billion by 2030). Within this, MicroGEO Satellites represent a fast-growing market segment. In 2023, three-quarters of GEOsat orders were in this category. The rise of reconfigurable satellites capable of performing both broadcast and broadband functions is expected to further grow the market and increase the number of GEOsat launches per year. The GEOsat segment remains the highest-revenue part of the commercial satellite industry and is evolving, with new mission profiles emerging.

The MicroGEO satellite (up to 500 kg) is defined by the high cost of orbit access typical of GEO, high reliability standards for long-duration missions (>15 years), but reduced development, manufacture costs and deployment time.

For example, MicroGEO sat launched in 2024 include those capable of orbit relocation to serve multiple customers. Due to maturity, MicroGEO satellite propulsion has defaulted to chemical and Hall Effect Thruster (HET) systems, both characterized by low specific impulse and limited lifetime. Gridded Ion Thrusters (GITs), historically HET competitors, are more complex and costly to develop but offer unmatched specific impulse and lifetime. This notable increase enables longer-duration GEO missions and therefore increased revenue for satellites in operation.

Analysis of Business segment and Mission needs

What are the proposed different business models?

- S/C owned by the TeleCOM entity vs S/C operated by the manufacturer to sell service on demand to different customers
- Fixed Geo orbit position vs orbital relocation to serve customers in different geographical locations
- Existing business cases: Astranis and Swiss12

Mission Needs

- Top level mission requirements: Mass/Volume (fairing allocation, n. of units)
- Redundancy -> multiple satellites
- Delta-V needs and mission lifetime
- EP propulsion class: power, thrust, ISP. HETs vs GIEs trade-off
- Size, mass, ISP Thrust, Thrust to power, lifetime

Impact on S/C system

- Impact on orbit raising (if needed), manoeuvring time, orbit relocation

- Impact on Business / service, cost of orbit raising and orbit relocation delay

Laizet, Sylvain Prof.

Project no: **SL01 ***

Project title: Quantum algorithms to solve PDEs - 2 projects available

Supervisor: Laizet, Sylvain Prof.

Co-

supervisor(s):

Category: Computational; Numerical; Theoretical

Software:

Confidential: No

Many scientific events can be explained by tracking the changes of a specific variable over space and time, which is typically represented by a partial differential equation (PDE). These equations are pervasive across various fields, from the study of plasmas and fluid dynamics to finance and biological systems, but they can be incredibly difficult to solve. Using quantum computers to solve PDEs more efficiently than current conventional methods could greatly hasten advancements in science. Despite the plethora of quantum algorithms suggested for tackling PDEs, the extent of their practical advantage remains uncertain. Quantum algorithms for tackling PDEs fall into two distinct groups: fully quantum algorithms, which utilise quantum circuits to manipulate the quantum state in accordance with the relevant PDE, and hybrid quantum-classical algorithms, in which a quantum computer plays a role in a broader classical computing process. In this project, various existing quantum algorithms will be implemented and tested to evaluate their performance for solving a wide range of PDEs.

Laizet, Sylvain Prof.

Project no: **SL02 ***

Project title: Learning to optimise wind farms with machine learning algorithms - 2 projects available

Supervisor: Laizet, Sylvain Prof.

Co-

supervisor(s):

Category: Computational; Numerical

Software:

Confidential: No

Wind farm optimisation presents a complex challenge involving multiple variables such as turbine placement, wind dynamics, and operational parameters. Machine learning algorithms offer promising solutions by analysing vast datasets to extract specific patterns. This research project aims to develop novel ML approaches that can maximise energy output while minimising environmental impact and operational costs. The project will explore reinforcement learning for real-time turbine control, deep learning for wind pattern prediction, and evolutionary algorithms for optimal spatial configuration. The goal will be to significantly increase renewable energy production efficiency, contributing to global sustainability goals and reducing the cost of wind energy. Practical implementations will be tested through simulations and published data at existing wind farms.

Laizet, Sylvain Prof.

Project no: SL03 *

Project title: Prediction models for gravity currents - 2 projects available

Supervisor: Laizet, Sylvain Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Numerical

Software:

Confidential: No

Gravity currents are common in nature, either in the atmosphere due to sea-breeze fronts, in avalanches of airborne snow or debris flows, or in the ocean due to turbidity currents and river plumes. They are created by the release of a fixed volume of a heavy suspension into a lighter ambient fluid. The aim of this project is to investigate various models to predict the extent of gravity current in various configurations. The gravity current will be modelled by either the single- or the two-layer shallow-water equations, the two-layer equations being necessary to include the effects of the overlying fluid, which are important when the depth of the current is comparable to the depth of the overlying fluid. The key element of the project is the implementation of numerical methods which can deal with discontinuities. Comparison will be made with existing experimental and numerical data. Previous experience using Matlab or Python would be very useful but is not mandatory.

Laizet, Sylvain Prof.

Project no: SL04 *

Project title: Wind farm power maximisation with analytical wake models - 2 projects available

Supervisor: Laizet, Sylvain Prof.

Co-

supervisor(s):

Category: Computational; Numerical; Theoretical

Software: FLORIS

Confidential: No

Modern large-scale offshore wind farms consist of multiple turbines clustered together usually in well-structured formations. Such a clustering exhibits a number of drawbacks during the operation of the wind farm, as some of the downstream turbines will inevitably have to operate within the wake of the upstream ones. A wind turbine operating within a wake field is an issue for two reasons. First, the apparent reduction of its power output due to the wind speed de-acceleration and second an increase of the fatigue loads due to experiencing the upstream wake-laden turbulence. Power losses due to wake effects were recently reported to be in the order of 10–25 % while the fatigue-related failures were reported to be around the same levels owing to a limited understanding of the offshore turbulence. To summarise, currently installed wind farms do not produce as much power as expected. Two common wind farm control strategies to deal with these issues are wake steering and axial induction control. Wake steering typically uses yaw misalignment of the turbines to redirect the wake around downstream turbines. In this project, optimisation campaigns will be performed for large-scale wind farms using FLORIS, which is a controls-focused wind farm simulation software incorporating steady-state engineering wake models into a performance-focused Python framework. The aim is to demonstrate the potential of wake steering to increase the power output of existing wind farms.

Lee, Koon-Yang Prof.

Project no: *KL01 **

Project title: Image recognition for conservation and preservation science - 2 projects available

Supervisor: Lee, Koon-Yang Prof.

Co-

supervisor(s): Panesar, Ajit Dr

Category: Computational; Survey; Numerical; Literature

Software:

Confidential: No

The Oddy test is a simple conservation screening method used to see whether a material releases harmful gases that could damage museum objects, especially metals, in enclosed display cases. It works by sealing the test material with silver, copper, and lead coupons under warm, humid conditions and checking the metal for corrosion after about 28 days. Its main problems are that the results can be subjective, poorly reproducible, and sensitive to small differences in procedure or evaluator experience. It is also non-specific, so it shows that something is being emitted but not exactly what the pollutant is or how risky it will be in real use. We believe that AI and image recognition can help reduce the subjective nature of Oddy-test interpretation by providing a more standardized, quantitative way to assess corrosion on the metal coupons. Instead of relying solely on a conservator's visual judgment, this project will train an image-analysis model to interpret Oddy test results to reduce the "subjectiveness" in the test. You are expected to be proficient at Python for this project. Project keywords: Oddy test, corrosion, convolutional neural networks Recommended reading: Long, E.R., Bone, A., Breitung, E.M. et al. Automated corrosion detection in Oddy test coupons using convolutional neural networks. Herit Sci 10, 150 (2022). <https://doi.org/10.1186/s40494-022-00778-3>

Lee, Koon-Yang Prof.

Project no: *KL02 **

Project title: How sustainable is sustainable materials? - 2 projects available

Supervisor: Lee, Koon-Yang Prof.

Co-

supervisor(s):

Category: Computational; Literature; Survey; Design

Software: Simapro for sure. Maybe Abaqus / 3DX is needed.

Confidential: No

If one were to do a quick search for "sustainable materials" on Google, one could find numerous sustainable material innovations that could potentially reduce our dependency on fossil derived resources. In fact, many of the companies or research groups who invented these innovations will boldly claim that their materials are truly environmental friendly without any proof because they are using "sustainable feedstock". This is very misleading as using sustainable feedstock as the raw material does not automatically imply the final finished product is truly sustainable. The environmental impact associated with the processing of these sustainable feedstock could be higher than processing fossil-derived resources. In this project, the students will be supporting existing research activities in Professor Koon-Yang Lee's group on vegan friendly sustainable leather, biodegradable espresso coffee pods, flexible paper packaging and many other wood-derived biobased products by conducting lifecycle assessment (LCA) for these different material innovations. The actual material and process to be modelled will be decided upon first meeting. Potential

students taking on this project must have taken or is taking AERO70002 Advanced Manufacturing.

Project keyword: Lifecycle assessment, sustainability, goal and scope definition, cradle-to-gate, cradle-to-grave, mid point and end point impact.

Recommended reading: Kooduvalli K, Vaidya UK, Ozcan S. Life Cycle Assessment of Compostable Coffee Pods: A US University Based Case Study. Sci Rep. 2020 Jun 8;10(1):9158. doi: 10.1038/s41598-020-65058-1.

Lee, Koon-Yang Prof.

Project no: *KL03 **

Project title: Materials Analysis Using Diffraction (MAUD): Can it be used for semi-crystalline materials? - 2 projects available

Supervisor: Lee, Koon-Yang Prof.

Co-supervisor(s):

Category: Literature; Numerical; Computational

Software: MAUD

Confidential: No

Crystallinity refers to the degree at which the atoms or molecules are arranged in an orderly pattern, forming a crystalline structure. The higher the crystallinity, the higher the mechanical performance. Generally speaking, crystallinity of a material can be quantified using Rietveld refinement. However, this refinement method possess a significant challenge to the analysis of semi-crystalline materials. In particular, Rietveld refinement requires information about crystal lattice structure but the presence of an amorphous phase, by the very definition, does not have a crystal lattice structures. In this project, students will be using developing a suitable protocol to use MAUD (<https://luttero.github.io/maud/>) to analyse the crystallinity of semi-crystalline materials, specifically cellulose.

Keywords: Bragg's law, microstructure analysis, crystallography and symmetry, unit cell, least-squares optimisation, cellulose I alpha, cellulose I beta

Recommended video: <https://www.youtube.com/watch?v=ygDWgvs1cg8&t=208s>

Levis, Errikos Dr

Project no: *EL01*

Project title: Design of Small UAVs for Joined Formation Flight - 2 projects available

Supervisor: Levis, Errikos Dr

Co-supervisor(s):

Category: Computational;Design

Software: matlab

Confidential: No

Current research into formation flying has concentrated on close or extended formation flying, which promises reductions of induced drag to a theoretical maximum of 50% for trailing aircraft. Joining multiple vehicles in flight would allow the wetted area and aspect ratio to be manipulated, optimising the swarm for the intended mission. For example joining the lifting surfaces of just two aircraft side by side could result in increases of up to 40% in endurance or 20% in range. Even greater improvements in range, and an increase in optimum cruise speed would be obtained if the wings of the aircraft were to overlap, reducing the overall wetted area and associated viscous drag and increasing the overall wing loading. Past work has shown substantial improvements in aerodynamic performance are possible but highlighted issues with vehicle stability and the added weight of joining mechanisms. This project will concentrate on the effect of different joining formations on the vehicle's overall mission performance.

Levis, Errikos Dr

Project no: *EL02*

Project title: Design and Optimization of Wing-In-Ground Effect vehicles - 1 project available

Supervisor: Levis, Errikos Dr

Co-supervisor(s):

Category: Computational;Design

Software: matlab

Confidential: No

The Wing-In-Ground Effect (WIG) concept has been around for some time and was investigated in depth by Soviet designers, resulting in number of vehicles known as Ekranoplans. Due to the potential reduction in drag achieved by flying in close proximity to the ground the concept has recently been revived by Boeing and other research institutes. Flying near the surface however does also complicate the aerodynamics and stability of the craft. The project will build on the work of previous students to design and optimize sample aircraft and evaluate the effect that size and operating points have on their performance and thus commercial viability.

Levis, Errikos Dr

Project no: *EL03*

Project title: Prediction of seaplane hull hydrodynamics - 1 project available

Supervisor: Levis, Errikos Dr

Co-supervisor(s):

Category: Computational;Theoretical

Software: Matlab, Python

Confidential: No

Predicting the hydrodynamic performance of planning hulls is a key step in determining the thrust requirement for seaplanes. While some semi-empirical methods have been developed in the past they fail to capture the hull's behaviour throughout the different regimes encountered during takeoff and landing. This project will thus aim to make use of machine learning tools to develop better design methods for seaplane hulls.

Levis, Errikos Dr

Project no: *EL04*

Project title: Hybrid-Electric/distributed propulsion for aircraft - 2 projects available

Supervisor: Levis, Errikos Dr

Co-supervisor(s):

Category: Computational;Design

Software:

Confidential: No

In June 2016, NASA announced that the X-57 research plane would be used to study the aerodynamic and propulsive efficiency improvements made possible by the use of hybrid electric distributed propulsion systems. Projects on this area will build on the work of past students and concentrate on:

- 1) Investigating the applicability of such technologies to aircraft of different types and sizes.
- 2) The design and optimization of a wing-distributed propulsion system.
- 3) The effects that using such systems may have on the stability and controllability of aircraft.
- 4) The design & optimization of hybrid electric aircraft for minimum cost vs CO₂ and the impact of extraneous factors

A good understanding of the principles of aircraft design and good coding skills are required.

Levis, Errikos Dr

Project no: *EL05*

Project title: Approximating real aircraft flight models using flight path data - 1 project available

Supervisor: Levis, Errikos Dr

Co-

supervisor(s):

Category: Computational; Numerical

Software:

Confidential: No

This project will aim to develop methods of approximating key performance metrics of aircraft using freely available ADS-B flight tracking data. Students applying for this project will require a strong understanding of aircraft performance and have an interest in learning about aircraft operations.

Levis, Errikos Dr

Project no: *EL06*

Project title: Development of a static Bleriot XI flight simulator with force feedback controls - 2 projects available

Supervisor: Levis, Errikos Dr

Co-

supervisor(s):

Category: Experimental: Manufacturing intensive; Computational; Design

Software:

Confidential: Yes

The department is working with a team intending to reproduce a historical air race, using simulators of the Bleriot XI aircraft. This project thus aims to develop a prototype of a fixed base flight simulator for a Bleriot XI. The project is composed of two sub-projects, namely the development of a flight model for the aircraft, and the design of the flight simulator hardware. The flight model will be developed using technical data from one (or more) of the surviving aircraft, coupled with analytical and computational methods to further characterise its aerodynamics and flight mechanics.

To develop the prototype flight simulator, students will have to consider the original aircraft's layout, as well as manufacturing and budgetary constraints in order to develop, prototype and test solutions for the simulator's control system. The development of a hardware solution will include developing an interface with the flight simulation software.

Li, Qianqian Dr

Project no: QL01

Project title: Nanoparticle reinforced lightweight metal composites via melt stirring (S)
– 2 projects available

Supervisor: Li, Qianqian Dr

Co-supervisor(s):

Category: Experimental;Experimental: Manufacturing intensive;Experimental: Ready-made experiment

Software:

Confidential: No

Nanoparticles such as nanocarbon and nano-SiC have attracted great attention as reinforcements in composites due to their excellent physical and mechanical properties. They are considered as ideal reinforcements to produce composites. Much research has been done on polymer composites; while metal composites have not been fully exploited due to the more difficult production process and more complicated reinforcing mechanisms, despite high potential of such composites. In this project, the student will produce nanoparticle reinforced magnesium composites via melting route and investigate the composites mechanical properties and microstructure. The production parameters will be investigated and optimised.

Li, Qianqian Dr

Project no: QL02

Project title: Nanocarbon based thermoplastic composites: manufacturing and characterisation (S) - 1 project available

Supervisor: Li, Qianqian Dr

Co-supervisor(s): Tagarielli, Vito Dr

Category:

Experimental;Experimental: Manufacturing intensive;Experimental: Ready-made experiment

Software:

Confidential: No

Nanocarbon such as carbon nanotubes and graphene based polymer composites have become a hot topic in composites in recent years. This is due to nanocarbon extraordinary physical and mechanical properties (very high Young's modulus, high electrical conductivity etc.), which makes it a highly interesting candidate for multifunctional composites. In this project, the student will optimise the production process and check the influence of the different parameters on the procedure (e.g. different concentration of graphene, different dispersion methods...) and manufacture such composites using casting and hot pressing. Characterisation of the mechanical properties, electrical and thermal conductivity of the composites will be a second focus point.

Li, Qianqian Dr

Project no: QL03

Project title: CFD simulation of nanoparticle dispersion in metal melt (S) - 1 project available

Supervisor: Li, Qianqian Dr

Co-supervisor(s): Peiro, Joaquim Prof.

Category: Computational; Analysis; Numerical

Software: The flow simulations will be performed using STARCCM+ and validated against experimental data. Initially we will consider SiC nano-whiskers (650 nm in diameter and 5 µm in length, rigid stick like) dispersing in magnesium melt at about 650 C. Then we will

Confidential: No

The use of carbon nanotubes (CNTs) in nano-composites is increasingly attracting interest due to their ability to increase the ductility and tensile properties of the composite. The focus of this project is on the use of CNTs in magnesium-based metal matrix composites (MMCs) which have extensive application in the automotive, aerospace and biomedical industries, especially for components working at temperatures above 150°C.

Our interest is on MMCs produced using methods involving molten metal. The major drawback of this type of methods is that CNTs tend to agglomerate resulting in poor interfacial bonding between the CNTs and the metal matrix after solidification, which deteriorates the mechanical properties of the nano-composite.

This project aims at investigating, through flow simulations, suitable configurations of the stirrer and crucible in the furnace to promote shear stresses that can be used to reduce or ideally prevent agglomeration. The flow simulations will be performed using STARCCM+ and validated against experimental data. Initially we will consider SiC nano-whiskers (650 nm in diameter and 5 µm in length, rigid stick like) dispersing in magnesium melt at about 650 C. Then we will proceed to simulate melts with nano-particles of smaller diameters such as multiwall carbon nanotubes (15 nm diameter and 1 µm length, softer tube shape). As part of

the investigation, we will assess a few furnace configurations with different nanoparticle concentration, energy input, stirring speed etc.

Li, Qianqian Dr

Project no: *QL04*

Project title: Elevated temperature mechanical behaviour of high-strength aluminium matrix composites (S) - 1 project available

Supervisor: Li, Qianqian Dr

Co-supervisor(s): Lewis, Peter (Industry partner)

Category: Experimental; Experimental: Ready-made experiment

Software:

Confidential: Yes

This project will investigate the mechanical performance of high-strength aluminium matrix composites (AMCs) across a range of temperatures relevant to aerospace and high-performance automotive applications. The focus will be on understanding how material composition and product form influence elevated-temperature tensile behaviour.

A systematic design of experiments will be developed in collaboration with an industrial partner (Materion UK), considering variables such as alloy matrix composition, reinforcement fraction (e.g. SiC content), and product form (e.g. billet, bar, plate, or sheet).

The student will carry out mechanical testing (primarily tensile testing) from room temperature up to 200 °C, alongside detailed microstructural characterisation (e.g. optical microscopy and SEM). The aim is to establish clear processing-structure-property relationships and identify key factors governing performance.

The project offers strong potential for generating publishable results and will provide the student with experience in both experimental materials engineering and industry-relevant research.

Liem, Rhea Dr

Project no: *RLI01 **

Project title: Sensitivity studies of aircraft sizing and cost to top level aircraft requirements (including environmental metrics) - 1 project available

Supervisor: Liem, Rhea Dr

Co-

supervisor(s):

Category: Computational;Theoretical;Literature

Software: Potentially use the open-source AeroFuse (written in Julia programming language -- very similar to MATLAB -- with Pluto notebook interface).

Confidential: No

Aircraft design is a highly multidisciplinary process, where changing one parameter in one discipline will change several others in other disciplines. This makes the process iterative (hence time consuming), since there is no closed-form solution in designing a system as complex as aircraft. There are some intuitions involved in the process (e.g., a larger aircraft will need a bigger engine and can typically fly faster) and some physicists have even shown that aircraft must obey some theoretical allometric rules, which put aircraft in the same category as birds and animals (e.g., a bigger bird also has bigger motive organs and also generally flies faster). However, some of the so-called intuitions are not easy to develop – as evident in the many students struggling with their aerospace vehicle design (AVD) projects! In this project, the student will work towards creating a “dashboard” that can show how aircraft sizing and cost are affected by top level aircraft requirements (TLARs) that drive aircraft design at the conceptual level. With the growing importance of environmental impact assessments in aviation, the TLARs might include some environmental metrics as well (e.g., fuel consumption, or even how contrail avoidance impact flight mission profile). Student can choose to focus on fundamental, theoretical studies (e.g., using the constructal law) or to use an existing open-source aircraft design framework AeroFuse (implemented using Julia programming with Pluto notebook interface) – or by combining both approaches. The deliverable of this project is expected to be able to help people (students, professionals, enthusiasts, hobbyists, etc.) have better intuition on aircraft design. More importantly, the dashboard can support decision-making processes in aircraft design.

Liem, Rhea Dr

Project no: *RLI02 **

Project title: Developing and using statistical physics modelling to study the underlying, unapparent behaviours of air transportation systems -2 projects available

Supervisor: Liem, Rhea Dr

Co-

supervisor(s):

Category: Computational;Theoretical;Analysis;Literature

Software: Python

Confidential: No

Complexity science, or complex systems science, was established in the mid-1980s as a response to the emergence of complex systems (be it in physical, biological, or social

systems). Complex systems are typically characterized by the presence of: (1) self-organisation, (2) interdependence and non-linearity, (3) connectivity, and (4) autonomy and adaptation. Understanding the connections and patterns of interaction between the components is key in understanding how the entire system behaves. However, with the above-mentioned characteristics, and the fact that the system can evolve, modelling and analysing complex systems are challenging.

Statistical physics modelling has emerged as a reliable method to study the macroscopic behaviour of a complex system by modelling the microscopic interactions between the components/elements in the model. In a way, it can be perceived as a form of agent-based modelling, with each component acts as an agent. While this modelling approach might not be able to realistically represent the real-world system, it has several benefits, including generality and its ability to model hypothetical situations (thanks to its datafree approach). In addition, statistical physics modelling is typically computationally efficient.

We can find some demonstrated success of statistical physics modelling in studying diverse fields such as ground traffic, pedestrian traffic, social networks, and population dynamics. The models can help identify phase transitions (e.g., from free-flow to congested in traffic-flow modelling) and critical density (e.g., when the traffic flow reaches its peak), which can be useful in designing relevant infrastructure. Despite its effectiveness in modelling ground and pedestrian traffic, its adoption to air traffic management (ATM) is still limited, owing to the higher level of complexity observed in aircraft movement. Some of these complexities are attributed to the ATM regulations and rules that aircraft must obey, as well as the reduced freedom of movement that aircraft encounter (e.g., an aircraft cannot freely turn like a pedestrian can, nor can it stop mid-air). Inspired and motivated by how statistical physics modelling has helped identify underlying behaviours in other traffic modes, this project is aimed to adapt statistical physics approach (e.g., cellular automaton [CA]) to suit aircraft traffic movement.

In this project, student will learn new topics such as statistical physics and air traffic. In addition to performing a literature review pertaining to these topics, the project will involve some theoretical work and a substantial amount of programming (mostly in Python).

Liem, Rhea Dr

Project no:	RLI03 *
Project title:	Investigation of the hydrodynamic, aerodynamic, and stability requirements for amphibious firefighting aircraft - 1 project available
Supervisor:	Liem, Rhea Dr
Co-supervisor(s):	
Category:	Theoretical;Computational;Analysis;Literature
Software:	Python/Julia

Confidential: No

More frequent forest fires and wildfires have increased firefighting fleets in the recent years, motivating the development of amphibious firefighting aircraft equipped with a scooping mechanism. Some recent developments in Europe include: SEAGLE by Roadfour, FREGATE-100 by HYNAERO, and 19-01. Amphibious aircraft are defined as air vehicles that can take-off and land from both ground and water, offering operational flexibility especially in areas with rough terrain or with limited areas for proper runways. Designing such vehicles is notably more complex than designing their conventional counterparts, primarily due to the trade-off between aerodynamic and hydrodynamic performance, watertightness requirements, and additional water stability requirements. Designing an amphibious firefighting aircraft is therefore not an easy feat – in addition to the challenges mentioned above, we now also need to ensure the effectiveness and quality of scooping mechanism and determine the optimum flight conditions and performance during scooping. In this project, student will investigate the aerodynamic, hydrodynamic, and stability performance requirements during the scooping procedure. The project will involve theoretical studies and developing some code modules (in Python or Julia) for these disciplines, which will be crucial in the development of aircraft design and performance analysis framework for this special-purpose air vehicle.

Liem, Rhea Dr

Project no: *RLI04 **

Project title: Investigating formation flight implementation in commercial aviation - 1 project available

Supervisor: Liem, Rhea Dr

Co-

supervisor(s):

Category: Computational; Literature; Analysis

Software: Python

Confidential: No

The aerodynamic benefits of formation flight have been demonstrated in literature, potentially reducing fuel consumption by 10% to 15%. However, their practical application in commercial aviation remains limited by current operational frameworks. Indeed, one of the major challenges in integrating formation flight strategy into the air transportation system is its coordination with the existing air traffic management system and route networks (i.e., flight schedules). Planned flight formations might not work as intended when some of the flights experience departure delays, which commonly occur. Existing studies are mostly limited to static flight missions, where mission parameters are determined pre-flight and without considering any air traffic management and airline operational constraints. This project goes beyond a feasibility study of formation flight implementation in commercial aviation. The student will incorporate operational constraints into the assessment of formation flight strategies suitable for commercial aviation and identify the trade-off between

aerodynamic benefits and operational penalties (e.g., fuel penalties incurred when aircraft loiter or adjust speeds). Ideally, we want to find a break-even point where aerodynamic gain outweighs tactical cost (or show that such a point does not exist, or only exists under specific conditions). To close the study, the student is expected to discuss the required infrastructure and technological advancements (e.g., inter-aircraft communication system, wake tracking systems, improved wing design) to bring formation flights closer to real-world implementation in commercial aviation -- and how they can do so.

Lubbock, Roderick Dr

Project no: *RLU01 **

Project title: Development of an assistive knee brace - experimental (S) - 1 project available

Supervisor: Lubbock, Roderick Dr

Co-

supervisor(s):

Category: Experimental;Experimental: Structures;Experimental: Manufacturing intensive;Design;Analysis

Software:

Confidential: No

Knee problems such as osteoarthritis and sports injuries to cartilage and ligaments affect millions of people around the world, and can have severe impacts on patients' physical and mental health. Assistive knee braces allow these patients to live a more normal active lifestyle. Numerous knee brace designs have been proposed [1] however the design problem is far from solved. The aim of this project is to design, build and test an assistive knee brace. Students interested in this project must have a solid understanding of structural mechanics and mechanical design and must be confident in designing, building and testing prototypes and carrying out laboratory experiments.

[1] L. Zhang, G. Liu, B. Han, Z. Wang, H. Li and Y. Jiao, "Assistive devices of human knee joint: A review", Robotics and Autonomous Systems, vol. 125, p. 103394, 2020. Available: 10.1016/j.robot.2019.103394.

Lubbock, Roderick Dr

Project no: *RLU02 **

Project title: A spatial impulse response technique for improving the frequency response of thermocouple probes (S) - 1 project available

Supervisor: Lubbock, Roderick Dr

Co-supervisor(s):

Category: Experimental;Theoretical;Analysis

Software:

Confidential: No

Thermocouple probes are ubiquitous in the field flow temperature measurement however they are limited by their low frequency response due to the large thermal inertia of the bead. This experimental project looks at improving the probe frequency response for a given flow by deriving the probe impulse response by traversing it through a spatially varying temperature field at different known velocities. This technique has already been shown to work for pressure probes [1].

The project was undertaken previously by an MSc student who obtained some promising results, however there were some significant shortcomings in his experiments. The aim of this project is to repeat these experiments and obtain improved data with the aim of a draft journal paper by the end of the project.

The project will make use of an existing experimental rig for traversing thermocouple probes across a hot flow from a heat gun at different velocities, along with a simple PC-based DAQ system. Theoretical aspects will include developing a solid understanding of thermocouple-probe based flow temperature measurements, whilst analytical aspects will include time-frequency analysis (impulse response, deconvolution, FFTs etc.).

[1] Burdett, Lubbock & Povey, 2018. AN IMPULSE RESPONSE TECHNIQUE TO IMPROVE THE EFFECTIVE FREQUENCY RESPONSE OF PRESSURE PROBES. XXIV Biennial Symposium on Measuring Techniques in Turbomachinery, Prague, Czech Republic, 29 - 31 August 2018. Paper No MTT2418B23. Available from: https://meastechturbo.com/paper-archives/2418-prague-2018/item/download/466_1bdd5c50c7ece7b207acee25244b0fac

Lubbock, Roderick Dr

Project no: *RLU03 **

Project title: Development of an open-source portable horizontal axis wind turbine - experimental - 1 project available

Supervisor: Lubbock, Roderick Dr

Co-supervisor(s):

Category: Experimental

Software:

Confidential: No

This project aims to develop an open-source horizontal axis wind turbine (HAWT) for charging personal devices such as laptops and mobile phones when e.g. camping/hiking in remote areas and during crisis situations (see e.g. mobile phone charging clubs in war-torn Ukraine). A closed-source device has recently been developed (<https://shineturbine.com>) however this is expensive and has relatively low power for its size. Amateur developers have also developed interesting open-source devices (<https://hackaday.io/project/185070-3d-printed-portable-wind-turbine>) however these are low power offering a lot of scope for development by an experienced engineer.

The project will primarily focus on the non-aerodynamic aspects of a HAWT device in order to develop a prototype that can easily accommodate various blade designs, and ideally can also be used as a test stand for continuous development of improved blades by the open-source community (this may however require a separate test vehicle).

Magri, Luca Prof.

Project no: *LMG01*

Project title: Precursors of extreme events and critical transitions: A data-driven deterministic approach (S) - 1 project available

Supervisor: Magri, Luca Prof.

Co-supervisor(s):

Category: Computational;Theoretical;Numerical;Machine learning

Software: Tensorflow, Python, Pytorch (maybe)

Confidential: No

This project will investigate how quantum-inspired machine learning can be designed and evaluated for short-horizon forecasting of low-to-mid dimensional chaotic systems. The main aim is to explore the relationship between reservoir architecture, noise, and forecasting performance, using generalized synchronization, echo state property, and dynamical stability as a conceptual guide [1]. The project will focus on one or two benchmark systems, such as the Lorenz, and compare a recurrence-free quantum reservoir architecture against a simple classical reservoir baseline. The goal is to scale this up to larger dimensional systems. The student will develop a reproducible simulation pipeline, test a limited range of reservoir hyperparameters, and measure prediction accuracy, forecast horizon, and robustness to injected noise. A further optional objective is to examine whether the reservoir can also capture qualitative properties of the underlying dynamics, such as attractor geometry or sensitivity to initial conditions.

The ideal candidate is a student who is willing to learn things that they have not studied in the Imperial course in Aero, do some pen and paper work (i.e., maths), and coding.

No CFD or large-scale simulations of complex turbulent flows will be part of this project.

[1] <https://royalsocietypublishing.org/rspa/article/481/2324/20250550/234346>

Magri, Luca Prof.

Project no: *LMG02 **

Project title: Quantum machine learning: Time series forecasting of nonlinear systems and extreme events - 1 project available

Supervisor: Magri, Luca Prof.

Co-supervisor(s):

Category: Computational;Theoretical;Literature

Software: Python or Matlab

Confidential: No

This project will study an efficient numerical method for a focused class of fluid-flow problems, using the paper as a guide for algorithm design, stability, and accuracy. The student will implement a simplified solver for the governing equations, test it on one or two benchmark cases, and compare performance against a standard baseline in terms of error, runtime, and robustness. The project remains flexible in the choice of discretisation, boundary conditions, and test cases, but specific enough to deliver a working code, quantitative results, and a short evaluation of when the method is most effective. The goal is to design an architecture that can learn a fast-slow decomposition of dynamical systems based on [1] and the theory from [2].

[1] <https://www.sciencedirect.com/science/article/pii/S0045782525008722>

[2] <https://arxiv.org/abs/2604.12869>

Magri, Luca Prof.

Project no: *LMG03 **

Project title: Reduced-order modelling of turbulent flows with quantized local reduced-order models - 1 project available

Supervisor: Magri, Luca Prof.

Co-supervisor(s):

Category: Computational

Software:

Confidential: No

This project investigates quantized local reduced-order modelling in time (ql-ROM) for nonlinear dynamical systems. Using clustering, the solution manifold is partitioned into local regions, and separate POD-Galerkin reduced models are built for each cluster. The method will be tested on the Kuramoto-Sivashinsky equation, comparing ql-ROM with a standard global ROM in terms of stability, short-term prediction error, and long-term statistical

accuracy. The project aims to determine whether local models provide improved performance for highly nonlinear, chaotic dynamics. Results will help assess the practicality of ql-ROM as a computationally efficient framework for reduced-order simulation and prediction of complex systems.

[1] <https://www.sciencedirect.com/science/article/pii/S0045782525006656>

[2] <https://arxiv.org/abs/2511.04586>

Mainini, Laura Prof.

Project no: *LMN01 **

Project title: Data-efficient early diagnostics and prognostics for sustainable aviation (S) - 2 projects available

Supervisor: Mainini, Laura Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Literature

Software: MATLAB and Simulink; modelling and simulation SW; possible access to HPC

Confidential: No

Next generation aircraft require the development and integration of a deal of innovative technologies to meet the ambitious sustainability goals set for aviation. This transformational effort is associated with a tremendous increase of the complexity of the onboard systems and their multiphysics coupled

behaviours. Additional tangles arise from the stringent requirements on safety, reliability and operational performances of aerospace systems which further complicate the introduction onboard of novel technologies. In this scenario, the rapid identification of incipient multimodal faults constitutes a significant opportunity to accelerate the actualization of sustainability ambitions. The project focuses on (i) study of power systems/architectures for low emissions and their fault modes; (ii) investigation of computational methods and active learning strategies for data-efficient diagnostics and reliability assessment of multi-physics onboard systems; (iii) study and assessment of the impact of advanced (early-stage, data-efficient) diagnostics and prognostics on the time-to-adoption/integration of new technologies. The thrusts build upon concepts of modelling and simulation, dynamic data driven methods, uncertainty quantification, data assimilation, predictive digital twins and scientific machine learning.

Mainini, Laura Prof.

Project no: *LMN02 **

Project title: Multisource learning for multidisciplinary design of aerospace vehicles (S)
projects available

Supervisor: Mainini, Laura Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Literature

Software: MATLAB (or Python); modelling and simulation SW; possible access to HPC

Confidential: No

The adoption of detailed models for the multidisciplinary design of aerospace vehicles permits to enhance the identification of superior design solutions, but would prohibitively rise the demand for computational resources and time. Multisource methods acknowledge the availability of multiple representations (models) of physical systems and phenomena that can be interrogated to inform the design process. Those representations can differ for their underlying governing assumptions, their resolution or level of captured details, the cost required to be evaluated or obtain responses, etc..

Based on the Box's principle of model usefulness, these models can be used in concertation to achieve major savings of computational time and resources. This project focuses on the study and development of computational methods to dynamically elicit different models to efficiently inform and guide the multidisciplinary design of aerospace vehicles. Specific applications are defined across (i) novel power architecture solutions and non-conventional configurations for sustainable aviation; or (ii) scalable benchmark problems for aeronautics and/or space applications.

The project requires interest in enriching the familiarity with the specific application and the associated technologies. The

studies build up on modelling and simulations, and leverage concepts of multidisciplinary design optimization, multifidelity methods for design optimization, physics based machine learning, active learning from multiple oracles.

Mainini, Laura Prof.

Project no: *LMN03 **

Project title: InfoSymbiotics for next generation aerospace vehicles' readiness and operations (S) - 2 projects available

Supervisor: Mainini, Laura Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Literature

Software: MATLAB (or Python); modelling and simulation SW; possible access to HPC

Confidential: No

Reliability and readiness of aerospace vehicles in future operational scenarios can be augmented through a dynamic reshaping of their mission profiles in response to the evolution of their health state and contingencies. The hazards might be hard to characterize a priori, but the impact onto the mission could result in catastrophic failure. The structural integrity or the nominal functioning of the systems could be dramatically jeopardized, eventually leading to the loss of the vehicle. The possibility to reconfigure the mission would allow the aerospace vehicle to complete the mission successfully and continue operating safely under degraded conditions, without over-replacing parts, oversizing components, and multiplying systems redundancies. This project focuses on computational methods to enable dynamic mission replanning: forms of artificial reasoning that (i) learn from sensor measurements and predict in real-time the evolving/residual capabilities, (ii) use those predictions to enable fast and trustworthy autonomous operational decisions, (iii) use sensor measurements, capability predictions and the updated decision space to dynamically refine data acquisition strategies. The thrusts build upon concepts of modelling and simulation, infosymbiotics and dynamic data-driven methods, predictive digital twins, model order reduction, scientific machine learning, data fusion and virtual sensing.

Mainini, Laura Prof.

Project no: *LMN04 **

Project title: Design of Deformable Wing Solutions for Improved Aircraft Efficiency (3 projects) (S) - 3 projects available

Supervisor: Mainini, Laura Prof.

Co-supervisor(s): Venetsanos, Demetrios Dr

Category: Computational; Design; Experimental: Structures

Software:

Confidential: No

This project aims to design and prototype a deformable wing system to address the limitations of conventional mechanical control structures in aircraft wings by developing a novel approach that integrates deformability into the entire wing structure. The primary objective of this project is to explore design alternatives and construct a deformable wing that can efficiently control an aircraft's motion through complaint shape changes. By utilizing lightweight materials and advanced manufacturing techniques, the goal is to achieve a significant reduction in weight while maintaining or improving aerodynamic performance. The design process will entail selecting appropriate materials and sizing the structural components, as well as determining the optimal assembly configuration to achieve the desired flexibility and aerodynamic properties. A deformable wing prototype will then be built to conduct structural and wind tunnel testing, characterize aerodynamic performance and structural integrity, and conduct comparative assessment to conventional wing structures. The ambitious research outcomes will contribute to advancing the field of aeronautical

engineering and may have potential applications in various industries, including aircraft manufacturing, robotics, and renewable energy.

Interested students will have the freedom to come up with morphing configurations as they deem suitable, e.g. with distributed control solutions. Also, should they wish to, interested students will have the opportunity to explore computational methods for the design (to include generative approaches), rather than go straight into the prototyping.

Mylvaganam, Thulasi Dr

Project no: *TM01 **

Project title: Autonomous collision avoidance for multi-agent systems
- 1 project available

Supervisor: Mylvaganam, Thulasi Dr

Co-

supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical;Literature;Survey

Software: MATLAB

Confidential: No

Teams of autonomous (ground or airborne) vehicles, often referred to as agents, can perform tasks with minimal human intervention. This is of interest to several applications, such as space exploration, search and rescue missions and structural monitoring and maintenance. Recently the problem of distributed autonomous collision avoidance has been considered using a game theoretic framework. The method is remarkable in that it comes with strong performance guarantees, while having low computational demands.

In this project, the student will familiarise himself/herself with the aforementioned approach, and perform a thorough comparative study with alternative techniques available in the literature. As such, the project will involve a thorough literature review, and gaining familiarity with various approaches, all of which involve elements of nonlinear dynamical systems and control. It is therefore crucial that the student has a strong background and interest in mathematics and control. As such, it will be beneficial for the student to follow the optional module on "Advanced Control".

Mylvaganam, Thulasi Dr

Project no: *TM02 **

Project title: Data-driven (approximate) solutions of nonlinear optimal control problems
- 1 project available

Supervisor: Mylvaganam, Thulasi Dr

Co-supervisor(s):

Category: Computational;Theoretical;Design;Analysis;Survey;Numerical;Literature

Software: MATLAB

Confidential: Yes

The solution of optimal control problems involving nonlinear systems is characterised by means of the so-called Hamilton-Jacobi-Bellman partial differential equation (PDE). Obtaining closed-form solutions to PDEs is, in general, not feasible and obtaining numerical solutions is, in general, computationally expensive. To circumvent this issue, we have developed a method to systematically construct approximate solutions to nonlinear optimal control problems. The method requires the solution of nonlinear algebraic equations (in place of PDEs). Recently, we presented a method to solve these using data in place of accurate model knowledge.

This project is centered about this data-driven approach. Starting from the basics, the student will delve into the details of the underlying theory, gain familiarity via the consideration of simple numerical examples and, subsequently, delve into meaningful simulation studies involving an application of the student's choice (e.g. related to robotic systems, biomedical systems, space systems, etc.). In doing so the student may be required to look into further theoretical aspects (yet to be formally established), e.g. to consider the impact of noise.

The project will inevitably involve topics in nonlinear control (related to the control method considered) and it will be of a highly mathematical nature. Thus, a strong background in mathematics and control is essential. It will be beneficial to follow the optional module "Advanced Control". The project will involve programming in MATLAB.

Mylvaganam, Thulasi Dr

Project no: *TM03 **

Project title: Control of robotic systems for space applications (S) - 1 project available

Supervisor: Mylvaganam, Thulasi Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical;Literature

Software: Matlab

Confidential: No

Robotic systems are ubiquitous, and space is no exception. In this project the student will consider robotic systems that are of interest to space applications (e.g. robotic arms). For certain types of missions, accurate models describing the behaviour of such robotic systems may be lacking (e.g. due to the robotic system operating in an unknown environment) and, simultaneously, efficient operation may be crucial (e.g. due to energy limitations).

Namely, robotic systems in space may be required to operate efficiently and reliably in unknown environments. Motivated by this, the student will consider data-driven and/or optimal control of robotic systems for space applications. The student will explore existing models describing the dynamics of such systems in “typical” environments and exploit these to build a “simulation environment” in Matlab. The student will then consider data-driven control (treating the dynamics as fully/partially unknown) and/or optimal control of the considered robotic system. Both objectives may necessitate the consideration of nonlinear control techniques. The performance of developed controllers will be studied via simulations.

The project will involve the consideration of nonlinear dynamical systems and advanced control techniques, as well as convex optimisation. For this reason, an excellent background and a keen interest in mathematics and control theory is absolutely essential. The project will involve a significant amount of programming in Matlab (and will require gaining familiarity with various toolboxes available for solving semidefinite programming problems).

Mylvaganam, Thulasi Dr

Project no: *TM04*

Project title: Data-driven control of snake-like robots (S) - 1 project available

Supervisor: Mylvaganam, Thulasi Dr

Co-

supervisor(s):

Category: Computational;Theoretical;Design;Literature;Numerical

Software: Matlab (and toolboxes such as cvx)

Confidential: No

Snake-like robots have gained interest for various applications, e.g. fire fighting, subsea inspection and space exploration. However, as they may operate in unknown environments, often models describing their behaviour are not known exactly. Recently, we have considered data-driven control of such systems and the application of the developed control method has been validated via relatively simple simulation studies. In this project the student will explore applications of the developed method further, specialising on one/two specific application avenues (potentially involving space applications). The work will involve gaining familiarity with the developed control framework, which is based on a simplified "control-oriented model". Having gained familiarity with the state-of-the art, the student will consider the application of the developed control method to a more accurate (and more complex) model of the snake robot, and they will perform detailed simulation studies (including the development of visual demonstrations) related to the chosen application avenues. This will involve undertaking a detailed literature review, to understand how to represent "typical" environmental conditions and robot configurations for the selected application avenues.

The project is of a highly mathematical nature and will, inevitably, involve consideration of nonlinear dynamical systems and nonlinear control. It is therefore essential that the student has an excellent background in mathematics and control engineering (the optional module "Advanced Control" will be beneficial). The project will involve significant programming in Matlab and the use of various toolboxes (e.g. cvx).

Mylvaganam, Thulasi Dr

Project no: *TM05 **

Project title: Autonomous collision avoidance for multi-agent systems (Experimental)
- 1 project available

Supervisor: Mylvaganam, Thulasi Dr

Co-supervisor(s):

Category: Experimental: Ready-made experiment

Software: MATLAB and C/C++/Python

Confidential: No

Teams of autonomous (ground or airborne) vehicles, often referred to as agents, can perform tasks with minimal human intervention. This is of interest to several applications, such as space exploration, search and rescue missions and structural monitoring and maintenance. Recently the problem of distributed autonomous collision avoidance has been considered using a game theoretic framework. The method is remarkable in that it comes with strong performance guarantees, while having low computational demands.

In this project, the student will familiarise himself/herself with the aforementioned approach. Once the student has gained sufficient familiarity with the theory, they will validate the results experimentally using off-the-shelf platforms available in the Aerial Robotics Lab (this includes small drones and ground robots). Note that any experimental work can take place only if/when sufficient understanding of the theory is in place. The student may then also be required to consider scenarios which require further developing the existing theory.

The project will inevitably involve elements of nonlinear dynamical systems and control. It is therefore crucial that the student has a strong background and interest in mathematics and control. As such, it will be beneficial for the student to follow the optional module on "Advanced Control". Considering that the project involves significant experimental work, it is also crucial that the student has an aptitude for experimental work and is motivated to undertake such work independently, gaining familiarity with the platforms used.

Mylvaganam, Thulasi Dr

Project no: *TM06 **

Project title: Data-driven control of nonlinear systems (S) - 1 project available

Supervisor: Mylvaganam, Thulasi Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical;Literature

Software: MATLAB (plus associated toolboxes, e.g. cvx)

Confidential: No

Modern dynamical systems are increasingly complex. As a consequence accurate models may be either unavailable or too complex for control design. Motivated by this, the student will consider recently developed data-driven control techniques for nonlinear systems. Specifically, the student will consider methods that consider techniques that build upon ideas that utilise Willems' Fundamental Lemma (for linear systems).

In the first part of the project the student will gain familiarity with the aforementioned theory. Subsequently, the student will perform thorough simulation studies (involving an application of the student's choice -- provided it is amenable with the considered control technique).

The project will inevitably involve aspects of nonlinear dynamic systems and control. It will therefore be beneficial for the student to follow the optional module on "Advanced Control".

The project will involve programming in MATLAB (and related toolboxes, eg. cvx). The project will be of a highly mathematical nature, so it is important that the student has an interest and aptitude for such work.

Nóvoa, Andrea Dr

Project no: *AN01*

Project title: Data-driven reduced order modelling in fluid mechanics –
1 project available

Supervisor: Nóvoa, Andrea Dr

Co-supervisor(s):

Category: Computational;Theoretical

Software: Python, Jax, MATLAB

Confidential: No

This project explores data-driven reduced-order modeling (ROM) in fluid mechanics to address the computational challenges of simulating complex flows. By leveraging linear and nonlinear encoding-decoding techniques, the project aims to construct efficient ROMs that capture dominant flow structures while minimizing computational costs. A key focus is to increase the ROM accuracy for turbulent and transient flows, where traditional methods

struggle due to nonlinear dynamics and missing scales. The project will exploit machine learning tools to perform tasks such as closure modelling, nonlinear manifold learning, or control. The application of the project will be relevant to aeronautic and environmental engineering.

Nóvoa, Andrea Dr

Project no: **AN02**

Project title: Real-time modelling with data assimilation - 2 projects available

Supervisor: Nóvoa, Andrea Dr

Co-

supervisor(s):

Category: Theoretical;Computational

Software: Python

Confidential: No

This project aims to introduce physical knowledge into real-time data assimilation methods to enable real-time physical modelling. Physics-constrained data-assimilation frameworks will be developed by integrating conservation laws, statistics, and Kalman filtering approaches. The challenge is to ensure that the models remain physically plausible while adapting to incoming data. Students will explore concepts from signal processing, constrained optimization and Bayesian theory. These methods will be applied to complex, real-world systems—such as aviation and weather prediction—where real-time decision-making under uncertainty is critical.

Nóvoa, Andrea Dr

Project no: **AN03**

Project title: Real-time modelling in high dimensional systems - 1 project available

Supervisor: Nóvoa, Andrea Dr

Co-

supervisor(s): Magri, Luca Prof.

Category: Theoretical

Software:

Confidential: No

This project aims to introduce physical knowledge into real-time data assimilation methods to enable real-time physical modelling. The challenge is to ensure that the models remain

computationally tractable in high-dimensional systems. Students will explore concepts from signal processing, Bayesian theory, and potentially quantum computing. These methods will be applied to complex, real-world systems—such as aviation and weather prediction—where real-time decision-making under uncertainty is critical.

Palacios, Rafael Prof.

Project no: *RP01 **

Project title: Control system design for wind tunnel demonstration of flutter suppression
1 project available

Supervisor: Palacios, Rafael Prof.

Co-

supervisor(s):

Category: Experimental;Computational;Design

Software: Matlab, Python, LabView

Confidential: No

This project is an investigation on aeroservoelastic vehicle design, in particular, the integration of control systems on wings to increase their flutter speed (an stability augmentation control problem). This can tilt the trade-off between aerodynamic and structural constraints towards much lighter wing designs, but at the expense of a much complex design evaluation. In this project, flutter suppression strategies will be first numerically investigated on an existing actuated composite wing. A flap driven by an Arduino card has been shown to produce efficient actuation for a wind-tunnel implementation with minimum delays, but the optimal layout of sensors (accelerometers and strain gauges) and the identification of the closed-loop system dynamics needs to be identified. The project will address those issues to identify and demonstrate effective aeroelastic stability augmentation strategies, both through simulation and experiments.

Palacios, Rafael Prof.

Project no: *RP02 **

Project title: Data-driven model for wing loads under gusts - 1 project available

Supervisor: Palacios, Rafael Prof.

Co-supervisor(s):

Category: Computational

Software: python, matlab

Confidential: No

A classical aerodynamic solution due to W. P. Sears describes the interactions between an airfoil and an incoming flow. It is however cumbersome to use directly for simulation and numerous approximations have been proposed, with mixed success. In this project we will explore data-driven solution using recently developed algorithms for sparse identification in nonlinear dynamics. We will further seek extensions of the approach to wings in atmospheric turbulence. This is a computational project that can will need both python and matlab.

Further reading: <https://doi.org/10.2514/1.J060863>

Palacios, Rafael Prof.

Project no: *RP03 **

Project title: Hinged winglets for aircraft load alleviation - 1 project available

Supervisor: Palacios, Rafael Prof.

Co-supervisor(s):

Category: Computational

Software: python

Confidential: No

Hinged winglets have been recently proposed to reduce adverse aerodynamic loading due to atmospheric gusts (<https://www.airbus.com/newsroom/stories/the-albatross-is-inspiring-tomorrows-next-generation-of-aircraft-wings.html>). As this is a passive mechanism, the challenge is to obtain a single design that shows good performance in different flight segments. In this project, this will be investigated using an in-house aeroelastic solver for flexible aircraft dynamics that includes a multibody implementation to model the hinge dynamics (www.imperial.ac.uk/aeroelastics/sharpy). The aeroelastic response of wings of increased flexibility and aspect ratio, under different hinge partitions will be explored aiming to identify any sweet spots in the design space for this setup. This project extensively uses python.

Further reading: <https://doi.org/10.2514/1.C035602>

Panesar, Ajit Dr

Project no: *AP01 **

Project title: Scientific ML in advanced manufacturing process simulations - 2 projects available

Supervisor: Panesar, Ajit Dr

Co-supervisor(s):

Category: Computational; Numerical

Software: Python, Ansys, Abaqus

Confidential: No

Scientific Machine Learning (SciML) in additive manufacturing (AM) process simulations is an emerging field that blends physics-based models with machine learning to improve simulation fidelity, reduce computational cost, and enable real-time or data-driven insights. The group's earlier efforts have included solving PDEs related to AM processes (e.g., heat conduction) using neural nets that embed physical laws to estimate the Temperature field in metal AM. The aspiration is to predict part distortions and utilise this capability for both inverse problem (process parameter identification) and design optimisation. ML models could include: Physics-Informed Neural Networks (PINNs); Gaussian Processes for uncertainty quantification; Autoencoders for dimensionality reduction; Neural operators (e.g., DeepONet, FNOs) for learning operators between function spaces (great for PDE-based simulations); and Bayesian optimization for tuning process parameters. For more context, refer <https://idea-lab-ic.github.io/#/README>

Panesar, Ajit Dr

Project no: **AP02 ***

Project title: ML enabled discovery of metamaterials for demanding engineering application - 1 project available

Supervisor: Panesar, Ajit Dr

Co-supervisor(s):

Category: Computational; Numerical; Analysis

Software: Python, Ansys, Abaqus

Confidential: No

ML is rapidly transforming how materials are designed for mechanical, thermal, acoustic, and electromagnetic functions. These smart materials can be tailored to perform specific engineering tasks and ML makes it possible to explore massive design spaces, optimize performance, and reduce cost/time in ways traditional methods can't. We will utilise inverse design coupled with topology optimisation to realise the multiscale solutions for end-use engineering applications. Multiple materials, material orthotropy and geometric non-linearities will be considered. Prior work in the group has explored Mixture Density Networks (MDNs) to enable one-to-many solutions which will be leveraged in this project to allow a portfolio of solutions to be obtained.

For more context, refer <https://idea-lab-ic.github.io/#/README>

Panesar, Ajit Dr

Project no: **AP03 ***

Project title: Metal additive manufacturing in Space (Experimental) (S) - 1 project available

Supervisor: Panesar, Ajit Dr

Co-supervisor(s): Knoll, Aaron Dr

Category: Experimental;Experimental: Manufacturing intensive;Experimental: Ready-made experiment;Numerical;Analysis

Software: Python, Ansys

Confidential: Yes

Maturing additive manufacturing in space involves developing space grade hardware and revisit the parameters required for an optimal print to succeed in microgravity, vacuum, and extreme thermal environments. These are radically different from Earth, so parameters needs adaptation — and in many cases, require support from numerical models. This project will focus on material extrusion based process but employ Martian/Lunar-regolith with Clay to explore the feasibility of a drying based consolidation method. Previous projects have documented knowhow on vacuum chamber testing, developing hardware for polymer blend extrusion, and computational simulation of such processes in-Space conditions. Maturing the manufacturing readiness levels of these technologies brings us one step closer to fabricate in Space environment.

For more context, refer <https://idea-lab-ic.github.io/#/README>

Panesar, Ajit Dr

Project no: **AP04 ***

Project title: Multimaterial topology optimisation for multifunctional applications (S) - 1 project available

Supervisor: Panesar, Ajit Dr

Co-supervisor(s):

Category: Computational;Design;Numerical;Analysis

Software: Python, Abaqus

Confidential: No

Multimaterial topology optimisation (MMTO) is an advanced computational design technique that determines not only the optimal shape of a structure but also how multiple materials should be distributed within it to achieve the best performance under given constraints.

Unlike traditional single-material methods, it enables the use of materials with different mechanical, thermal, or functional properties—making it ideal for high-performance applications in aerospace, automotive, biomedical implants/devices, and additive manufacturing. Using methods like SIMP, level-set, or phase-field models, and often enhanced with machine learning or surrogate models, designers can optimize structures for stiffness, weight, thermal resistance, and manufacturability simultaneously. This approach is especially powerful when combined with multi-material 3D printing, allowing for the creation of highly efficient, functionally graded, or hybrid-material components. This project will explore utilisation of MMTO for both small (cm) and large (m) scale parts considering appropriate manufacturing routes therefore progressing towards process-aware design-optimisation.

For more context, refer <https://idea-lab-ic.github.io/#/README>

Panesar, Ajit Dr

Project no: *AP05 **

Project title: Foundational models for multifunctional component design - 1 project available

Supervisor: Panesar, Ajit Dr

Co-

supervisor(s):

Category: Literature;Numerical;Computational;Analysis;Theoretical

Software: Python, Abaqus, CAD

Confidential: No

Foundational models (e.g. LLMs) are revolutionising complex engineering multifunctional CAD design by enhancing the ability to create and optimise designs that meet diverse performance requirements across multiple engineering domains. By analysing vast amounts of design data, LLMs can assist in automating the generation of CAD models, optimise for multiple objectives (such as mechanical strength, thermal efficiency, and material usage), and offer real-time design suggestions. These models enable multi-disciplinary integration, facilitating the design of components or systems with complex, interdependent functions. LLMs also streamline the iterative design process, reducing the time spent on adjustments and improving collaboration across different engineering disciplines, making them invaluable tools for industries like aerospace, automotive, and industrial design. This project explores the value added through these models in both the iterative design-optimisation stage and the design-to-fabrication workflow. There is flexibility to take a more systems approach or deep-dive into a more specific computational aspect of the study.

For more context, refer <https://idea-lab-ic.github.io/#/README>

Panesar, Ajit Dr

Project no: **AP06 ***

Project title: Topology optimisation of shape morphing lenses for Space applications (S)
1 project available

Supervisor: Panesar, Ajit Dr

Co-

supervisor(s):

Category: Design;Numerical;Computational;Experimental: Ready-made
experiment

Software:

Confidential: No

This project focuses on designing flexible (compressible/twistable), jointless multiscale structure that perform specific electromagnetic functions — such as lenses or antennas — while operating under combined thermal and mechanical loading conditions, mainly relevant to launch and in-orbit conditions. Methods like SIMP, level-set, and phase-field are commonly employed to determine the optimal material layout, which will be leveraged to identify superior multi-objective performance. There is an expectation that prototypes will be fabricated to demonstrate the mechanical and where possible the thermal response. Possibility to undertake electromagnetic characterisation at partner institution will be explored. The primary objective of this project will be to develop a computational framework that can reveal manufacture-ready performance optimal solutions.

Papadakis, George Prof.

Project no: **GP01 ***

Project title: Application of machine learning to flow control. (S) - 1 project available

Supervisor: Papadakis, George Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Numerical

Software: MATLAB

Confidential: No

The aim of this project is to combine machine learning and more traditional techniques to control non-linear flow systems that exhibit chaotic behaviour. The method will be based on a recent development by the supervisor that allows the forecasting of a chaotic system from sparse data. The performance of the new approach will be assessed in two test cases, (a) non-linear model that describes the self-sustaining process in near-wall turbulence and (b)

the Kuramoto Sivashinsky equation that exhibits chaotic behaviour for a certain set of parameters. The project will examine the time that it takes to synthesise the controller and its performance with respect to fully traditional methods.

Papadakis, George Prof.

Project no: GP02 *

Project title: Flow and parameter estimation from sparse data (S) - 1 project available

Supervisor: Papadakis, George Prof.

Co-supervisor(s):

Category: Computational;Theoretical

Software: Open foam

Confidential: No

Data Assimilation techniques allow the incorporation of sparse measurement data into a computational model in order to extract velocity or pressure in the whole domain. It is also possible to extract unknown parameters, such as the approaching velocity. For example, we can predict the flow around an aerofoil using only pressure and shear stress measurements at the surface of the aerofoil, without any other information, such as the approaching velocity or the angle of attack. In this project, we will extend and apply an algorithm recently developed by the supervisor to find the lift and drag coefficients using only surface data on the aerofoil. A critical question is the optimal sensor location, are there locations that are better than others? Which variable provides more information, pressure or shear stress? There are numerous applications in many other fields, such as environmental flows, cardiovascular flows etc.

Papadakis, George Prof.

Project no: GP03 *

Project title: Foundational models in fluid dynamics. (S) - 1 project available

Supervisor: Papadakis, George Prof.

Co-supervisor(s):

Category: Computational;Theoretical

Software: OPEN FOAM

Confidential: No

The central objective of the project is to explore foundational models in fluid dynamics. More specifically we aim to predict an unsteady flow without performing a simulation, only using prior training data. We will apply the idea to the unsteady flow in a bifurcation. Simulations will be performed for different bifurcation angles to build a database, and the flow in another angle will be predicted using only information from the database. The key challenge is the geometric change of the new bifurcation with respect to the members of the database and how this can be handled. There are numerous applications of this technology in all areas of aerospace and beyond, for example bioengineering flows, environmental flows etc.

Papadakis, George Prof.

Project no: GP04 *

Project title: Flow super-resolution from under-resolved data. (S) - 1 project available

Supervisor: Papadakis, George Prof.

Co-

supervisor(s):

Category: Computational;Theoretical

Software: OPEN FOAM

Confidential: No

The idea of the project is the super-resolution of a flow using under-resolved data. For example, experimental PIV data resolve the flow at limited resolution, can we couple the PIV data with the governing equations to resolve the flow at scales not seen by the PIV? The supervisor believes that the answer is YES! We will apply an algorithm proposed recently by the supervisor to real PIV data to extract features not seen in the data. There are numerous applications of the algorithm to denoise experiments and extract information that cannot be obtained from the equations or the experiments alone. It is only through the combination of these two methods that we can get very rich information about the flow.

Papadakis, George Prof.

Project no: GP05 *

Project title: Forecasting of 3D flows using sparse data. (S) - 1 project available

Supervisor: Papadakis, George Prof.

Co-

supervisor(s):

Category: Computational

Software: MATLAB and OPEN FOAM

Confidential: No

Forecasting the future evolution of a dynamical system from present data has received considerable attention in meteorology but not in fluid dynamics. The ability to estimate the future state of a flow from current data has many applications, examples include urban safety (prediction of toxic pollutant trajectory than can guide evacuation measures), prediction of extreme events (such as heat waves) ahead of time etc. The central aim of the project is the forecastic of turbulent flows using a methodolgy that was developed recently by the supervisor. The methodology will be employed to flow problems that exhibit strong rare events that in practise can cause severe damage. Can we predict the occurence of these events before they occur? What is the lead time? What is the probability to predict correctly the event at a given future time? These are crucial questions that will be considered during the project.

Papadakis, George Prof.

Project no: GP06 *

Project title: Turbulence modelling from DNS data (S) - 1 project available

Supervisor: Papadakis, George Prof.

Co-supervisor(s):

Category: Computational

Software: MATLAB

Confidential: No

The objective of this project is to extract a turbulence model from DNS data. In particular the project will focus on extracting an eddy viscosity model that accounts for non-local effects. Current approaches are very time consuming, and in the project a new approach will be trialled. The new idea will be applied to the flow between parallel flat plates (channel flow) where reference data exist. Extraction of a turbulence model from DNS data has many applications. The non-locality of the model also allows the extraction of new physical information, that can be exploited for the synthesis of new control and estimation strategies.

Peiro, Joaquim Prof.

Project no: JP01

Project title: Simulation of air flow and heat transfer in networks of tunnels - 2 projects available

Supervisor: Peiro, Joaquim Prof.

Co-

supervisor(s):

Category: Computational;Theoretical;Design

Software: Matlab, C++, Fortran or python for projects 1, 3, and 4; StarCCM+ license for projects 2 and 5.

Confidential: No

Passenger comfort and safety are the main concerns in the design of tunnels. A train travelling at high speed through a tunnel generates a pressure wave that might cause discomfort to the passengers. Similarly, the transit of trains in a network of tunnels generates heat that affects the temperature environment. These are a few of the projects proposed under this theme:

1. Aerodynamic interaction of a train entering a tunnel: A 1D code for the modelling of pressure waves in tunnels will be used to select the optimal tunnel section so that the increase in pressure does not go beyond the threshold of discomfort.
2. Analysis and design of a tunnel ventilation system in the event of a fire. Fires increase the resistance to air flow due temperature increases which also lead to significant buoyancy effects for large fires. The CFD code STARCCM+ will be used to model this "throttling" effect of the fire and flow stratification due to heat-induced buoyancy.
3. Long term evolution of the temperature in an underground network: To ensure that the fluctuations of temperature within the network are within acceptable limits for passenger comfort. The major challenge is the slow release of heat through the tunnel walls that often requires to model temperature fluctuations over several years.
4. Modeling of tunnel boom cause by micro-pressures waves generated by the entry in the tunnel of trains traveling at high speeds (above 300 km/h). The sound produced at the exit of the tunnel is a function of the pressure history. This will be calculated by the 1D code and the results processed using a suitable model of sound generation from pressure fluctuations.
5. CFD modelling of the pressure wave generated by the entry of a train in a tunnel using StarCCM+. For short tunnels three-dimensional effects are important and the one-dimensional approximation is no longer accurate. This project will compare CFD simulations with those of the 1D and experimental. These comparisons will be used to validate the CFD simulation and investigate the range of validity of the one-dimensional approach.

Peiro, Joaquim Prof.

Project no: JP02

Project title: Simulation of flow in physiological networks - 3 projects available

Supervisor: Peiro, Joaquim Prof.

Co-

supervisor(s):

Category: Computational;Numerical;Literature

Software: Matlab, Python, and C++ or Fortran compilers

Confidential: No

Arterial, venous, lymphatic and respiratory networks are made up of millions of vessels so 3-D flow simulations are just not feasible. Reduced 1-D models of such physiological systems based on the area-averaged governing equations provide a computationally affordable and reasonably accurate simulation capability to study the flow in networks of vessels. These are a few of the projects proposed under this theme:

1. A 1-D model of the venous system: To adapt an existing 1-D solver for the arterial system to the venous system. This will require investigating suitable constitutive laws to represent the elastic behaviour of veins and the effect of their valves and implement these by means of an appropriate pressure-area relation in the existing solver.
2. A 1-D model of the respiratory network: To adapt an existing 1-D solver for the arterial system to the respiratory system. This will require devising suitable constitutive pressure-area relations to represent the distensibility of the airways and implement these in the existing solver.
3. Modelling auto-regulation in the arterial system: To investigate the mechanisms of regulation of blood flow in the arterial system when it is subject to external actions such as, for instance, changes in temperature, pressure or posture. To devise suitable models and to implement them in an existing 1-D solver for the arterial system.
4. A 1-D model of the lymphatic system: To devise a set governing equations to model of the lymphatic system that connects tissues to the bloodstream. The model of the collecting lymphatic vessels will consist of lymphangions, deformable vessels which contract and propel lymph, enclosed between valves which promote unidirectional flow. To develop a numerical method of solution of the set of differential equations.

Peiro, Joaquim Prof.

Project no: *JP03*

Project title: Uncertainty quantification in simulation (S) - 2 projects available

Supervisor: Peiro, Joaquim Prof.

Co-

supervisor(s):

Category: Computational; Numerical

Software: Python, Matlab, or computer language of choice.

Confidential: No

There many sources of uncertainty in modelling any type of physical phenomena:

operational, geometric,

and model inadequacy. We will look at a number of methods to investigate the effect of

operational and geometrical uncertainty:

1. Markov Chain and Multi-level Monte Carlo;
2. Intrusive polynomial chaos;
3. Non-intrusive polynomial chaos;

4. Surrogate models.

These can be used in a to compute probabilistic expectations in a number of application areas such as aerofoil performance, financial modelling, vehicle trajectories, epidemic modelling, and predictive microbiology, to name but a few.

Peiro, Joaquim Prof.

Project no: *JP04 **

Project title: Aerodynamic modelling and simulation of flapping flight - 1 project available

Supervisor: Peiro, Joaquim Prof.

Co-supervisor(s):

Category: Computational

Software: STARCCM+ and/or computer language of choice: matlab, python, C++, etc.

Confidential: No

The outstanding performance of animal flapping flight at low Reynolds numbers has sparked an interest in the design, development and deployment of bio-inspired flapping-wing micro-air vehicles. However, the aerodynamic characteristics of flapping flight are significantly different from those of conventional-flight vehicles due to their unsteady nature and low Reynolds number. This project will investigate the aerodynamics of flapping flight, and analyse and implement computer models for the simulation of aerodynamic forces and, possibly flow control, with a range of fidelities: from reduced models to, more complex, CFD models. The ultimate aim is the rapid prediction of aerodynamic performance in preliminary design, control, and optimization of flapping-wing micro-air vehicles.

Pinho, Silvestre Prof.

Project no: *SP01*

Project title: Structural simulation of a hydrogen-powered aircraft dry-wing - 1 project available

Supervisor: Pinho, Silvestre Prof.

Co-supervisor(s): Wang, Yvonne

Category: Computational; Design; Analysis

Software: Abaqus

Confidential: No

Hydrogen is the most promising technology to allow aviation to meet NetZero by 2050. Airbus carried out an extensive programme (ZEROe) where they proposed three aircraft concepts that could meet NetZero requirement for different market segments. Two of these solutions include dry-wing solutions. In the UK, the Aerospace Technology Institute recently concluded another project (FlyZero) with similar aims where they proposed three new concepts, all of which including dry-wings. In dry-wing solutions, the fuel is not carried inside the wings, and these can therefore be thinner. A thinner, more highly loaded wing is not only harder to size for strength — it is also less forgiving of impact damage, and the bird-strike and damage-tolerance requirements set out in CS-25 are not relaxed simply because the fuel has changed.

In this project, you will start by reviewing the outcome of the FlyZero project and a previous FYP in which the CAD and FEM model of a dry wing for one of the FlyZero concepts was developed. You will then use that model to assess the damage tolerance of the design against the relevant CS-25 requirements: running quasi-static load cases up to ultimate failure, simulating bird-strike events using explicit dynamics, and evaluating the residual strength of the structure after damage. The objective is not simply to determine when the wing breaks, but to understand how and why, and what that implies for the design of the next generation of hydrogen-powered aircraft.

Pinho, Silvestre Prof.

Project no: SP02

Project title: Physics-informed AI and Bayesian optimisation - 1 project available

Supervisor: Pinho, Silvestre Prof.

Co-supervisor(s): Li, Runze

Category: Computational; Numerical

Software: Abaqus, TensorFlow

Confidential: No

Currently, material models for advanced composite materials used in aircraft structures are based on sophisticated analytical idealisations of the material at very small scales. This physical basis gives engineers some confidence that their structural designs will be sound. However, then manufactured at large scales, materials are often different to those manufactured at small scales. There is therefore a need for developing material models from information arising at larger scales.

Digital Image Correlation allows us to obtain the strain field for a large component being tested, including during failure, all while measuring the external work done by the applied forced.

In this project, you will design, train and test a physics-informed AI model to extract the constitutive response (including during damage growth) from the strain field and external work in one or more simple components. Rather than use experimental strain data (which

would be the ultimate aim but is more time consuming), you will create FE models of these simple components to generate the training strain data. You will then train your model and iterate the model architecture until your model is able to predict the constitutive response. Once your ML model is built and validated, you will use Bayesian Optimisation to find out the configuration of the FE model (shape, layup, features such as holes, etc) that maximise the predictive capability of the ML model.

The outcome of this project will be a proposed architecture for the ML model, as well as the definition of the corresponding experimental test specimens that should be manufactured and tested to apply your model in a real scenario.

Pinho, Silvestre Prof.

Project no: SP03

Project title: Bio-inspired structures manufactured by Automated Fibre Placement - 1 project available

Supervisor: Pinho, Silvestre Prof.

Co-supervisor(s): Kazemi, Erfan Dr

Category: Experimental

Software:

Confidential: No

We recently acquired an AFP machine suitable for manufacturing continuous fibre reinforced composites which provides us with unique manufacturing capabilities extremely relevant for industry.

AFP can be used to design and automatically manufacture complex CFRP parts that would otherwise be overly labour intensive or even impossible.

In this project, you will design, simulate, manufacture (using AFP) and test a bio-inspired structure, such as a skin-stiffener structure.

Pinho, Silvestre Prof.

Project no: SP04

Project title: Numerical simulation of impact on critical aircraft structures (S) - 3 projects available

Supervisor: Pinho, Silvestre Prof.

Co-supervisor(s): Del Rosso, Stefano Dr

Category: Numerical;Computational

Software: Abaqus

Confidential: No

The rapid proliferation of uncrewed aerial systems has created a new class of impact threat for which civil aviation has no established certification framework. Bird-strike requirements in CS-25 have been refined over decades and are grounded in a well-characterised impactor: a soft, roughly homogeneous body of known mass. A drone is a fundamentally different projectile (a mix of stiff plastic, metal motors, lithium-ion batteries and carbon-fibre frames) and both experimental and numerical studies, notably the FAA-sponsored ASSURE programme and subsequent EASA work, have shown that drone impacts can be significantly more damaging than bird strikes of equivalent mass. The problem is particularly acute for the next generation of composite-intensive airframes: thin-walled dry wings for hydrogen aircraft, composite hydrogen pressure vessels, and carbon/epoxy fan blades, where the response to localised, hard-body impact is governed by damage mechanisms that conventional failure criteria do not capture.

In this project, you will use advanced composite failure models to simulate the impact of projectiles and representative drone components on critical aircraft structures: e.g. a wing leading edge, an H2 tank wall, or an engine fan blade. You will build the explicit dynamics model from geometry to validated output, calibrate the material response against published experimental data, and use the simulations to investigate which impact scenarios cause critical damage and which do not. This project requires a keen interest in modelling and finite elements, and you can expect to finish it with a genuinely deep proficiency in Abaqus/Explicit and in the progressive damage modelling of composite structures under impact.

Pinho, Silvestre Prof.

Project no: SP05

Project title: Graph Neural Networks for strength prediction (S) - 1 project available

Supervisor: Pinho, Silvestre Prof.

Co-supervisor(s): Patrignani, Luca

Category: Computational;Numerical

Software: Abaqus, Tensorflow

Confidential: No

Predicting the strength of open-hole composite laminates remains a standard challenge in aerostructure design. Classical criteria (Whitney–Nuismer, finite fracture mechanics) capture first-order behaviour but do not fully account for the combined effect of stacking sequence, hole size, hole shape, and interactions between multiple holes. Progressive damage finite element models are accurate but too expensive to sweep across the thousands of cases a preliminary design demands, and conventional machine learning surrogates force the laminate and geometry into a flat feature vector that limits

generalisation. Graph Neural Networks (GNNs) are a more natural fit: plies and their interfaces form a graph, the finite element mesh is a graph, and holes embed as features on it.

In this project, you will build a progressive damage FE model in Abaqus for an open-hole CFRP specimen and generate a dataset of failure loads across stacking sequences, hole sizes, and hole shapes. You will then design, train and iterate a GNN architecture that encodes both the laminate and the specimen geometry, benchmarking predictions against published experimental data (Green, Wisnom and Hallett; Camanho and co-workers). Finally, you will use Bayesian Optimisation to identify the FE configurations that best probe the limits of your model and thus most deserve to be tested experimentally.

The outcome will be a proposed GNN architecture for open-hole strength prediction in CFRP laminates, together with a recommended set of physical test specimens for experimental validation.

Quino Quispe, Gustavo Dr

Project no: GQQ01 *

Project title: Testing 2.0 of high-performance materials - 1 project available

Supervisor: Quino Quispe, Gustavo Dr

Co-

supervisor(s):

Category: Analysis;Computational;Numerical

Software: Matlab, Abaqus, DIC software, Dragonfly

Confidential: No

Conventional mechanical testing methods require direct measurement of force and deformation to obtain the material's constitutive response (e.g. stress versus strain). While these methods are reliable, they can be costly and time-consuming. "Testing 2.0" represents a new approach to material characterisation, combining computational simulations and non-conventional data-rich tests to achieve a more complete understanding of complex behaviour and properties.

This project aims to utilise finite element modelling updating/virtual field method to characterise the mechanical response of high-performance materials. A case of study will be the property identification of high-performance polymers or composites based on full field surface or 3D volumetric measurements. The researcher will utilise both experimental full-field strain data and numerical modelling simulations to develop an optimization framework for characterising constitutive material parameters. The project will involve coding, analysis of experimental data, Digital Image Correlation, and FE software such as Abaqus.

Quino Quispe, Gustavo Dr

Project no: GQQ02 *

Project title: Degradation/Durability of advanced materials - 1 project available

Supervisor: Quino Quispe, Gustavo Dr

Co-

supervisor(s):

Category: Experimental;Experimental: Structures

Software:

Confidential: No

Composite materials are extensively used in various industries due to their numerous advantages, including lightweight, high strength, and corrosion resistance. However, in-service exposure to environmental factors such as heat and humidity can adversely affect the mechanical performance of these materials over time, posing significant challenges in designing safe and durable structures.

This project seeks to explore the impact of humidity exposure on the mechanical properties of advanced materials. The experimental approach will involve subjecting the materials to humidity pre-conditioning followed by mechanical testing. The student will utilise specialized testing rigs and employ advanced techniques such as microscopy, image analysis, and digital image correlation to analyse the results.

Quino Quispe, Gustavo Dr

Project no: GQQ03 *

Project title: Impact behaviour of polymers and polymer based composites - 3 projects available

Supervisor: Quino Quispe, Gustavo Dr

Co-

supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

Real engineering structures and components may be subjected to dynamic or impact loads, such as hail or bird strikes. In addition, polymers and polymer-based composites are known to be strain rate dependent, meaning their mechanical behaviour can change according to the rate of deformation. In this project, the researcher(s) will investigate how the mechanical behaviour of high-performance materials changes when loads are applied up to one million times faster than in standard quasi-static experiments.

This project will involve the design and execution of mechanical impact experiments.

Experiments will be conducted using specialised rigs and advanced techniques, including split Hopkinson bar, microscopy, high-speed imaging, and digital image correlation. The

findings from this project will contribute to the development of strain rate dependent models for polymers and polymer-based composites.

Quino Quispe, Gustavo Dr

Project no: **GQQ04 ***

Project title: Micromechanical modelling of advanced composites
- 2 projects available

Supervisor: Quino Quispe, Gustavo Dr

Co-

supervisor(s):

Category: Computational;Analysis;Numerical

Software: Python
Matlab
Abaqus

Confidential: No

The development of new materials, or the improvement of existing ones, often requires extensive manufacturing and testing iterations, which can be both costly and time-consuming. Micromechanical modelling stands as a powerful tool for this, enabling the rapid exploration of how microstructural features, such as constituents, defects, and microarchitecture, affect the expected macroscopic mechanical response of heterogeneous materials.

This project will focus on the creation of micromechanical models to investigate the behaviour of composite materials. Implicit or explicit finite element simulations will be performed on unit cells, utilising a semi-automated or fully automated approach. The outcomes of this work will contribute to the development of strain rate-dependent models for polymers and polymer-based composites, thereby supporting and accelerating material development.

Experience with finite element analysis (FEA) and programming would be advantageous for this project.

Ribera Vicent, Maria Dr

Project no: **MRV01**

Project title: Helicopter rotor wake/fuselage aerodynamic interaction
- 1 project available

Supervisor: Ribera Vicent, Maria Dr

Co-

supervisor(s):

Category: Computational;Analysis

Software: FORTRAN, MATLAB

Confidential: No

A free-vortex rotor wake model can improve the aerodynamic modelling of a helicopter and its predictions of the dynamic response using a comprehensive flight dynamics model. The proposed project will follow up on two previous projects studying the aerodynamic model of the UH-60 helicopter using a panel method and modelling the rotor wake with free vortex wake, modified to account for the presence of the fuselage. The combined models will be used to study and validate the performance of the helicopter in low to mid speed flight.

Ribera Vicent, Maria Dr

Project no: *MRV02*

Project title: Modelling of a rotor wake distortion due to the presence of the fuselage - 1 project available

Supervisor: Ribera Vicent, Maria Dr

Co-

supervisor(s):

Category: Computational;Analysis

Software: FORTRAN, MATLAB

Confidential: No

The aerodynamic environment of a helicopter is very complex, with the main rotor wake interacting with the fuselage, empennage and tail rotor wake. A comprehensive helicopter flight dynamics model is available which has been validated against the UH-60 flight test program. This code includes a time-marching rotor vortex wake model which is capable of capturing the wake dynamics during manoeuvres and allows the wake vortices to distort under their mutual influence. The free wake model, however, does not currently distort due to the presence of the fuselage. The proposed project will investigate approaches to modelling the distortion of the wake filaments and implement the most suitable one into the free wake code. The model will be compared to the current approach (without distortion) and to flight test data for the UH-60 Blackhawk helicopter.

Ribera Vicent, Maria Dr

Project no: *MRV03*

Project title: Helicopter tail rotor modelling effect on the yaw performance of a UH-60 helicopter - 1 project available

Supervisor: Ribera Vicent, Maria Dr

Co-

supervisor(s):

Category: Computational

Software: FORTRAN, MATLAB

Confidential: No

The aerodynamic environment of a helicopter is very complex, with the main rotor wake interacting with the fuselage, empennage and tail rotor wake. A comprehensive helicopter flight dynamics model is available which has been validated against the UH-60 flight test program. While the flight dynamics model correlates well with the flight test data in general, the tail rotor control is poorly predicted. This project aims to explore the current simplified tail rotor aerodynamic model and propose improvements to augment the yaw performance of the helicopter simulation model.

Ribera Vicent, Maria Dr

Project no: *MRV04*

Project title: Exploration and augmentation of the downwash modelling on a helicopter flight dynamics simulation model - 1 project available

Supervisor: Ribera Vicent, Maria Dr

Co-

supervisor(s):

Category: Computational

Software: FORTRAN, MATLAB

Confidential: No

The aerodynamic environment of a helicopter is very complex, with the main rotor wake interacting with the fuselage, empennage and tail rotor. A comprehensive helicopter flight dynamics model is available which has been validated against the UH-60 flight test program. While the flight dynamics model correlates well with the flight test data in general, the pitch attitude and longitudinal control predictions in the low to mid speed range, where the main rotor wake interacts with the fuselage and empennage, could be improved. This project will explore the current simplified implementation of the downwash effects and explore, using any tools necessary, ways to validate and improve the existing downwash model.

Ribera Vicent, Maria Dr

Project no: *MRV05*
Project title: Multifidelity helicopter blade design using NeuralFoil - 1 project available

Supervisor: Ribera Vicent, Maria Dr

Co-supervisor(s):

Category: Computational;Design

Software: FORTRAN, MATLAB, NEURALFOIL

Confidential: No

A comprehensive helicopter flight dynamics code, capable of modelling both steady-state and transient manoeuvres for a range of flight conditions is available as a baseline for this project. The code includes data for the UH-60 helicopter, although much of the data corresponds to the baseline models and there is room to improve the helicopter performance by optimising the blades.

The proposed project will perform a parametric design optimisation of the UH60 Blackhawk helicopter blades. It is proposed to use NeuralFoil to explore different aerofoil configurations.

Rigas, Georgios Dr

Project no: *GR01*

Project title: Wind Tunnel Exploration of Flapping and Rotating Wing Aerodynamics - 1 project available

Supervisor: Rigas, Georgios Dr

Co-supervisor(s):

Category: Experimental;Analysis

Software: Labview, RobotStudio, Matlab

Confidential: No

Unlike fixed wings, flapping wings generate lift through a suite of unsteady aerodynamic mechanisms—such as wake capture, delayed stall, and rotational lift—which play a vital role in the flight of birds and insects. These effects, although complex, are key to achieving the agile, efficient flight seen in nature.

This experimental project will investigate how these unsteady mechanisms differ between fixed and flapping wing configurations, using a state-of-the-art wind tunnel setup. At its core is a newly acquired ABB IRB-140 six-axis robot arm, which will control the wing's position and motion in 3D, enabling the execution of complex, bird-inspired trajectories such as the figure-eight path observed in hovering flight.

Key aspects of the project include:

- Conducting wind tunnel tests to explore the impact of parameters like aspect ratio, Reynolds number, and flapping frequency
- Measuring and analyzing lift, drag, and flow instabilities linked to leading-edge vortices and vortex shedding
- Programming and operating the industrial robot using RobotStudio
- Rapid prototyping of flapping/rotating wing designs

This is an exciting opportunity to combine hands-on experimentation, robotics, and fluid mechanics to better understand one of nature's most effective forms of locomotion. Students with interests in aerodynamics, experimental techniques, or bio-inspired engineering are encouraged to apply. Some experience in any of the following areas—robotics, wind tunnel testing, or unsteady aerodynamics—will be advantageous.

Rigas, Georgios Dr

Project no: **GR02 ***

Project title: Drag and Flow Instabilities in the Wake of Realistic Car Geometries - 1 project available

Supervisor: Rigas, Georgios Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis

Software: FreeFem++, StarCCM, FLuent or other CFD software

Confidential: No

The unsteady wake behind bluff bodies—such as cars and trucks—is a major contributor to aerodynamic drag, structural fatigue, and noise. These effects arise from hydrodynamic instabilities like vortex shedding, which are often challenging to predict and control in realistic, three-dimensional flows.

In this project, you will explore the fundamental mechanisms of these instabilities using advanced computational tools applied to geometries of direct industrial relevance and flow control techniques to improve performance. Building on prior work in simplified settings, we will now investigate the wake dynamics of the Ahmed body—a canonical car model widely used in the automotive industry to study aerodynamics or the Generic Truck Model.

The focus will be on identifying and analyzing three-dimensional bifurcations and global instabilities in the wake, providing insight into how complex flow features contribute to drag and flow unsteadiness. Another opportunity will be to optimise the geometry of additive aerodynamic devices like boat-tails/rear-flaps.

You will gain experience in:

- High-fidelity numerical simulations
- Global stability analysis and bifurcation theory
- Applied aerodynamics with industrially relevant geometries and flow control.

This project is ideal for students with a strong interest in fluid dynamics, applied mathematics, or CFD, and is especially relevant to those considering careers in the automotive or aerospace industries. Prior exposure to numerical methods or fluid simulations will be helpful but not strictly required.



Rigas, Georgios Dr

Project no: **GR03 ***

Project title: Reinforcement Learning and Behavioural Flow Control for Road-Vehicle Aerodynamics - 1 project available

Supervisor: Rigas, Georgios Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis

Software: Tensorflow/Pytorch, Python or equivalent software for ML/AI

Confidential: No

Many high-performance street and F1 cars have recently added moving surfaces (moving spoilers and wings) for adjusting the downforce at various speed regimes. The position of the surface is determined for static conditions based on a look-up table. In this study, we will exploit modern data driven control techniques (reinforcement learning & behavioural control) for discovering optimal control laws and dynamically adapting the position of the control surfaces during attitude changes in the car (yaw, speed). The demonstration will be performed using simplified 2D Direct Numerical Simulations, and will guide the current wind tunnel experiments.

RL: <https://arxiv.org/abs/1808.07664>

Behavioral control: <https://imarkovs.github.io/tutorial.pdf>

Experiment: <https://doi.org/10.1017/jfm.2016.495>

This project is ideal for students interested in AI for engineering, flow control, and automotive aerodynamics. Prior experience in machine learning, CFD, or control systems will be an asset.

Rigas, Georgios Dr

Project no: **GR04 ***

Project title: Experimental Reinforcement Learning for Road-Vehicle Flow Control - 1 project available

Supervisor: Rigas, Georgios Dr

Co-

supervisor(s):

Category: Experimental;Computational

Software:

Confidential: No

Active aerodynamic surfaces—like movable spoilers and wings—are increasingly used in high-performance and Formula 1 vehicles to optimize downforce across different speed regimes. Traditionally, these surfaces are controlled using precomputed look-up tables, which do not account for real-time changes in vehicle attitude such as yaw and speed. This project breaks new ground by experimentally applying reinforcement learning (RL) to aerodynamic flow control. For the first time, RL algorithms will be trained in the loop with wind tunnel experiments to dynamically adjust control surfaces based on live flow conditions. The aim is to discover and implement control strategies that adapt in real time, improving aerodynamic efficiency far beyond what static logic allows.

Project highlights:

- Implementation of modern RL techniques (ref below)
- Exploration of behavioural control concepts (tutorial below)
- Real-time control of active surfaces in wind tunnel tests (related experiment

below)

- Hands-on experience with flow measurement and experimental control systems

This is a cutting-edge project at the intersection of AI, fluid mechanics, and experimental aerodynamics. It is ideal for students eager to explore reinforcement learning in a physical, high-impact setting. Prior experience with RL, control systems, or experimental setups is beneficial.

RL: <https://arxiv.org/abs/1808.07664>

Behavioral control: <https://imarkovs.github.io/tutorial.pdf>

Experiment: <https://doi.org/10.1017/jfm.2016.495>

Rigas, Georgios Dr

Project no: **GR05 ***

Project title: Wake Steering in Wind Farms: Improving Aerodynamic Efficiency Using Experimental Wind-Tunnel Data - 1 project available

Supervisor: Rigas, Georgios Dr

Co-supervisor(s): Magri, Luca Prof, Laizet Sylvain Prof.
Category: Computational;Theoretical
Software:
Confidential: No

Wind farms often operate below their optimal aerodynamic efficiency because wind turbines interact through their wakes. When an upstream turbine extracts energy from the wind, it generates a lower-momentum wake that can reduce the power output and increase the unsteady loading of downstream turbines. A promising strategy to mitigate these losses is wake steering, where upstream turbines are intentionally yawed relative to the incoming wind direction so that their wakes are deflected away from downstream turbines. Although this may slightly reduce the power of the yawed turbine, it can increase the total power output of the wind farm.

This project will investigate wake steering using experimental wind-tunnel data from scaled wind-farm configurations. The student will analyse measurements of turbine power, flow fields, and wake behaviour under different yaw angles and turbine arrangements. The aim is to understand how wake deflection affects downstream turbine performance and to identify operating conditions that improve overall farm efficiency.

The project will combine elements of fluid mechanics, data analysis, and aerodynamic optimisation. Depending on progress, the student may compare experimental trends with simplified wake models, quantify uncertainty in the measurements, and explore data-driven approaches for predicting optimal yaw settings.

The main objectives of the project are to:

1. Analyse experimental wind-tunnel data for wind-turbine wake interactions.
2. Quantify the effect of yaw misalignment on wake deflection, wake recovery, and downstream turbine performance.
3. Evaluate the trade-off between power loss at the yawed upstream turbine and power gain at downstream turbines.
4. Identify yaw-control strategies that maximise total wind-farm power output.
5. Compare experimental results with simple analytical or engineering wake-steering models, where appropriate.

Robinson, Paul Prof.

Project no: *PR01*
Project title: FE investigation of thermoplastic interleaved composites short strip composites - 1 project available

Supervisor: Robinson, Paul Prof.
Co-supervisor(s):
Category: Computational

Software: ABAQUS, MATLAB

Confidential: No

A considerable amount of waste carbon-fibre epoxy prepreg (carbon fibres in a partially cured resin) is generated in the UK. Currently waste treatment of these materials consists of landfilling, incineration or pyrolysis to recover the fibres. This project will investigate a new proposal to convert prepreg waste into mouldable fully cured panels and mouldable mats made from thermoplastic-coated short prepreg strips. The thermoplastic coating of the strips enables the moulding capability in the cured composite and also provides the potential for closed loop recycling. This project will investigate FE simulation to guide optimisation of the new material and the associated manufacturing process.

Robinson, Paul Prof.

Project no: *PR02 **

Project title: Investigation of an interleaved strategy for easy repair of impact damage in composites - 1 project available

Supervisor: Robinson, Paul Prof.

Co-supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

Impact damage in a laminated composite often consists of interlaminar fracture between the laminae and repair of this damage usually involves removal of the damaged material by machining and adhesive bonding of a repair patch. Recent work at Imperial has explored an easy repair concept in which a thermoplastic interleaf is added to the composite to provide a repairable 'weak link'. The properties of the interleaf are chosen such that on impact the damage occurs preferentially within the interleaf. The composite can then be heated and the damage repaired. This project will perform further experimental investigations to help develop and assess this easy repair concept.

Robinson, Paul Prof.

Project no: *PR03*

Project title: Finite element investigation of strategies for improving the resistance of laminated composites to delamination - 1 project available

Supervisor: Robinson, Paul Prof.

Co-

supervisor(s):

Category: Computational

Software: Abaqus

Confidential: No

Laminated polymer matrix composites are susceptible to delamination (i.e. crack growth between the laminae). In this project strategies to improve the resistance to delamination will be investigated using finite element modelling. For example, some composites possess a high resistance to delamination because they develop a considerable amount of fibres which bridge between the delaminated surfaces in the wake of the crack tip. FE modeling will be used to investigate methods for modifying the composite to ensure that extensive fibre bridging will occur in the event of a delamination. The student must be familiar with Abaqus.

Robinson, Paul Prof.

Project no: *PR04*

Project title: Experimental investigation of strategies for improving the resistance of laminated composites to delamination - 1 project available

Supervisor: Robinson, Paul Prof.

Co-

supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

Laminated polymer matrix composites are susceptible to delamination (i.e. crack growth between the laminae). In this project strategies to improve the resistance to delamination will be investigated. For example, some composites possess a high resistance to delamination because they develop a considerable amount of fibres which bridge between the delaminated surfaces in the wake of the crack tip. This experimental study will investigate methods for modifying the composite to ensure that extensive fibre bridging will occur in the event of a delamination.

Robinson, Paul Prof.

Project no: *PR05*

Project title: Experimental investigation of the compression performance of unidirectional composites - 1 project available

Supervisor: Robinson, Paul Prof.

Co-

supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

The compression strength of unidirectional composites is low compared to the tensile strength. NextCOMP is a current research project being undertaken jointly by Imperial and Bristol University and is exploring strategies to improve the compression properties of high performance composites. In this project, composite rod specimens will be manufactured and tested to evaluate the effectiveness of an eccentric strut buckling test for measurement of compression strength.

Robinson, Paul Prof.

Project no: *PR06*

Project title: Experimental investigation of the effect of lamina prestress on the compression performance of composite laminates - 1 project available

Supervisor: Robinson, Paul Prof.

Co-

supervisor(s):

Category: Experimental

Software:

Confidential: No

Fibre reinforced composite laminates commonly fail by kink band formation when subjected to in-plane compression. This project will use thermoplastic interleaved composite laminates in which the prestress can be changed after curing. Unidirectional and multidirectional laminates will be manufactured and tested to investigate how changes in the lamina prestress levels influence the compression behaviour.

Robinson, Paul Prof.

Project no: *PR07*

Project title: Experimental investigation of thermoplastic interleaved composites short strip composites - 1 project available

Supervisor: Robinson, Paul Prof.

Co-

supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

A considerable amount of waste carbon-fibre epoxy prepreg (carbon fibres in a partially cured resin) is generated in the UK. Currently waste treatment of these materials consists of landfilling, incineration or pyrolysis to recover the fibres. This project will investigate a new proposal to convert prepreg waste into mouldable fully cured panels and mouldable mats made from thermoplastic-coated short prepreg strips. The thermoplastic coating of the strips enables the moulding capability in the cured composite and also provides the potential for closed loop recycling.

This project will investigate processes for production of the interleaved strip-based mouldable cured panel and mouldable bonded mat ensuring the required strip distribution and orientation, and will evaluate the mechanical performance and moulding capability of the new composite material.

Robinson, Paul Prof.

Project no: *PR08*

Project title: Experimental investigation of interlaminar creep fracture growth in composites - 1 project available

Supervisor: Robinson, Paul Prof.

Co-

supervisor(s):

Category: Experimental: Structures

Software:

Confidential: No

Laminated polymer matrix composites are susceptible to interlaminar fracture (i.e. crack growth between the laminae). The critical energy release rate, G_c , of a laminated composite is a measure of its resistance to interlaminar fracture and is measured under quasi static load conditions. However, interlaminar crack growth can occur at an applied energy release rate lower than G_c ; the threshold energy release rate for interlaminar crack growth in fatigue is substantially less than the G_c . This project will investigate whether creep growth of interlaminar cracks will occur at an applied energy release rate less than G_c and

will measure the associated crack growth rates. Techniques to perform the creep tests will be devised to perform the creep fracture tests but will include the use of an existing constant energy release rig fixture.

Santer, Matthew Prof.

Project no: *MSA01 **

Project title: Multiphysics Metamaterials for Space (S) - 2 projects available

Supervisor: Santer, Matthew Prof.

Co-supervisor(s):

Category: Computational

Software: Python including Firedrake

Confidential: No

Metamaterials are materials that are designed and optimized with features over multiple length scales. This can access novel and unexpected properties that would not otherwise exist in nature. In addition, different physics can be coupled to achieve highly multifunctional systems. For example, coupling electro-magnetic and structural properties can enable novel antenna architectures; coupling optical and structural properties can enable cloaking and low visibility; coupling thermal and structural properties can enable integrated thermal protection and management systems; and so on.

Space is an application environment where the properties enabled by metamaterials are particularly attractive. The specific objectives (including the application and the physics to be coupled) will be defined at the beginning of the project based on the intersection of the research state of the art and the interests of the student.

At the core of the work will be the use of the Firedrake python-based PDE solver which enables the coupled constitutive equations to be straightforwardly defined. This builds on several research activities already carried out within the Department. A strong proficiency in python is a prerequisite.

Santer, Matthew Prof.

Project no: *MSA02 **

Project title: Deployable Structures and In-space Assembly and Manufacture (S) - 2 projects available

Supervisor: Santer, Matthew Prof.

Co-supervisor(s):

Category: Experimental;Computational;Design

Software: Depending on the project could include: python, matlab, abaqus

Confidential: No

A perennial challenge in the design of large space systems is that the volume required for operation in space can substantially exceed the available volume within the launch vehicle. This has been partially addressed by the development of deployable structures based on mechanisms and other less conventional approaches, although there are still significant challenges remaining to be overcome, particularly with respect to precision, robustness and efficiency. However, recently a new class of space mega structure has been proposed which will enable observatories, power stations, data centres, habitation modules and more. This will require the assembly in space, potentially based on deployable units, but may also potentially be manufactured and assembled in orbit. How to achieve this is not a solved problem.

Specific objectives for this project will be defined at the beginning of the project based on the intersection of the research state of the art and the interests of the student. It is expected that the project will involve the design, analysis and potentially bench-top experimental validation of deployable and/or in-space assembled structural concepts.

The requirements of the project will depend on the agreed objectives but is likely to include the use of finite element analysis in addition to coding with python or matlab. Taking the Spacecraft Structures module either before or during the project is a requirement.

Santer, Matthew Prof.

Project no: *MSA03 **

Project title: Aero-thermo-structural Optimization of Hypersonic Vehicles (S)
- 2 projects available

Supervisor: Santer, Matthew Prof.

Co-supervisor(s):

Category: Computational

Software: Python; Firedrake

Confidential: Yes

The design of hypersonic vehicles is immensely challenging due to the need to manage simultaneously extremely high heat loads, thermal stresses, and flight loads. Solving this problem is essential for both space — atmospheric re-entry — and defence applications. Over the last decade in the Department of Aeronautics we have developed and validated a python-based engineering-level simulation capability which couples modified Newtonian

aerodynamic solver with a nonlinear thermo-structural finite element solver and is at the forefront of European capability. In addition, the solver has been coupled with an interior point optimizer to enable design optimization.

This project will contribute to the continued development of the code and augment its capabilities. The specific objectives — and the application domain — will be defined based on the current requirements and the student's interests but are likely to focus on the thermo-structural aspects of design and optimization.

The project will require a strong proficiency with python and numerical methods.

Seifikar, Masoud Dr

Project no: *MSE01 **

Project title: Investigation of Piezoelectric Materials for Health Monitoring, Sensors, and Energy Harvesting in Aerospace Applications - 3 projects available

Supervisor: Seifikar, Masoud Dr

Co-

supervisor(s):

Category: Computational;Theoretical

Software: ABAQUS, MATLAB

Confidential: No

Piezoelectric materials generate an electrical charge when subjected to mechanical stress, making them useful in various applications, including in the field of aeronautics. In recent years, piezoelectric materials have played a vital role in the aerospace industry, particularly in health monitoring, sensors, and energy harvesting. The objective of this project is to investigate the potential use of piezoelectric materials in these applications and explore effective methodologies for simulating and analysing their electrical and structural properties.

The project will employ the finite element method to make precise numerical models of piezoelectric materials when dielectric, piezoelectric, and mechanical properties are known. We will investigate the static and dynamic responses of piezoelectric material using commercially available finite element software ABAQUS. The project will also design and test various types of sensors, specifically tailored to meet the requirements of the aerospace industry. We will evaluate the performance of these sensors and optimize their design for maximum efficiency. Furthermore, we will explore energy harvesting techniques and investigate how piezoelectric materials can be used to harvest energy from ambient sources. We will compare and evaluate various harvesting techniques to determine the most efficient and cost-effective methods.

Seifikar, Masoud Dr

Project no: *MSE02 **

Project title: Developing dilute nitride multi-junction solar cells for space applications · projects available

Supervisor: Seifikar, Masoud Dr

Co-

supervisor(s):

Category: Computational;Theoretical

Software: MATLAB/ Python

Confidential: No

Achieving a clean energy society through renewable sources hinges on the advancement of high-performance solar cells. Space applications, with their demanding power needs and extended missions, require solar technology of the highest efficiency. Multi-junction solar cells have emerged as superior to traditional silicon-based cells, and the integration of dilute nitride materials holds the promise of further enhancing their performance in the challenging space environment.

To improve the efficiency of multi-junction solar cells beyond existing triple-junction architectures, a sub-cell with a bandgap of around 1 eV is essential. The development of dilute nitride GaInNAs material system offers a compelling solution. This material system, initially proposed for 1 eV lasers and subsequently for 1 eV solar cells, shows promise when lattice-matched to GaAs. Current triple-junction solar cells based on GaInP/GaInAs/Ge have achieved efficiencies up to 41.6%. By incorporating a middle junction with a band gap around 1 eV, utilizing dilute nitride materials such as GaInNAs(Sb), even higher efficiencies are predicted while maintaining lattice-match compatibility with GaAs or Ge. Additionally, replacing Ge with a 0.9 eV GaInNAs(Sb) subcell in a GaInP/GaAs/GaInNAs configuration is forecasted to achieve efficiencies up to 44.5%.

This project aims to theoretically investigate the photovoltaic (PV) performance of GaInNAs and GaInNAsSb diodes, previously developed for photodetectors, to assess their viability for integration into multi-junction solar cells for space applications. This project is anticipated to provide insights into the potential of dilute nitride materials, particularly GaInNAs and GaInNAsSb, for improving the efficiency and performance of multi-junction solar cells in space applications. The theoretical investigation will lay the foundation for future experimental studies and the development of high-efficiency space solar technology.

Seifikar, Masoud Dr

Project no: *MSE03 **

Project title: Critical Assessment of Piezoelectric Wind Energy Harvesting (S) - 2 projects available

Supervisor: Seifikar, Masoud Dr

Co-

supervisor(s):

Category: Theoretical;Computational;Analysis;Experimental;Literature;Experimental: Structures

Software: 3DExperience, Abaqus, StarCCM+, MATLAB

Confidential: No

This project aims to critically assess the potential and limitations of piezoelectric wind energy harvesting (PWEH) systems, particularly for low-power aerospace applications such as structural health monitoring or autonomous sensor nodes. Piezoelectric energy harvesting relies on converting mechanical vibration—typically caused by wind-induced mechanisms such as flutter, vortex-induced vibration (VIV), or galloping—into electrical energy through materials like PVDF or PZT.

The first part of the project will involve a comprehensive literature review of recent developments and benchmark studies, including both modelling and experimental data. The student will evaluate various PWEH configurations, comparing their performance, operational bandwidth, and scalability. This will lead to the identification of key limitations in material selection, mechanical robustness, energy output, and interface electronics.

In the second phase, the student will develop analytical or finite element models of a selected PWEH configuration, using simplified representations of flutter or VIV systems. These models will help explore parameter sensitivities—such as geometry, wind speed, and piezo material properties—and be compared against experimental results.

A possible practical component will complement the computational work. A simple bench-scale prototype will be constructed using a piezoelectric beam and a bluff body (e.g., flag-style or cantilever setup). Output voltage and power will be measured under varying wind conditions using a controlled fan or small wind tunnel. Basic circuit rectification and load testing will be performed to assess real-world efficiency.

The project will conclude with a critical analysis of the results and a set of design recommendations for future scalable PWEH systems suitable for aerospace applications.

Shamsuddin, Siti Ros Dr

Project no: *SRS01 **

Project title: Damping properties of thermoplastic film interleaved thermosetting composite - 1 project available

Supervisor: Shamsuddin, Siti Ros Dr
Co-supervisor(s):
Category: Experimental;Experimental: Manufacturing intensive;Theoretical;Analysis

Software:

Confidential: No

Advanced fibre reinforced polymer composite materials are often used in load-bearing and structural applications due to their high strength, high modulus and low density. Interleaved composites have been studied extensively as composite toughening or composites with controllable stiffness. Some thermoplastic materials are also compatible with epoxies (such as PSU, PES, PS, PA, etc.). In this study, the damping properties of thermoplastic interleaved CFRP with various interleaved architectures (0deg, 0/90 deg and 90 deg) will be studied and compared with conventional CFRP through DMA analysis. The experimental results will also be compared to existing theoretical models.

Shamsuddin, Siti Ros Dr

Project no: *SRS02 **

Project title: Predicting mechanical properties of interleaved CFRP composites using Machine Learning - 2 projects available

Supervisor: Shamsuddin, Siti Ros Dr

Co-

supervisor(s):

Category: Computational;Theoretical;Design;Numerical;Analysis;Literature

Software: Deep understanding of MATLAB/Python and ABAQUS is required

Confidential: No

Carbon fibre reinforced polymer composites (CFRP) are becoming increasingly relevant in lightweight structures due to its high specific strength and stiffness properties. However, the nature of composite materials having many constituents can lead to complex behaviour under load, which makes prediction of its mechanical properties very challenging. The brittleness of traditional composite materials can also lead to catastrophic failure which is undesirable. To increase toughness of CFRP, interleaving techniques can be employed but choosing the appropriate interleaving material is also crucial to guarantee perfect adhesion between all the constituents. With recent development in artificial intelligent tools such as Linear Regression, Artificial Neural Network, the prediction of mechanical responses of composites can be achieved. This project will compare previous experimental results and the predicted mechanical properties using machine learning tools to study the adaptability of these tools in predicting the mechanical properties of interleaved CFRP. The integration of machine learning approaches in composite design can reduce experimental costs while enhancing material optimisation.

Shamsuddin, Siti Ros Dr

Project no: **SRS03 ***

Project title: Training deep learning for adhesion properties with single fibre fragmentati test on carbon fibre reinforced thermoplastic composites - 1 project available

Supervisor: Shamsuddin, Siti Ros Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical;Literature

Software: Deep understanding of MATLAB/Python is required

Confidential: No

The fibre-matrix adhesion is critical in any fibre reinforced thermoplastic composite materials for optimising the mechanical performance of the composite. This is because stress transfer between fibres happens through the interface and subsequently the matrix. The single fibre fragmentation test is a widely used experimental method to determine the interfacial shear strength (IFSS), which is translated to the adhesion property at the fibre-matrix interface. This study will utilise the use of artificial neural network (ANN) method to predict the IFSS between various carbon fibres and matrices. The data set will be extracted from published data and used in the prediction.

Shamsuddin, Siti Ros Dr

Project no: **SRS04 ***

Project title: Machine learning methods on predicting fracture toughness of interleaved composites - 2 projects available

Supervisor: Shamsuddin, Siti Ros Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis;Numerical;Literature

Software: Deep understanding of MATLAB/Python is required

Confidential: No

Thermoplastic interleaves are widely used to improve the delamination resistance of fibre-reinforced composites, but the resulting fracture behaviour depends on a complex interaction between material properties, laminate configuration and processing conditions. Conventional experimental characterisation is time-consuming and costly, making data-

driven prediction an attractive alternative. This project investigates the use of machine learning to predict the interlaminar fracture toughness of composite materials containing thermoplastic interleaving. The study will develop machine learning models using input features that may include interleaf type, thickness, fibre volume fraction, laminate lay-up, matrix properties and processing parameters. Experimental data collected from literature and previous laboratory testing will be used to train and validate regression models such as random forests, support vector regression, or neural networks. Model performance will be evaluated using standard statistical metrics and compared with simpler empirical or analytical approaches.

Shamsuddin, Siti Ros Dr

Project no: **SRS05 ***

Project title: Interlaminar fracture toughness of interleaved composites: FEA vs experimental data analysis - 2 projects available

Supervisor: Shamsuddin, Siti Ros Dr

Co-supervisor(s):

Category: Computational;Theoretical;Design;Analysis;Numerical;Literature

Software: Deep understanding of MATLAB/Python and ABAQUS is required

Confidential: No

CFRP laminates are prone to delamination due to its anisotropic and heterogenous nature thus limiting their use in certain structural applications. To overcome this as well as the brittle nature of thermoset matrices, a new way of manufacturing interleaved composites is introduced. Interleaves refers to adding other materials between the prepreg layers to control delamination and enhance interfacial properties. One of the easiest interleaving methods is to use thermoplastic films. This study will focus on FEA analysis of perforated and unperforated PES film interleaved CFRP for interlaminar fracture toughness. An FEA analysis will be conducted, and the results will be compared and analysed against experimental values.

Sharif Khodaei, Zahra Prof.

Project no: **ZSK01**

Project title: Fault/anomaly detection (S) - 1 project available

Supervisor: Sharif Khodaei, Zahra Prof.

Co-supervisor(s):

Category: Experimental: Ready-made experiment;Computational;Analysis

Software: Matlab, Python, Possibly Abaqus

Confidential: No

Structural Health monitoring is based on diagnosing the health state of a structure based on the recorded response from permanently installed sensors. However, the reliability of the diagnosis is directly linked to the health of the sensors and the quality of the recorded data. This project focuses on developing methods for real-time sensor fault detection and data credibility assessment in structural monitoring systems. By analyzing signals from sensors (e.g. acceleration, strain, or guided waves) , the project aims to distinguish between structural anomalies and sensor malfunctions. Advanced statistical analysis, redundancy checks, and machine learning algorithms can be applied to evaluate data integrity, isolate faulty sensors, and ensure the reliability of measurements. The outcome will enhance the robustness of structural health monitoring (SHM) frameworks, ensuring accurate decision-making and reducing maintenance risks. A library of experimental data are already available which can be used for verification and validation of the algorithms. There could be a possibility of small scale tests carried out in the lab to simulate a faulty sensor readings. In the absence of lab access, numerical simulation with Abaqus can be carried out to support the developments.

Sharif Khodaei, Zahra Prof.

Project no: ZSK02 *

Project title: Digital Twin Updating via Modal Data Assimilation for Structural Integrity Assessment (S) - 2 projects available

Supervisor: Sharif Khodaei, Zahra Prof.

Co-supervisor(s):

Category: Experimental: Structures;Computational;Analysis;Numerical

Software: Abaqus, Matlab or Python

Confidential: No

This project aims to develop and demonstrate a physics-based digital twin (DT) framework for structural integrity assessment through modal data assimilation, with a particular focus on aeronautical structures. The core objective is to establish a robust methodology for reconciling discrepancies between finite element (FE) model predictions and measured vibration characteristics, thereby enabling reliable detection and quantification of structural degradation. The motivation stems from a key limitation in current SHM practice: modal features such as natural frequencies and mode shapes are widely used due to their global sensitivity and ease of measurement, yet their effectiveness for damage identification is often constrained by model uncertainty, environmental variability, and limited sensor

density. Addressing these challenges is central to advancing certifiable digital twin concepts for aircraft structures.

Technically, the project will involve constructing a high-fidelity FE model of a representative structure (e.g. plate, beam, or simplified wing section) and generating baseline modal properties under both pristine and damaged conditions. Synthetic “measured” datasets will be created by introducing controlled perturbations to account for noise and modelling discrepancies. A sequence of model updating strategies will then be implemented, starting from deterministic approaches (e.g. sensitivity-based or least-squares updating of stiffness parameters) and progressing towards simplified data assimilation techniques (e.g. Kalman filtering concepts or regularised inverse formulations). The student will quantify the sensitivity of modal observables to different damage scenarios, assess identifiability limits, and investigate the influence of sensor placement and sparsity on update accuracy. The final outcome will be a validated computational workflow demonstrating how modal data can be systematically integrated into a digital twin to improve state estimation, reduce uncertainty, and support structural health and usage monitoring in an aerospace context.

Sharif Khodaei, Zahra Prof.

Project no: ZSK03 *

Project title: Real-Time Usage Monitoring of Aircraft Structures via Strain-Based Digital Twin Reconstruction (S) - 2 projects available

Supervisor: Sharif Khodaei, Zahra Prof.

Co-

supervisor(s):

Category: Experimental: Ready-made
experiment;Computational;Numerical;Analysis

Software:

Confidential: No

This project aims to develop a computational framework for real-time usage monitoring of aircraft structures based on sparse strain measurements, forming the basis of a strain-driven digital twin. The central objective is to infer the full-field structural response from strain data recorded at a limited number of sensor locations, enabling continuous estimation of structural states under realistic operational conditions. The motivation arises from a fundamental constraint in practical Structural Health and Usage Monitoring (SHUM) systems: while strain sensors (e.g. strain gauges or fibre optic systems) provide high-fidelity local measurements, they do not directly capture the global structural response required for robust assessment of load distribution and structural behaviour. Bridging this gap is essential for scalable, sensor-efficient digital twin implementations in aerospace applications. Technically, the project will begin with the development of a finite element (FE) model of a representative aircraft structure (e.g. wing panel, fuselage section, or stiffened plate), which

serves as the physics-based backbone of the digital twin. Synthetic or provided strain data at discrete locations will be used to emulate onboard sensing. The student will then implement methodologies to reconstruct the full-field strain distribution from sparse measurements, exploring approaches such as interpolation techniques (e.g. radial basis functions), reduced-order modelling, or inverse methods based on FE model superposition. Following strain reconstruction, the corresponding stress field will be derived using appropriate constitutive relations, accounting for material behaviour (isotropic or composite). The reconstructed stress or strain fields will then be used to characterise structural usage in real time, for example through load spectrum reconstruction, identification of critical load events, or statistical descriptors of operational exposure. The project may further investigate the sensitivity of reconstruction accuracy to sensor placement, noise, and model-form uncertainty. The final outcome will be a computational workflow that demonstrates how sparse strain sensing can be systematically transformed into actionable structural usage information within a digital twin framework, supporting next-generation SHUM systems for aerospace structures.

Sharif Khodaei, Zahra Prof.

Project no: ZSK04 *

Project title: Structural Health Monitoring of Composite Structures under Operational and Environmental Variability (S) - 2 projects available

Supervisor: Sharif Khodaei, Zahra Prof.

Co-supervisor(s):

Category: Literature;Computational;Analysis;Experimental: Ready-made experiment

Software:

Confidential: No

This project aims to investigate the state of the art in structural health monitoring of composite aerospace structures, with a specific focus on how damage detection and characterisation methods perform under realistic operational and environmental conditions. The motivation is that composite structures are increasingly used in aircraft due to their high specific stiffness and strength, but their damage mechanisms—such as delamination, matrix cracking, fibre breakage, impact damage and debonding—can be difficult to detect and quantify using conventional inspection methods. A key challenge is that SHM signals are strongly influenced not only by damage, but also by temperature, loading condition, boundary condition variation, moisture, ageing and sensor variability. Distinguishing damage-sensitive features from variability-induced changes is therefore essential for reliable and certifiable SHM systems.

Technically, the project will begin with a thorough literature review covering guided waves, vibration/modal methods, strain-based monitoring, acoustic emission, data-driven SHM, and

hybrid approaches for composite structures. Particular attention will be given to methods that address environmental and operational variability, including baseline compensation, baseline-free approaches, transfer learning, statistical pattern recognition, physics-informed modelling, and uncertainty-aware damage diagnosis. The computational part of the project will use either existing experimental datasets, analytical signals, or finite-element-generated responses to develop and test new damage detection and characterisation techniques. Depending on data availability, the student may analyse guided wave signals, modal parameters, strain fields, or other SHM observables from pristine and damaged composite configurations. The work will involve signal processing, feature extraction, damage-sensitive index formulation, classification or regression models, and assessment of robustness under simulated or measured variability. The final outcome will be a critical review of current SHM capability for composites, together with a prototype computational methodology demonstrating how existing or simulated data can be used to improve damage detection, localisation, and severity estimation under realistic aerospace operating conditions.

Sherwin, Spencer Prof.

Project no: SS01

Project title: Computational fluid dynamics using spectral/hp element methods (S) - 3 projects available

Supervisor: Sherwin, Spencer Prof.

Co-supervisor(s):

Category: Computational;Analysis;Numerical

Software: Nektar++VisIt or Paraview

Confidential: No

Projects are available in the development and application of high order finite element methods known as spectral/hp element discretisations. These project will be based around the library, Nektar++, details of which can be found under (www.nektar.info). The primary focus of the projects will be the development and/or application of these techniques to incompressible and compressible flow problems similar to those shown under www.nektar.info. Although there are no specific prerequisites for the projects computing and coding skills are preferable. Interest in development of GPU based solvers is also attractive.

Sherwin, Spencer Prof.

Project no: SS02 *
Project title: Pulse wave modelling of arterial networks. (S) - 1 project available

Supervisor: Sherwin, Spencer Prof.

Co-supervisor(s):

Category: Computational;Theoretical;Numerical

Software: Nektar++
 Paraview

Confidential: No

The 1D Euler equations which commonly arise in Aeronautics are the equations that govern pulse wave propagation in the human arterial system, indeed Euler's original work was related to blood flow rather than gas dynamics. In this project we will develop an arterial pulse wave model to reproduce the waves in the human cardiovascular system on GPUs. This will leverage on-going developments in the Nektar++ redesign project for GPU and vectorized capable solvers. Additional development may also include a new visco-elastic formulation and an implicit in time discretisation for greater efficiency.

Sherwin, Spencer Prof.

Project no: SS03 *
Project title: Software Engineering in open source CFD package Nektar++ - 2 projects available

Supervisor: Sherwin, Spencer Prof.

Co-supervisor(s): Cantwell, Chris Dr

Category: Computational;Numerical

Software: Nektar++Paraview/VisIt

Confidential: No

Software engineering is an integral part of any computational package. For example, to improve the robustness and flexibility of the Nektar++ (www.nektar.info) flow solver and to help provide an easier way to understand the fundamentals of the spectral/hp element methods we have been developing a python interface. We also maintain a Continual Integration environments within gitlab (gitlab.nektar.info) which helps us ensure that previous developments are robust with respect to new developments. However we do not currently have a performance monitoring framework. This project is therefore focussed on the software engineering aspects of the code development and can be tailored depending on mutual discussion on your interests and the needs of the project.

Steiros, Kostas Dr

Project no: *KS01*

Project title: Machine Learning of Hidden Symmetries in Turbulence
- 1 project available

Supervisor: Steiros, Kostas Dr

Co- Zhu, Kaixin

supervisor(s):

Category: Computational;Theoretical

Software: Matlab, StarCCM, Python

Confidential: No

Many theoretical results in fluid dynamics, as well as in other areas of mechanics, rely on identifying symmetries in the governing field equations. Classic examples in the field of turbulent flows include the logarithmic law of the wall in turbulent boundary layers and Kolmogorov's $-5/3$ scaling law for the turbulent kinetic energy cascade, both of which are derived using symmetry arguments.

A rigorous mathematical framework for identifying such symmetries is provided by Lie group theory. However, this approach is generally limited to highly simplified and idealized problems, such as the Blasius boundary layer that we studied in Aerodynamics 3.

Despite these limitations, if symmetries are present in a physical system, they must also manifest in observable quantities, such as experimental or numerical data. This suggests that, in principle, symmetries could be identified using data-driven techniques.

The aim of this project is to develop a neural network-based algorithm capable of extracting symmetries (or self-similarities) from turbulent velocity fields. As a first step, the method will be applied to a relatively simple flow configuration obtained from a RANS solver, for example a bluff body wake generated by the student. If successful, the approach will then be extended to real experimental datasets.

This is a challenging project that requires a strong foundation in both the underlying mathematics and fluid mechanics.

Steiros, Kostas Dr

Project no: *KS02*

Project title: Analytical and Numerical modelling of Wind Turbines Under Dynamically Yawed Conditions - 1 project available

Supervisor: Steiros, Kostas Dr

Co- Thompson, Craig

supervisor(s):

Category: Computational;Theoretical;Numerical

Software:

Confidential: No

There is currently significant interest in dynamically optimizing the yaw angle of wind turbine rotors, as recent studies suggest that this strategy can substantially improve the overall efficiency of wind farms. However, there are presently no simplified models capable of accurately predicting the aerodynamic behaviour of wind turbines or entire wind farms under time-varying yaw conditions.

The aim of this project is to develop a time-dependent analytical wake model that can capture these dynamic yaw effects. To support this effort, the student will perform unsteady simulations—using either URANS or LES—of a single wind turbine or a full wind farm operating under realistic conditions, including variations in incoming turbulence, wind direction, and turbine control settings.

A key objective will be to identify and characterize a convection parameter governing wake evolution, which will then be incorporated into a two-dimensional, time-dependent analytical wake model. There is scope to extend this framework further by including wake superposition methods, three-dimensional wake formulations, and atmospheric boundary layer inflow conditions.

Finally, the project will investigate how the developed model can be used to optimize wind farm performance under time-varying operating strategies, potentially leveraging techniques such as reinforcement learning.

This project requires a strong interest in analytical modelling, scientific computing, and wind energy systems.

Steiros, Kostas Dr

Project no: *KS03*

Project title: Wind tunnel measurements of aerofoils - 1 project available

Supervisor: Steiros, Kostas Dr

Co-

supervisor(s):

Category: Experimental;Design

Software:

Confidential: No

We have recently developed a revised formulation of the classical Blade Element Momentum (BEM) theory for wind turbines and aircraft propellers. The key innovation in this framework is the explicit modelling of rotor wake pressure, which is not accounted for in the original BEM formulation. Further work is now required to validate and extend this model through laboratory experiments.

BEM relies on accurate lift and drag data for aerofoils over a range of angles of attack and Reynolds numbers. However, such data are often incomplete or unavailable in the literature. To address this, the student will carry out aerofoil force-balance measurements in the T2 wind tunnel at Imperial College London. This project is well suited to a student with a strong interest in experimental fluid mechanics and measurement techniques.

Steiros, Kostas Dr

Project no: *KS04*

Project title: Aerodynamics of Selective Laser Melting - 1 project available

Supervisor: Steiros, Kostas Dr

Co-supervisor(s): Panesar, Ajit Dr

Category: Computational; Numerical

Software:

Confidential: No

Selective Laser Melting (SLM) is an advanced additive manufacturing technology that is gaining increasing attention due to its ability to produce components that are not feasible using traditional methods, particularly complex geometries at low production volumes. In SLM, a thin layer of powder is spread over a build platform and selectively melted using a heat source, typically a Nd:YAG laser. Successive layers are then deposited and fused, with the process repeated until the full geometry is formed.

A critical feature of SLM systems is the presence of a jet of inert gas, usually Argon, which is used to remove by-products of the melting process such as spatter and smoke. However, this gas flow is also believed to generate large-scale vortical structures and stagnation regions within the build chamber. These flow features may contribute to part defects that are not yet fully understood.

The aim of this project is to investigate and demonstrate the existence of these flow structures using RANS simulations of the internal flow within an SLM machine. The numerical results will be complemented and validated using a dataset provided by the industrial partner Alloyed.

Steiros, Kostas Dr

Project no: **KS05**

Project title: Manufacturing and testing of a wind tunnel active grid system - 1 project available

Supervisor: Steiros, Kostas Dr

Co-supervisor(s): Thompson, Craig

Category:

Experimental;Experimental: Manufacturing intensive;Design

Software:

Confidential: No

While wind tunnels are well suited for generating steady, laminar inflows, many practical applications require controlled unsteady or gusty flow conditions. Examples include the aerodynamics of tall buildings, wind turbines, and a wide range of fundamental studies in turbulence. However, such “gusty” wind tunnel capabilities are rare, and currently do not exist at Imperial College London.

To address this limitation, this project will focus on the design and manufacture of an “active grid”—a system of multiple paddles that oscillate in a controlled manner to generate tailored turbulent flow conditions. The paddles will be 3D-printed and actuated using miniature stepper motors and microcontrollers, enabling precise control over the properties of the generated turbulence.

This project is ideally suited to a student with a strong interest in experimental design, electronics, and manufacturing, particularly in the context of fluid mechanics.

The successful development of this system will provide a valuable experimental capability for several research groups within the department, enabling new investigations into unsteady aerodynamics and turbulence.

Tagarielli, Vito Dr

Project no: **VT01 ***

Project title: Research topics in experimental solid mechanics (S) - 2 projects available

Supervisor: Tagarielli, Vito Dr

Co-supervisor(s):

Category: Experimental: Structures

Software: -

Confidential: Yes

Select this project if you want to perform research in one of the areas below; the detailed topic is to be agreed in the future, based on your inclinations and skills, as well as demand. Please be aware that attending campus and working hard will be essential, from October to June. Please discuss informally with the supervisor the details of the possible activities.

Activities this year might include:

- Development of new mechanical testing techniques, including multiaxial tests on highly instrumented specimens.
- Experiments in fracture mechanics of composites and high performance materials.
- Dynamic mechanical testing using the Hopkinson bar setup.
- Manufacturing (by bacteria) and testing of biomaterials like bacterial nanocellulose.
- Analysing real x-ray scans of composites microstructures and developing advanced digital image/volume correlation and machine learning to generate virtual microstructures.

These activities will involve operating delicate equipment and dangerous materials, sometimes substantial coding (x-ray project), keeping clear logs of all measurements and processing, presenting the associated data, but they will also require an understanding of the theoretical basis of the phenomena observed, a review of the existing literature and some degree of inventive. The experimental activities can be tailored, to some limited degree, to the inclinations of the student and might be complemented by theoretical/numerical modelling and applications of machine learning for students interested in these additional themes.

These experimental projects are harder, not easier than average! Note that this project proposal dates to April 2026 and my interests might have slightly changed by October.

Tagarielli, Vito Dr

Project no: VT02 *

Project title: Research topics in experimental fluid mechanics (S) - 2 projects available

Supervisor: Tagarielli, Vito Dr

Co-supervisor(s):

Category: Experimental

Software:

Confidential: Yes

Select this project if you want to perform research in one of the areas below; the detailed topic is to be agreed in the future, based on your inclinations and skills, as well as student demand. Please be aware that attending campus and working hard will be essential, from October to June. Please discuss informally with the supervisor the details of the possible activities. These are likely to be finalised closer to the project start, however this year they might include:

- Development of deflagration and deflagration-to-detonation transition experiments for flammable gaseous mixtures, following and expanding the work done in previous years in this area in the department.

- Development (from the ground up) of an experimental setup and technique to drive oscillation, cavitation and collapse of bubbles in rubbery media (this involves playing with rubbers, hydrogels, liquid metals and high voltages).
- Development (from the ground up) of an experimental setup and technique to measure the laminar/turbulent transition in liquid metals and polymers, including mercury, Galinstan and a polymeric resin at room temperature.

These activities will involve operating delicate equipment and dangerous materials, keeping clear logs of all measurements and processing/presenting the associated data, but they will also require an understanding of the theoretical basis of the phenomena observed, a review of the existing literature and some degree of inventive. The experimental activities can be tailored, to some limited degree, to the inclinations of the student and might be complemented by theoretical/numerical modelling and applications of machine learning for students interested in these additional themes. These experimental projects are harder, not easier than average! Note that this project proposal dates to April 2026 and my interests might have slightly changed by October.

Tagarielli, Vito Dr

Project no: VT03 *

Project title: Applications of machine learning to problems in mechanics (S)
- 3 projects available

Supervisor: Tagarielli, Vito Dr

Co-supervisor(s):

Category: Computational

Software: abaqus

Confidential: Yes

Select this project if you want to perform research in this area. This might require knowledge of the basics of the numerical techniques used in FEM, CFD and coupled multiphysics simulations, basic knowledge of machine learning techniques, coding and software skills, ability to read and digest difficult papers containing advanced physics and maths. Some activities might require using external online tools (eg Nvidia Physics Nemo). The project will be 100% computational and regular attendance to campus is not essential. Please discuss informally with the supervisor the details of the possible research. These are likely to be finalised closer to the project start, and this year they might include:

- Novel machine learning algorithms for numerical simulations: conceptual studies and theoretical development (math/coding enthusiast(s) sought)
- Machine learning in solid mechanics: new algorithms for scaling-up and generalisation problems (FE/coding enthusiast sought)

- Machine learning in fluid mechanics and fluid-structure interaction. Coupled CFD/FE Eulerian/Lagrangian simulation of explosive events and development of fast surrogate models (FE/CFD/coding enthusiast sought)
 - Ensemble learning, knowledge-based ML, data fusion, and techniques to integrate and homogenise information from multiple sources into intelligent agents (math/coding enthusiast sought)
 - Surrogate models of particle-based simulations and their coupling with CFD and FE models (math/coding enthusiast sought).
- All projects will include a review of the existing literature. The research and its supervision will be tailored to some limited degree to the inclinations and skills of the student, and allocated to specific students after preliminary discussions.

Tagarielli, Vito Dr

Project no: VT04 *

Project title: Agent-based simulation of physical or socioeconomic systems (S)
- 1 project available

Supervisor: Tagarielli, Vito Dr

Co-

supervisor(s):

Category: Computational

Software:

Confidential: Yes

This project requires full-time efforts and a good degree of initiative and confidence. It can be undertaken fully remotely. The project aims at developing agent-based modelling frameworks of complex systems. In such frameworks, the system is analysed by modelling the response of individual agents, who operate following simple rules, and interacting with each other and the external world, while following or reacting to some system-level directions, which are in turn affected by the collective behaviour of the agents. An example could be a colony of bacteria (agents) living and reproducing in a liquid medium, affected by their mutual interactions but also by the concentration of oxygen and nutrients in the liquid (high-level system dynamics), which is in turn affected by the collective behaviour of the bacteria (look at my paper in Cellulose).

This year i'd like to select the type of project based on inclination and conversation with the allocated student. I am interested in physical sytems (eg applications to solid mechanics, biology, applied mechanics) but also in political and socioeconomic systems (where agents are typically individuals or groups of individuals, and policies represent system-level dynamics).

The student will gain experience in computational modelling, simulation design, data integration, and the analysis of complex adaptive systems. From my perspective, this project

is an excuse to get closer to emergence in complex systems and so I would like the chosen topic to be able to observe some emergent phenomena. By October, I might have additional and more detailed topic proposal for this activity.

Tagarielli, Vito Dr

Project no: VT05 *

Project title: Training lightweight LLMs (S) - 4 projects available

Supervisor: Tagarielli, Vito Dr

Co-supervisor(s):

Category: Computational

Software: -

Confidential: Yes

This project is purely computational and can be undertaken fully remotely. It is obviously heavy in machine learning, and ideally the students should be inclined towards computation and should have already had some exposure to training of neural networks (or be very confident to pick that up quickly).

The objectives are:

- identify open-source LLM platforms that are sufficiently light in computational terms, to be able to run on mobile phones or ordinary desktops/laptops;
- learn how to train these LLMs from scratch, retrain them with new data, perform transfer learning, add and train additional layers, etc.

We would like to use LLM-based agents to control applications running on mobile phones or ordinary computers. We would like to impart different specific bits of sectorial knowledge to different LLM agents, and explore collective decision making by multiple LLM agents.

The project could suit a team of 2-3 students working in parallel, as it requires expertise in software, computation and machine learning. Get in touch with the supervisor if you have ideas about who the members of that team could be.

Venetsanos, Demetrios Dr

Project no: DV01 *

Project title: Geometric Tolerancing effects on the axial crush performance of Additive Manufactured strut-based lattice architectures (S) - 2 projects available

Supervisor: Venetsanos, Demetrios Dr

Co-supervisor(s): Vosniakos, George C Prof., Benardos, Panorios Assoc Prof. (NTUA)

Category: Computational;Design;Analysis;Numerical

Software: Abaqus or Ansys Workbench or LS-Dyna or any available industry-standard CAE software (to include a solver for explicit dynamics); Python or Matlab.

Confidential: No

A lattice structure is a space-filling unit cell that can be tessellated along any axis with no gaps between cells. These structures are of primary importance in aerospace engineering, as they can combine, in an optimum manner, weight, energy and advanced manufacturing time reduction.

The aim of this project is to investigate the effect of size and geometric tolerances on the mechanical performance of additively manufactured strut-based lattice structures, when an axial crushing force is applied. To this end, lattice unit cells, such as (but not limited to) Body-Centred Cubic (BCC), Face-Centred Cubic (FCC), Simple Cubic (SC), and Kelvin Unit (KU), will be considered. Also, well-known distributions of tolerances, taken from the literature, will be introduced. Furthermore, industry-standard software will be used to simulate the application of the axial crushing load.

This project is within an on-going collaboration with the National Technical University of Athens (Greece) – Section of Manufacturing Technology.

References

Li, D., Qin, R., Chen, B., Zhou, J. (2021) Analysis of mechanical properties of lattice structures with stochastic geometric defects in additive manufacturing. Materials Science and Engineering: A, vol. 822, 141666, ISSN 0921-5093. DOI: 10.1016/j.msea.2021.141666.

Cao., X., et al. (2020) Compression experiment and numerical evaluation on mechanical responses of the lattice structures with stochastic geometric defects originated from additive-manufacturing. Composites Part B: Engineering, vol. 194, 108030, ISSN 1359-8368; DOI:10.1016/j.compositesb.2020.108030.

Lozanovski, B., et al. (2019) Computational modelling of strut defects in SLM manufactured lattice structures. Materials & Design, vol.171, 107671, ISSN 0264-1275. DOI:10.1016/j.matdes.2019.107671

Venetsanos, Demetrios Dr

Project no: DV02 *

Project title: Geometric Tolerancing effects on the oblique crush performance of Additively Manufactured strut-based lattice structures (S) - 2 projects available

Supervisor: Venetsanos, Demetrios Dr

Co-supervisor(s): Vosniakos, George C Prof., Benardos, Panorios Assoc Prof. (NTUA)

Category: Computational;Design;Analysis;Numerical

Software: Abaqus or Ansys Workbench or LS-Dyna or any available industry-standard CAE software (to include a solver for explicit dynamics); Python or Matlab.

Confidential: No

This project is similar to the project titled "Geometric Tolerancing effects on the axial crush performance of Additive Manufactured strut-based lattice structures", the difference being that an oblique crushing load, instead of an axial one, is considered. All other information is the same.

References

Li, D., Qin, R., Chen, B., Zhou, J. (2021) Analysis of mechanical properties of lattice structures with stochastic geometric defects in additive manufacturing. Materials Science and Engineering: A, vol. 822, 141666, ISSN 0921-5093. DOI: 10.1016/j.msea.2021.141666.

Cao., X., et al. (2020) Compression experiment and numerical evaluation on mechanical responses of the lattice structures with stochastic geometric defects originated from additive-manufacturing. Composites Part B: Engineering, vol. 194, 108030, ISSN 1359-8368; DOI:10.1016/j.compositesb.2020.108030.

Lozanovski, B., et al. (2019) Computational modelling of strut defects in SLM manufactured lattice structures. Materials & Design, vol.171, 107671, ISSN 0264-1275. DOI:10.1016/j.matdes.2019.107671

Venetsanos, Demetrios Dr

Project no: *DV03 **

Project title: Crashworthiness of impact attenuators under axial crushing for electric Vertical Take-Off and Landing (eVTOL) vehicles - 1 project available

Supervisor: Venetsanos, Demetrios Dr

Co-supervisor(s): Vasileiou, Anastasia Dr (University of Manchester)

Category: Computational;Design;Analysis;Numerical

Software: Abaqus or Ansys Workbench or LS-Dyna or another equivalent industry standard CAE software, to include explicit dynamics; Python or Matlab

Confidential: No

For the emerging electric Vertical Take-Off and Landing (eVTOL) vehicle industry, as well as for formula cars, there is a need for crash mitigation systems, which will be able to attenuate impact energy during a crash. One solution is to introduce impact attenuators, in the form of tubular structures suitably attached to the vehicle. The impact energy these systems should absorb are described in specifications (e.g. Federal Aviation Administration, for aerial vehicles; FAA for formula cars).

This project concerns the design of impact attenuators in the form of thin-walled unitubular configurations of variable wall thickness, with longitudinal and/or circumferential stiffeners,

of variable thickness and profile, under the application of an axial crushing load. It is assumed that the tubular configurations are manufactured using an Additive Manufacturing process (e.g. 3D-printing), thus the stiffeners will, seamlessly and simultaneously, be manufactured along with the main body of the configurations (i.e. the stiffeners will not be added at a later stage as external components).

The novelty of this study is the consideration of stiffeners whose both thickness and profile may vary along the longitudinal and/or circumferential direction.

To measure the said crashworthiness, typical metrics will be used, e.g. Total Absorbed Energy (TEA), Specific Absorbed Energy (SEA), Peak and Average Crushing Force, Peak and Average Acceleration, Crushing Force Efficiency, etc. For the FE analyses required, an industry-standard CAE software (e.g. Abaqus, Ansys Workbench, LS-Dyna) will be used. For the optimisation, it is possible to use a scheme found in one of the Matlab Optimisation toolboxes, to include Artificial Intelligence-powered approaches. Alternatively, a Taguchi approach may be utilised. For the description of the non-uniform stiffeners, it is possible to use splines or similar geometric entities.

References

- Littell, J. D. (2019) Challenges in Vehicle Safety and Occupant Protection for Autonomous electric Vertical Take-off and Landing (eVTOL) Vehicles. AIAA/IEEE Electric Aircraft Technologies Symposium (EATS), pp. 1-16. DOI: 10.2514/6.2019-4504.
- Federal Register. "Special Conditions: Airbus A350-900 Airplane; Crashworthiness, Emergency Landing Conditions." Vol 79. No. 43. Friday July 25, 2014.
- Song, J., Xu, S., Zhou, J., Huang, H., Zou, M. (2021) Experiment and numerical simulation study on the bionic tubes with gradient thickness under oblique loading. Thin-Walled Structures, vol.163, 107624, ISSN 0263-8231; doi: 10.1016/j.tws.2021.107624.

Venetsanos, Demetrios Dr

Project no: DV04 *

Project title: Crashworthiness of impact attenuators under oblique crushing for electric Vertical Take-Off and Landing (eVTOL) vehicles (S) - 1 project available

Supervisor: Venetsanos, Demetrios Dr

Co-supervisor(s): Vasileiou, Anastasia Dr (University of Manchester)

Category: Computational;Design;Analysis;Numerical

Software: Abaqus or Ansys Workbench or LS-Dyna or another equivalent industry standard CAE software, to include explicit dynamics; Python or Matlab

Confidential: No

This project is similar to the project titled "Optimised crashworthiness of variable thickness impact attenuators under axial crushing", the difference being that an oblique crushing load, instead of an axial one, is considered. All other information is the same.

References

Littell, J. D. (2019) Challenges in Vehicle Safety and Occupant Protection for Autonomous electric Vertical Take-off and Landing (eVTOL) Vehicles. AIAA/IEEE Electric Aircraft Technologies Symposium (EATS), pp. 1-16. DOI: 10.2514/6.2019-4504.

Mohamed, M.N., Kumar, A.P. (2017) New Insight to Improve Energy Absorption Characteristics of Long Circular Tubes with Stiffeners as Controllable Energy-dissipating Devices. Procedia Engineering, Vol. 173, pp. 1399-1406, ISSN 1877-7058. DOI: 10.1016/j.proeng.2016.12.199.

Federal Register. "Special Conditions: Airbus A350-900 Airplane; Crashworthiness, Emergency Landing Conditions." Vol 79. No. 43. Friday July 25, 2014.

Song, J., Xu, S., Zhou, J., Huang, H., Zou, M. (2021) Experiment and numerical simulation study on the bionic tubes with gradient thickness under oblique loading. Thin-Walled Structures, vol.163, 107624, ISSN 0263-8231; doi: 10.1016/j.tws.2021.107624.

Qi, C. & Yang, S. (2014) Crashworthiness and lightweight optimisation of thin-walled conical tubes subjected to an oblique impact. Int. J. Crashworthiness 19, 334–351.

Tran, T. & Baroutaji, A. (2018) Crashworthiness optimal design of multi-cell triangular tubes under axial and oblique impact loading. Engineering Failure Analysis, vol. 93, pp. 241-256, ISSN 1350- 6307; doi: 10.1016/j.engfailanal.2018.07.003.

Abramowicz, W., & Jones, N. (1984). Dynamic axial crushing of circular tubes. International Journal of Impact Engineering, 2(3), 263–281. DOI: 10.1016/0734-743X(84)90010-1

Zhu, G., Sun, G., Yu, H., Li, S. & Li, Q. (2018) Energy absorption of metal, composite and metal/composite hybrid structures under oblique crushing loading. Int. J. Mech. Sci., 135, 458–483.

Venetsanos, Demetrios Dr

Project no: DV05 *

Project title: Student's own project proposal (S) - 3 projects available

Supervisor: Venetsanos, Demetrios Dr

Co-supervisor(s):

Category: Computational;Design;Analysis

Software:

Confidential: No

This project recognises that the most motivated research often begins not with an assigned title, but with a question a student has already been living with. Candidates who have identified a substantive and original research problem, whether through industrial experience, prior study, or independent curiosity, are encouraged to propose it as their Final Year or MSc Project.

Proposals should be grounded in structural design, analysis, or optimisation, though genuinely multidisciplinary projects may be considered, where project supervision will be shared with another member of staff.

Students returning from an industrial placement are particularly, though not exclusively, encouraged to draw on the technical challenges they encountered in practice.

Wynn, Andrew Dr

Project no: **AW01**

Project title: Data-driven modelling of turbulent flows using PDEs and Neural Networks - 1 project available

Supervisor: Wynn, Andrew Dr

Co-

supervisor(s):

Category: Computational;Theoretical;Analysis

Software: Matlab and Python.

Confidential: No

Creating reduced-order models for fluid flows opens the door to rapid numerical simulations, drives understanding of their physics, allow estimates of a flow's state and, ultimately, leads to flow control solutions for improved performance. Low-cost models are important, since resolved numerical simulations of the Navier-Stokes require a large computational. For turbulent flows, this typically prevents a full investigation of a their behaviour. A significant challenge is to create low-cost models that are accurate across a wide range of flow conditions. This question is central to data-driven modelling where it can be easy to train a model which performs well on a specific data-ensemble, yet performs poorly when flow conditions, such as Reynolds number, change.

The aim of this FYP is to use powerful AI methods to train simple PDEs which can underpin nonlinear reduced-order models of fluid flows. Technically, this will be achieved by embedding elliptic PDEs in an encoder-PDE-decoder mapping using deep neural networks. The idea is that PDEs can create generalisable models as flow conditions change and can also have low-computational cost if the class of model is chosen carefully. The FYP will train models, then use them to attempt to solve problems related to flow sensing, that is, estimating the current state of a fluid flow from a small number of sensor measurements. This project can be taken by two students.

Wynn, Andrew Dr

Project no: **AW02**

Project title: Mathematical analysis of fluid flows - 2 projects available

Supervisor: Wynn, Andrew Dr

Co-supervisor(s):

Category: Computational;Theoretical;Analysis

Software: MATLAB

Confidential: No

Due to their complexity, there are very few known closed-form solutions to the Navier-Stokes equations. You would certainly not expect to be able to work out, by hand, the velocity field for a turbulent flow! Given this, it is surprising that it can be possible to prove rigorous bounds on the time-averaged properties of turbulent flows via mathematical analysis of the Navier-Stokes equations. For example, one can prove statements such as “the average drag or heat transfer must be less than a certain value”. This gives potentially important information about the performance of a flow, without the need to construct, or simulate, its solution. Two FYP projects are available. One will look at flows of non-Newtonian fluids such as the Oldroyd-B or FENE-P models for polymeric flows. The other will consider bounds on the skin friction for turbulent boundary layers. In both cases, computational optimization may be used to inform the mathematical analysis of the underlying PDE models. These projects are for students who enjoy the mathematical aspects of fluid dynamics and PDEs. Elective models on Turbulence and Mathematics 3 are not strict prerequisites, but are complementary to some of the methods that will be used in this project.

Wynn, Andrew Dr

Project no: **AW03**

Project title: Nonlinear frequency response analysis using computational optimization - 1 project available

Supervisor: Wynn, Andrew Dr

Co-supervisor(s):

Category: Theoretical;Computational

Software: MATLAB

Confidential: No

The frequency response function is a fundamental tool with which to understand the forced response of a control system and is vital for many aspects of control design. Computing the frequency response for linear systems is easy, but much less so for nonlinear systems. This project will investigate methods of bridging this gap using tools from computational optimization, specifically using Lyapunov methods, linear matrix inequalities and

semidefinite programming – topics which are covered on the Advanced Control module (see also the paper with doi: 10.23919/ACC.2004.1383957). Target applications involve the forced response of highly flexible structures of importance to large wind turbines and next-generation high-aspect-ratio wing design.

Wynn, Andrew Dr

Project no: **AW04**

Project title: How to design fluid flows for optimal heat and contaminant transport - 2 projects available

Supervisor: Wynn, Andrew Dr

Co-

supervisor(s):

Category: Computational;Theoretical

Software: MATLAB

Confidential: No

The transport of scalars such as hot air, cold air, pollutants, particulates, or airborne viruses, is of central importance to building ventilation, urban pollution, thermal management, and transport processes across geophysics. Understanding how to best use fluid flows to transport contaminants is extremely difficult, both mathematically and practically, due to the complexity of the Navier-Stokes questions (see for example the paper <https://doi.org/10.1137/17M1150220>). This project will investigate a recent approach to finding optimal flows for scalar transport using tools from computational optimization, particularly Semi-Definite Programming (students taking the Advanced Control module will see an introduction to this concept). The aim is to convert the question “how to find a fluid flows which removes heat/contaminants as efficiently as possible” into a convex optimization problem that can be solved at a reasonably computational cost. This will involve aspects of PDE analysis, control systems, finite element analysis, and computational optimization.

Wynn, Andrew Dr

Project no: **AW05**

Project title: Estimation and sensing of fluid flows using sliding mode observers
- 1 project available

Supervisor: Wynn, Andrew Dr

Co-supervisor(s):

Category: Computational;Theoretical

Software:

Confidential: No

A fundamental and practically important question in fluid mechanics is how to best estimate the current state of a fluid flow, given only sensor readings such as boundary pressure measurements, or planar PIV data. This question is hard due to both the dynamic complexity of fluid flows and the relative sparsity of typical sensor information in practical situations. This project will investigate how to use the nonlinear state estimation technique of sliding mode observers to try to solve the fundamental problem in fluid mechanics. Estimator design is covered in 3rd year control systems for linear systems, while this project will consider nonlinear extensions of this method required to solve nonlinear problems in fluid mechanics. Examples from turbulent channel flows will be considered as part of the project.